Real Analysis

Module 2RA

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Chapter 0

Preliminaries

In this chapter we give a short overview of basic concepts on sets and number systems. We expect that you are familiar with these concepts. We will not discuss them in class. Please make sure that you read this chapter and that you are comfortable with the concepts.

0.1 Sets

Intuitively, a *set* is a collection of objects. Given a set A and an object x, we write $x \in A$ if x is in the set A and say that x is an *element* of A. Otherwise, we write $x \notin A$ and say that x is not an element of A. The *empty set* $\varnothing$ is the set that contains no elements.

We say that two sets A and B are *equal* and write $A = B$ if every element of A is an element of B and every element of B is an element of A, i.e.

$$\forall\, x : x \in A \iff x \in B.$$

We say that a set A is a *subset* of a set B and write $A \subseteq B$ or $B \supseteq A$ if every element of A is also an element of B, i.e.

$$\forall\, x : x \in A \implies x \in B.$$

Clearly, $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

For every set A we have $\varnothing \subseteq A$ and $A \subseteq A$. If $B \subseteq A$ and $B \neq A$, then B is called a *proper subset* of A. The *union* of two sets A and B is the set

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

The *intersection* of two sets A and B is the set

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

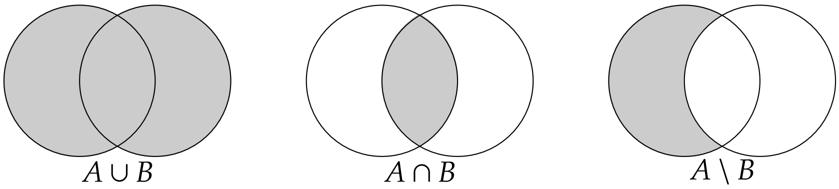

Figure 0.1: $A \cup B$, $A \cap B$, $A \setminus B$

The *difference* of two sets A and B is the set

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

Two sets A and B are called *disjoint* if $A \cap B = \varnothing$.

Notice that $A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$ and that the sets $A \setminus B$, $A \cap B$ and $B \setminus A$ are pairwise disjoint.

The notions of unions and intersections can be extended to more than two sets: If $A_1, \dots, A_n$ is a *collection* of sets (or *family* of sets), we define

$$\bigcup_{k=1}^{n} A_k = \{x : x \in A_k \text{ for some } k \in \{1, \dots, n\}\},$$
$$\bigcap_{k=1}^{n} A_k = \{x : x \in A_k \text{ for all } k \in \{1, \dots, n\}\}.$$

Similarly, we will consider unions and intersections of countably many sets.

Intervals of real numbers are defined as follows. If $a, b \in \mathbb{R}$ with $a \le b$, we define

$$[a,b] = \{x \in \mathbb{R} : a \le x \le b\},$$
$$(a,b) = \{x \in \mathbb{R} : a < x < b\},$$
$$(a,b] = \{x \in \mathbb{R} : a < x \le b\},$$
$$[a,b) = \{x \in \mathbb{R} : a \le x < b\}.$$

There are also infinite intervals

$$[a,\infty) = \{x \in \mathbb{R} : x \ge a\},$$
$$(a,\infty) = \{x \in \mathbb{R} : x > a\},$$
$$(-\infty,b] = \{x \in \mathbb{R} : x \le b\},$$
$$(-\infty,b) = \{x \in \mathbb{R} : x < b\}.$$

Note that "$-\infty$" and "∞" are not real numbers, and the notation "($-\infty$" and "∞)" means that the corresponding intervals are infinite.

0.2 Number systems

In this module we will work with sets of numbers. The most important sets of numbers are:

- The *natural numbers* $\mathbb{N} = \{1, 2, 3, 4, 5, \ldots\}$.

  Notice that we do not include 0 to be a natural number. Some authors do, so you must be careful when you work with literature.

- The *integers* $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$.

- The *rational numbers* $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}$.

- The *real numbers* $\mathbb{R}$.

We have $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$; each number system is a proper subset of a larger one.

The natural numbers $\mathbb{N}$ are the numbers that we use for counting. They can be defined axiomatically (e.g., via the Peano Axioms). We do not go into details in this module.

Within the number system $\mathbb{N}$, the operation of addition works well. However, a difference of two natural numbers is not always a natural number, for example, $2 - 3 = -1$ is not a natural number. This motivates us to extend the number system $\mathbb{N}$ to a larger number system $\mathbb{Z}$ where also subtraction can be performed. This is done by adding the number 0 and a copy of $\mathbb{N}$ with a negative sign in front of each number.

The operations of addition, subtraction and multiplication can be performed within the number system $\mathbb{Z}$, but a quotient of two integers is not always an integer, for example, $2 \div 3 = \frac{2}{3}$ is not an integer. We enlarge $\mathbb{Z}$ to $\mathbb{Q}$ by considering fractions of integers. Some care is needed at this step because each rational number has infinitely many representations as a fraction due to the fact that $\frac{p}{q} = \frac{pk}{qk}$ for $p \in \mathbb{Z}$, $q, k \in \mathbb{Z} \setminus \{0\}$. This issue can be solved by considering an equivalence relation for pairs $(p, q) \in \mathbb{Z} \times \mathbb{Z}$, namely, (p, q) is equivalent to (m, n) if $pn = qm$. We do not go into details.

Now all four arithmetic operations $+, -, \times, \div$ can be performed within the number system $\mathbb{Q}$. Moreover, $\mathbb{Q}$ builds an algebraic structure called a *field*. This means that the operations of addition and multiplication satisfy the known properties such as commutativity, associativity, the existence of an identity, the existence of an inverse element, and obey the distribution property $a(b+c) = ab + ac$. Furthermore, there is a natural *order* defined on $\mathbb{Q}$: for any two numbers $x, y \in \mathbb{Q}$, exactly one of the following relations is true

$$x < y, \quad x = y \quad \text{or} \quad x > y.$$

Moreover, the order is consistent with the arithmetic operations. We call such a system an *ordered field*. We do not go into further details.

However, $\mathbb{Q}$ lacks an important property of completeness. We will discuss it in Sections 1.1 and 1.2. For example, there is no rational number x with $x^2 = 2$. We extend $\mathbb{Q}$ to a larger number system $\mathbb{R}$ which is a complete ordered field containing $\mathbb{Q}$ as a subfield.

It is highly nontrivial to construct $\mathbb{R}$ from $\mathbb{Q}$. We will not discuss this in our module; if you are interested, please look at Section 8.6 in Abbott's textbook "Understanding Analysis". On a practical side, there are several models of real numbers. For example, you can think of real numbers as decimal fractions, or as points on the real line.

You are probably familiar with a larger number system $\mathbb{C}$ of complex numbers. We will not discuss it in this module.

Chapter 1
The real numbers

1.1 The well-ordering principle and induction

The set $\mathbb{N}$ has a very important property that distinguishes it from the sets $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$:

Axiom 1.1 (*The Well-Ordering Principle*) *Every non-empty subset of $\mathbb{N}$ has a smallest element.*

$\triangle$

Obviously, this property does not hold for the systems $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. For example, the set $\{x \in \mathbb{Z} \text{ (or } \mathbb{Q},\text{ or } \mathbb{R}) : x < 0\}$ is non-empty, but has no smallest element.

The Well-Ordering Principle has many important consequences. One of them is the Principle of Mathematical Induction that is well familiar to you.

Theorem 1.2 (*The Principle of Mathematical Induction*) *For each $n \in \mathbb{N}$, let $P(n)$ be a logical proposition depending on n (i.e., $P(n)$ is a statement that is either true or false, depending on the value of n). If $P(1)$ is true and for each $k \in \mathbb{N}$ we have that $P(k) \implies P(k+1)$, then $P(n)$ is true for all $n \in \mathbb{N}$.*

$\triangle$

Proof. Consider the set $S = \{n \in \mathbb{N} : P(n) \text{ is false}\}$. We have to show that $S = \varnothing$.

Suppose by contradiction that $S \neq \varnothing$. By the Well-Ordering Principle (Axiom 1.1), S has a smallest element, say m. Observe that $m \neq 1$, since $P(1)$ is true. But then $m = k + 1$ with some $k \in \mathbb{N}$. Since m is a smallest element in $\mathbb{N}$ and $k < m$, we have that $k \notin S$. In other words, $P(k)$ is true. But this implies for $m = k + 1$ that also $P(m)$ is true, which is a contradiction. The contradiction shows that $S = \varnothing$ as desired.

$\square$

In fact, the Well-Ordering Principle and the Principle of Mathematical Induction are logically equivalent. We will not discuss this in detail.

Let us consider an example that shows a possible application of the Principle of Mathematical Induction.

Example 1.3 Consider the sequence $(a_n)_{n=1}^{\infty}$ defined recursively by

$$a_1 = \frac{1}{2},$$
$$a_{n+1} = a_n^2 + \frac{1}{5}, \quad n \in \mathbb{N}.$$

We want to show that

$$0 \le a_{n+1} \le a_n \le \frac{1}{2} \quad \text{for all } n \in \mathbb{N}.$$

We use the Principle of Mathematical Induction. Let $P(n)$ be the statement that the above inequalities hold true for a particular $n \in \mathbb{N}$. For the induction base, we have to show that $P(1)$ is true. Noting that $a_2 = \left(\frac{1}{2}\right)^2 + \frac{1}{5} = \frac{9}{20}$, we observe that

$$0 \le \underbrace{\frac{9}{20}}_{=a_2} \le \underbrace{\frac{1}{2}}_{=a_1} \le \frac{1}{2}$$

as desired.

For the induction step, we assume that $0 \le a_{k+1} \le a_k \le \frac{1}{2}$ for a certain $k \in \mathbb{N}$. We have to show that $0 \le a_{k+2} \le a_{k+1} \le \frac{1}{2}$. We know that the function $f(x) = x^2$ is increasing for $x \ge 0$. Thus, the inequality

$$0 \le a_{k+1} \le a_k \le \frac{1}{2}$$

implies, by taking a square of each term, that

$$0 \le a_{k+1}^2 \le a_k^2 \le \frac{1}{4}.$$

Adding $\frac{1}{5}$ to each term now gives

$$\frac{1}{5} \le a_{k+1}^2 + \frac{1}{5} \le a_k^2 + \frac{1}{5} \le \frac{1}{4} + \frac{1}{5},$$

and since $0 \le \frac{1}{5}$, $a_{k+1}^2 + \frac{1}{5} = a_{k+2}$, $a_k^2 + \frac{1}{5} = a_{k+1}$ and $\frac{1}{4} + \frac{1}{5} = \frac{9}{20} \le \frac{1}{2}$, we arrive at

$$0 \le a_{k+2} \le a_{k+1} \le \frac{1}{2}.$$

By the Principle of Mathematical Induction we conclude that

$$0 \le a_{n+1} \le a_n \le \frac{1}{2} \quad \text{for all } n \in \mathbb{N}.$$

△

We will now use the Well-Ordering Principle to prove that the set of rational numbers $\mathbb{Q}$ is not "complete". We will show that there exist lengths along the real line that are not rational.

Theorem 1.4 *There is no rational number r with $r^2 = 2$. In other words, $\sqrt{2} \notin \mathbb{Q}$.*

△

Proof.[1] Suppose, by contradiction, that there is a rational number r with $r^2 = 2$. Since r and $-r$ are simultaneously solutions of $r^2 = 2$, we assume that $r > 0$ and denote $r = \sqrt{2} \in \mathbb{Q}$. Then there exist $p, q \in \mathbb{Z}$, $q \neq 0$, such that $\sqrt{2} = \frac{p}{q}$. We may assume that $q > 0$ (if this is not the case, we change the signs of both p and q). Thus, $q \in \mathbb{N}$ and $p = \sqrt{2}q \in \mathbb{Z}$.

Consider the set $S = \{q \in \mathbb{N} : \sqrt{2}q \in \mathbb{Z}\}$. By our assumption, $S \neq \emptyset$. Therefore, by the Well-Ordering Principle (Axiom 1.1), S has a smallest element, say $q_0 \in \mathbb{N}$. The obvious inequality $1 < 2 < 4$ implies, by taking square roots, $1 < \sqrt{2} < 2$. Subtracting 1, we obtain $0 < \sqrt{2} - 1 < 1$. Now put

$$q_1 = (\sqrt{2} - 1)q_0.$$

We have $0 < q_1 < q_0$. Since $q_0 \in S$, we have that $\sqrt{2}q_0 \in \mathbb{Z}$, and thus $q_1 = \sqrt{2}q_0 - q_0 \in \mathbb{Z}$ as a difference of two integers. Since $q_1 > 0$, we conclude that $q_1 \in \mathbb{N}$. Moreover,

$$\sqrt{2}q_1 = \sqrt{2}(\sqrt{2} - 1)q_0 = 2q_0 - \sqrt{2}q_0 \in \mathbb{Z}.$$

It follows that $q_1 \in S$. But $q_1 < q_0$ and q_0 is a smallest element in S, which is a contradiction. The contradiction shows that our assumption that $\sqrt{2} \in \mathbb{Q}$ was wrong. We conclude that $\sqrt{2} \notin \mathbb{Q}$.

□

[1]Notice that a different proof of this theorem is given in Section 1.1 of Abbott's textbook "Understanding Analysis". Read it!

Theorem 1.4 says that there are "holes" in the set of rational numbers $\mathbb{Q}$. The set of real numbers $\mathbb{R}$ is an extension of $\mathbb{Q}$ that does not have such "holes". This will be discussed in the next section.

1.2 The completeness axiom

As we mentioned in Section 0.2, $\mathbb{R}$ is an ordered field containing $\mathbb{Q}$ as a subfield. Technically, there are many different ordered fields containing $\mathbb{Q}$ (for example, there is $\mathbb{Q}[\sqrt{2}]$, the smallest field containing $\mathbb{Q}$ and $\sqrt{2}$, and further similar fields — we are not going into details). Among all fields containing $\mathbb{Q}$, the set of real numbers $\mathbb{R}$ is characterized by an important property that distinguishes $\mathbb{R}$ from all other possible extensions of $\mathbb{Q}$. This property is the Axiom of Completeness (see Axiom 1.8) and says that $\mathbb{R}$ has no "holes". Before we can formulate it, we need a couple of definitions.

Definition 1.5 Let $A \subseteq \mathbb{R}$ be a nonempty set.

(1) We say that A is *bounded above* if there exists $M \in \mathbb{R}$ such that $a \leq M$ for all $a \in A$. A number M with this property is called an *upper bound* for A.

(2) Similarly, we say that A is *bounded below* if there exists $m \in \mathbb{R}$ such that $m \leq a$ for all $a \in A$. A number m with this property is called a *lower bound* for A.

(3) We say that A is *bounded* if A is bounded below and above. In other words, there exist $m, M \in \mathbb{R}$ such that $m \leq a \leq M$ for all $a \in A$.

△

Upper and lower bounds are not unique, and they need not belong to A.

Example 1.6 Consider the set

$$A = \{x \in \mathbb{R} : x^2 < 2\}.$$

Since $0 \in A$, A is nonempty. The set A is bounded above. For example, 3 is an upper bound for A, i.e. $x \leq 3$ for all $x \in A$. Indeed, if $x > 3$, then $x^2 > 9 > 2$, and therefore $x \notin A$. It follows that for each $x \in A$ we have $x \leq 3$. Similarly, 2 is an upper bound for A: if $x > 2$, then $x^2 > 4 > 2$, and so $x \notin A$. In fact, any $M \geq 0$ with $M^2 \geq 2$ is an upper bound for A.

△

Definition 1.7 Let $A \subseteq \mathbb{R}$ be a nonempty set.

(1) We say that $M \in \mathbb{R}$ is a *least upper bound* (a *supremum*) of A if

  (I) M is an upper bound for A,

  (II) if N is any upper bound for A, then $M \le N$.

  The supremum of A is denoted by $M = \sup A$.

(2) Similarly, we say that $m \in \mathbb{R}$ is a *greatest lower bound* (a *infimum*) of A if

  (I) m is a lower bound for A,

  (II) if n is any lower bound for A, then $m \ge n$.

  The infimum of A is denoted by $m = \inf A$.

△

The least upper bound of a set, if it exists, is unique. Indeed, suppose that M_1 and M_2 are two least upper bounds for A. Then by the second property in (1) in Definition 1.7 we have $M_1 \le M_2$ and, on the other hand, $M_2 \le M_1$. It follows that $M_1 = M_2$. By a similar argument, the greatest lower bound of a set, if exists, is unique.

Now we are in position to formulate the Completeness Axiom that is a defining property of the set $\mathbb{R}$.

Axiom 1.8 (*The Completeness Axiom*) *Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.*

△

Theorem 1.9 *There exists a real number x with $x^2 = 2$. In other words, $\sqrt{2} \in \mathbb{R}$.*

△

Proof. Consider the set

$$A = \{t \in \mathbb{R} : t^2 < 2\}.$$

We discussed in Example 1.6 that A is nonempty and bounded above. By the Completeness Axiom (Axiom 1.8) A has a least upper bound. Put $x = \sup A$. Clearly, $x \ge 0$, since $0 \in A$. We will show that $x^2 = 2$ by proving that both possibilities $x^2 < 2$ and $x^2 > 2$ cannot occur.

Suppose first that $x^2 < 2$. Since 2 is an upper bound for A and x is the least upper bound, we have $x \le 2$. Take $h \in (0,1)$, then $h^2 < h$ and

$$(x+h)^2 = x^2 + 2\underbrace{x}_{\le 2}h + \underbrace{h^2}_{<h} < x^2 + 4h + h = x^2 + 5h.$$

Choose $h \in (0,1)$ so small that $5h < 2 - x^2$ (this is possible because, by our assumption, $x^2 < 2$). Then we have

$$(x+h)^2 < x^2 + 5h < x^2 + 2 - x^2 = 2,$$

i.e. $(x+h)^2 < 2$. This implies that $x + h \in A$. But $x + h > x$, and this contradicts the fact that x is an upper bound for A. The contradiction shows that it is impossible that $x^2 < 2$.

Now suppose that $x^2 > 2$. For $h > 0$ we have

$$(x-h)^2 = x^2 - 2\underbrace{x}_{\le 2}h + \underbrace{h^2}_{>0} > x^2 - 4h.$$

Choose $h > 0$ so small that $4h < x^2 - 2$ and $h < x$. Then

$$(x-h)^2 > x^2 - 4h > x^2 - (x^2 - 2) = 2.$$

It follows that $x - h$ is an upper bound for A because for any $a \ge x - h > 0$ we would have $a^2 \ge (x-h)^2 > 2$, so that $a \notin A$. But $x - h < x$ and this contradicts the fact that x is the least upper bound for A. The contradiction shows that it is impossible that $x^2 > 2$.

The above implies that $x^2 = 2$, i.e. $x = \sqrt{2} \in \mathbb{R}$.

$\square$

Remark The Completeness Axiom does not hold for $\mathbb{Q}$. For example, the set

$$B = \{r \in \mathbb{Q} : r^2 < 2\}$$

is nonempty and bounded above, but it does not have a least upper bound in $\mathbb{Q}$. Indeed, the same argument as above shows that a least upper bound $M \in \mathbb{Q}$, if it exists, cannot satisfy $M^2 < 2$ and $M^2 > 2$. But we showed in Theorem 1.4 that there is no rational number $M \in \mathbb{Q}$ with $M^2 = 2$. Thus, B does not have a least upper bound in $\mathbb{Q}$.

$\triangle$

We continue this section by discussing some properties of least upper bounds and greatest lower bounds.

Let us consider intervals on the real line. If $a, b \in \mathbb{R}$, $a < b$, then

$$\sup\,[a,b] = \sup\,(a,b) = \sup\,(-\infty,b] = \sup\,(-\infty,b) = b,$$
$$\inf\,[a,b] = \inf\,(a,b) = \inf\,[a,\infty) = \inf\,(a,\infty) = a.$$

We see that $\sup A$ and $\inf A$ may or may not belong to A. This leads us to the following definition.

Definition 1.10 Let $A \subseteq \mathbb{R}$ be a nonempty set. We say that M is a *maximum* of A if $M \in A$ and $a \leq M$ for all $a \in A$. We denote the maximum of A by $M = \max A$. Similarly, m is a *minimum* of A if $m \in A$ and $a \geq m$ for all $a \in A$. We denote the minimum of A by $m = \min A$.

△

If A is nonempty and bounded above, then we know by the Completeness Axiom that $\sup A$ exists. However, $\max A$ may or may not exist. For example, $\max [a, b] = b$, but $\max (a, b)$ does not exist. If $\max A$ exists, then $\max A = \sup A$. If $\sup A \in A$, then $\sup A = \max A$. Similar statements hold, of course, for $\inf A$ and $\min A$.

A very useful reformulation of the second property in Definition 1.7 (1) is given in the following statement.

Proposition 1.11 (1) *Let $M \in \mathbb{R}$ be an upper bound of a nonempty set $A \subseteq \mathbb{R}$. Then $M = \sup A$ if and only if for any $\varepsilon > 0$ there is an element $a \in A$ with $a > M - \varepsilon$.*

(2) *Let $m \in \mathbb{R}$ be a lower bound of a nonempty set $A \subseteq \mathbb{R}$. Then $m = \inf A$ if and only if for any $\varepsilon > 0$ there is an element $a \in A$ with $a < m + \varepsilon$.*

△

Proof. We prove (1), the proof of (2) is left as an exercise.

A. For the forward direction, assume that $M = \sup A$. Take any $\varepsilon > 0$, then $M - \varepsilon$ is not an upper bound for A, since $M - \varepsilon < M$ and M is the smallest upper bound. Thus, there must exist an element $a \in A$ with $a > M - \varepsilon$.

B. Now assume that M is an upper bound for A and that for any $\varepsilon > 0$ there is an element $a \in A$ with $a > M - \varepsilon$. Let $N < M$. Then $N = M - \varepsilon$ with $\varepsilon = M - N > 0$. By our assumption, there exists $a \in A$ with $a > N$. It follows that N is not an upper bound for A. We conclude that for any upper bound N of A we must have $N \geq M$. But this is exactly the second property in Definition 1.7 (1). This proves that $M = \sup A$.

□

Suprema and infima obey a number of rules. A typical example is the following rule: If $A \subseteq \mathbb{R}$ is a nonempty set that it bounded above and $B = \{x + a : a \in A\}$, where $x \in \mathbb{R}$, then

$$\sup B = x + \sup A.$$

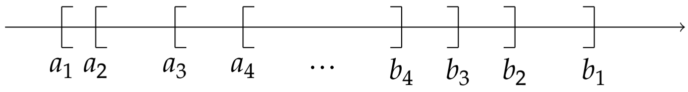

Figure 1.1: Nested intervals

We leave the proof as an exercise.

If $A \subseteq \mathbb{R}$ is a nonempty set that is bounded above, then

$$-A = \{-a : a \in A\}$$

is nonempty and bounded below. It is not difficult to show using the definition that

$$\inf(-A) = -\sup A.$$

Using this property, properties of smallest upper bounds can be reformulated as properties of greatest lower bounds. In particular, we could formulate the Completeness Axiom in terms of greatest lower bounds: Every nonempty subset of $\mathbb{R}$ that is bounded below has a greatest lower bound in $\mathbb{R}$.

1.3 Consequences of completeness

The following statement shows again that there are no "holes" in $\mathbb{R}$.

Theorem 1.12 *(The Nested Interval Property) Let $\left([a_n, b_n]\right)_{n=1}^{\infty}$ be a sequence of nonempty closed bounded intervals in $\mathbb{R}$ such that*

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq [a_3, b_3] \supseteq \cdots.$$

Then $\bigcap_{n=1}^{\infty} [a_n, b_n] \neq \emptyset$.

△

The name of the theorem comes from the fact that each interval is contained in all its predecessors (see Figure 1.1).

Before proving the theorem, we consider an example.

Example 1.13 Consider the intervals

$$[a_n, b_n] = \left[\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right], \quad n \in \mathbb{N}.$$

By the Nested Interval Property, $\bigcap_{n=1}^{\infty}\left[\sqrt{2}-\frac{1}{n}, \sqrt{2}+\frac{1}{n}\right] \neq \emptyset$. It is easy to see this directly. Indeed, $\sqrt{2} \in \left[\sqrt{2}-\frac{1}{n}, \sqrt{2}+\frac{1}{n}\right]$ for all $n \in \mathbb{N}$, and therefore $\sqrt{2} \in \bigcap_{n=1}^{\infty}\left[\sqrt{2}-\frac{1}{n}, \sqrt{2}+\frac{1}{n}\right]$. In fact, $\bigcap_{n=1}^{\infty}\left[\sqrt{2}-\frac{1}{n}, \sqrt{2}+\frac{1}{n}\right] = \left\{\sqrt{2}\right\}$, we will prove this shortly in Example 1.16.

△

Proof of Theorem 1.12. Put $A = \{a_n : n \in \mathbb{N}\}$. The set A is bounded above by b_1. By the Completeness Axiom (Axiom 1.8), A has a least upper bound M. We claim that $M \in \bigcap_{n=1}^{\infty}[a_n, b_n]$.

Indeed, since M is an upper bound for A, we have $a_n \leq M$ for all $n \in \mathbb{N}$. But each b_n, $n \in \mathbb{N}$, is also an upper bound for A. Since M is the least upper bound, $M \leq b_n$, $n \in \mathbb{N}$. Thus, $M \in [a_n, b_n]$ for all $n \in \mathbb{N}$, and therefore $M \in \bigcap_{n=1}^{\infty}[a_n, b_n]$.

□

Notice that the set of natural numbers $\mathbb{N}$ does not have a maximum element. Indeed, for each $n \in \mathbb{N}$ we can take $n+1 \in \mathbb{N}$ which is larger than n. It is less obvious that $\mathbb{N}$ is not bounded above as a subset of $\mathbb{R}$, that is, there is no $x \in \mathbb{R}$ that is larger than all natural numbers. This result is established in the next statement.

Theorem 1.14 (*The Archimedean Property*) *Given any $x \in \mathbb{R}$, there exists a natural number $n \in \mathbb{N}$ with $n > x$.*

△

Proof. Suppose, by contradiction, that there exists a number $x \in \mathbb{R}$ such that $x \geq n$ for all $n \in \mathbb{N}$. This means that $\mathbb{N}$ is bounded above. By the Completeness Axiom (Axiom 1.8), $\mathbb{N}$ has a least upper bound M.

The number $M-1$ is not an upper bound for $\mathbb{N}$, because $M-1 < M$ and M is the least upper bound. This means that there is a number $n \in \mathbb{N}$ with $n > M-1$. But then $n+1 > M$ and $n+1 \in \mathbb{N}$, so that M is not an upper bound for $\mathbb{N}$. The contradiction shows that our assumption was wrong and $\mathbb{N}$ is not bounded above in $\mathbb{R}$.

□

Corollary 1.15 *For each $\varepsilon > 0$ there exists $n \in \mathbb{N}$ with $0 < \frac{1}{n} < \varepsilon$.*

△

Proof. Take $x = \frac{1}{\varepsilon} \in \mathbb{R}$ in Theorem 1.14. So there is $n \in \mathbb{N}$ with $n > \frac{1}{\varepsilon} > 0$. This is equivalent to $0 < \frac{1}{n} < \varepsilon$. □

Example 1.16 This is a continuation of Example 1.13. We are now in a position to prove that

$$\bigcap_{n=1}^{\infty} \left[\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right] = \left\{\sqrt{2}\right\}.$$

Let $x \in \bigcap_{n=1}^{\infty} \left[\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right]$. Then $\left|x - \sqrt{2}\right| \leq \frac{1}{n}$ for each $n \in \mathbb{N}$. By Corollary 1.15 this is only possible if $\left|x - \sqrt{2}\right| = 0$, i.e., $x = \sqrt{2}$.

△

Let us turn our attention to the Nested Interval Property (Theorem 1.12) again.

Remark In Theorem 1.12 we require that the nested intervals $[a_n, b_n]$ are closed and bounded. These assumptions are needed for the theorem to hold. Taking onto account Theorem 1.14 and Corollary 1.15, we can consider the following examples.

The intervals $[n, \infty)$, $n \in \mathbb{N}$, are nested and closed, but unbounded. We have $\bigcap_{n=1}^{\infty}[n, \infty) = \varnothing$.

On the other hand, the intervals $\left(0, \frac{1}{n}\right)$, $n \in \mathbb{N}$, are nested and bounded, but they are not closed. Again we have $\bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right) = \varnothing$.

Of course we also cannot remove the assumptions that the intervals are nested. For example, the intervals $[n, n+1]$, $n \in \mathbb{N}$, are closed and bounded, but not nested, and $\bigcap_{n=1}^{\infty}[n, n+1] = \varnothing$.

△

Exactly as the Completeness Axiom (Axiom 1.8), the Nested Interval Property does not hold if we replace $\mathbb{R}$ by $\mathbb{Q}$. This can be demonstrated by the following example.

Example 1.17 Consider the intervals in $\mathbb{Q}$

$$I_n = \mathbb{Q} \cap \left[\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right] = \left\{r \in \mathbb{Q} : \sqrt{2} - \frac{1}{n} \leq r \leq \sqrt{2} + \frac{1}{n}\right\}, \quad n \in \mathbb{N}.$$

Their intersection is empty: $\bigcap_{n=1}^{\infty} I_n = \varnothing$. For,

$$\bigcap_{n=1}^{\infty} I_n \subseteq \bigcap_{n=1}^{\infty} \left[\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right] = \left\{\sqrt{2}\right\},$$

but $\sqrt{2} \notin \mathbb{Q}$. Thus, the Nested Interval Property in Theorem 1.12 does not hold if we replace $\mathbb{R}$ by $\mathbb{Q}$.

△

We will now use the Archimedean Property to show that $\mathbb{Q}$ is *dense* in $\mathbb{R}$. This means that any interval $(a, b) \subseteq \mathbb{R}$ contains rational numbers.

Theorem 1.18 *Given any two real numbers $a, b \in \mathbb{R}$ with $a < b$, there is a rational number $r \in \mathbb{Q}$ satisfying $a < r < b$.*

△

Proof. It suffices to prove the statement for $0 < a < b$. Indeed, if $a \leq 0$, then by the Archimedean Property (Theorem 1.14) there exists $k \in \mathbb{N}$ with $k > -a$. Then the numbers $a + k$ and $b + k$ satisfy $0 < a + k < b + k$. If we can find $\tilde{r} \in \mathbb{Q}$ such that $a + k < \tilde{r} < b + k$, then for the number $r = \tilde{r} - k$ we have $r \in \mathbb{Q}$ and $a < r < b$ as desired.

Therefore, assume that $0 < a < b$. By Corollary 1.15, there exists $n \in \mathbb{N}$ with $0 < \frac{1}{n} < b - a$. Put $S = \{m \in \mathbb{N} : an < m\}$. By the Archimedean Property (Theorem 1.14) $S \neq \varnothing$, and by the Well-Ordering Principle (Axiom 1.1) S has a smallest element $m_0 \in \mathbb{N}$. Put $r = \frac{m_0}{n}$. Since $m_0 \in S$ and $m_0 - 1 \notin S$, we have $m_0 - 1 \leq an < m_0$. This implies $r - \frac{1}{n} \leq a < r$. The first inequality gives $r \leq a + \frac{1}{n} < a + (b - a) = b$, and altogether we obtain $a < r < b$.

□

Given two real numbers a, b with $a < b$, we can find a number $r \in \mathbb{Q}$ such that $a + \sqrt{2} < r < b + \sqrt{2}$. Then $x = r - \sqrt{2}$ is an irrational number satisfying $a < x < b$. This proves the following statement.

Corollary 1.19 *Given any two real numbers $a, b \in \mathbb{R}$ with $a < b$, there is an irrational number x satisfying $a < x < b$.*

△

Thus, the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.

1.4 Cardinality

We have seen in the previous section that both the rational and the irrational numbers are dense in $\mathbb{R}$. In another sense, there are far more irrational numbers than rational numbers. We will discuss this in this section.

Recall that a *function* is a rule that assigns to every element x of a given set X a unique element y in a given set Y. We write $f : X \to Y$. We denote the element $y \in Y$ that is assigned to $x \in X$ by $f(x)$ and call $y = f(x)$ the *value* of f at x. The set X is called the *domain* of f, and Y is called the *codomain*.

Definition 1.20 Let $f : X \to Y$ be a function.

(1) f is called *injective* (a *one-to-one function*), if

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

for all $x_1, x_2 \in X$.

(2) f is called *surjective* (a function *onto*), if for each $y \in Y$ there is an element $x \in X$ with $f(x) = y$.

(3) f is called a *bijection* (a *one-to-one correspondence*), if f is both injective and surjective.

△

Whether or not a function $f : X \to Y$ is surjective depends on its codomain: f is surjective if and only if $Y = \{f(x) : x \in X\}$. On the other hand, whether or not $f : X \to Y$ is injective depends on its domain: f is injective if and only if for each $y \in Y$ there is at most one $x \in X$ with $f(x) = y$. Combining these two properties we state that f is a bijection if and only if for each $y \in Y$ there is exactly one $x \in X$ with $f(x) = y$.

You know that if $f : X \to Y$ is a bijection, then there exists the inverse function $f^{-1} : Y \to X$ such that $f^{-1} \circ f(x) = x$ for all $x \in X$, $f \circ f^{-1}(y) = y$ for all $y \in Y$. If f is a bijection, then f^{-1} is also a bijection.

Definition 1.21 We say that two sets A and B *have the same cardinality* if there exists a bijection $f : A \to B$. In this case we write $A \sim B$.

△

Since $f : A \to B$ is a bijection if and only if $f^{-1} : B \to A$ is a bijection, it does not matter whether we construct a bijection from A to B or from B to A. It also follows that $A \sim B \iff B \sim A$. Since the identity function $A \to A$ is a bijection, we always have $A \sim A$. Finally, since a composition of two bijections is again a bijection, we have $A \sim B$ and $B \sim C \implies A \sim C$. (These three properties say, in fact, that $\sim$ is an equivalence relation — we do not go into details here.)

Definition 1.22 A nonempty set A is called *finite* if $A \sim \{1, 2, \dots, n\}$ with some $n \in \mathbb{N}$. In this case we call n the *cardinality* of A. The empty set $\varnothing$ is regarded as being finite with cardinality zero.

△

It is quite easy to show that two finite sets have the same cardinality if and only if they have the same number of elements. The concept of cardinality was introduced as a way to describe sizes of infinite sets.

Definition 1.23 (1) A set A is called *countable* if $A \sim \mathbb{N}$.

(2) A set that is neither finite nor countable is called *uncountable*.

△

Notice that according to our terminology a countable set is infinite, i.e. finite sets are not regarded as countable. Sets that are either finite or countable are called *at most countable*. The terminology here is not consistent, so please be careful when using other books.

We will now spend some time discussing countable sets. Definition 1.23 says that A is countable if there exists a bijection $f : \mathbb{N} \to A$. This means that the elements of A can be listed as a sequence $f(1), f(2), f(3), \dots$.

Example 1.24 $\mathbb{Z}$ is countable because we can list its elements as a sequence:

$$0, 1, -1, 2, -2, 3, -3, \dots .$$

△

Example 1.25 The set of even natural numbers is countable because we can list its elements as a sequence $2, 4, 6, 8, \dots$. Similarly, the set of prime numbers is countable: $2, 3, 5, 7, \dots$.

△

In fact, any infinite subset of $\mathbb{N}$ is countable. This follows from a more general statement.

Proposition 1.26 *Any subset of a countable set is either finite or countable.*

△

Proof. Exercise. □

The intersection of two countable sets is a subset of both sets, so it must be either finite or countable. (It can indeed be finite: for example, the intersection of the countable set of even natural numbers with the countable set of prime numbers consists of one element 2.)

Proposition 1.27 *The union of two countable sets is countable.*

△

Proof. Let A and B be two countable sets. We can list their elements as sequences:

$$\begin{array}{rcl} A & : & a_1, a_2, a_3, a_4, \dots, \\ B & : & b_1, b_2, b_3, b_4, \dots. \end{array}$$

Now we can list the elements of $A \cup B$ as follows: first write

$$A \cup B \quad : \quad a_1, b_1, a_2, b_2, a_3, b_3, \dots,$$

and then remove the repeats. This shows that $A \cup B$ is countable.

□

Using induction, we can show that the union of finitely many countable sets is countable.

Let us now discuss the sets $\mathbb{Q}$ and $\mathbb{R}$. The following result is perhaps surprising.

Theorem 1.28 *The set $\mathbb{Q}$ is countable.*

△

Proof.[2] First we show that we can list positive rational numbers as a sequence. In the first step we arrange numbers of the form $\frac{m}{n}$, $m, n \in \mathbb{N}$, as an infinite square table:

$$\begin{array}{ccccccccc}
\frac{1}{1} & & \frac{1}{2} & & \frac{1}{3} & & \frac{1}{4} & \cdots \\
& \swarrow & & \swarrow & & \swarrow & & \\
\frac{2}{1} & & \frac{2}{2} & & \frac{2}{3} & & \frac{2}{4} & \cdots \\
& \swarrow & & \swarrow & & \swarrow & & \\
\frac{3}{1} & & \frac{3}{2} & & \frac{3}{3} & & \frac{3}{4} & \cdots \\
& \swarrow & & \swarrow & & \swarrow & & \\
\frac{4}{1} & & \frac{4}{2} & & \frac{4}{3} & & \frac{4}{4} & \cdots \\
\vdots & & \vdots & & \vdots & & \vdots & \ddots
\end{array}$$

[2]Notice that a somewhat different proof of this theorem is given in Section 1.5 of Abbott's textbook "Understanding Analysis".

Now we list these numbers as a sequence along south-west diagonals:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{2}{2}, \frac{3}{1}, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}, \frac{1}{5}, \frac{2}{4}, \frac{3}{3}, \frac{4}{2}, \frac{5}{1}, \dots .$$

Note that this sequence contains all positive real numbers. However, each number comes more than once. We delete the repeats and obtain the desired sequence

$$1, \frac{1}{2}, 2, \frac{1}{3}, 3, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, 4, \frac{1}{5}, 5, \dots .$$

In this way, we have listed all positive rational numbers as a sequence:

$$\{q \in \mathbb{Q} : q > 0\} = \{q_1, q_2, q_3, q_4, \dots\}.$$

To finish the proof, we list $\mathbb{Q}$ as the sequence

$$\mathbb{Q} = \{0, q_1, -q_1, q_2, -q_2, q_3, -q_3, \dots\}.$$

□

Combining ideas from the proof of Proposition 1.27 and Theorem 1.28, one can prove that the union of countable many countable sets is countable. We will not discuss this in the module, but try to prove it.

Now we turn our attention to the set of real numbers $\mathbb{R}$. We will show that it is much larger than $\mathbb{Q}$.

Theorem 1.29 *The set $\mathbb{R}$ is uncountable.*

△

Proof. Suppose, by contradiction, that $\mathbb{R}$ is countable. If this is the case, we are able to list its elements as a sequence:

$$\mathbb{R} = \{x_1, x_2, x_3, x_4, \dots\}.$$

Consider x_1. Take a closed and bounded interval $[a_1, b_1]$ with $a_1 < b_1$ such that $x_1 \notin [a_1, b_1]$. Now we split the interval $[a_1, b_1]$ into three thirds $[a_1, c_1]$, $[c_1, d_1]$, $[d_1, b_1]$. At least one of these three intervals does not contain x_2.[3] We denote this interval as $[a_2, b_2]$.

[3]Notice that it is not enough to divide $[a_1, b_1]$ into two halves: if x_2 happens to be exactly the midpoint of $[a_1, b_1]$, it will belong to both halves!

Now we have $a_1 \le a_2 < b_2 \le b_1$, $x_1 \notin [a_1, b_1]$, $x_2 \notin [a_2, b_2]$. We continue inductively. At the n-th step we divide the interval $[a_n, b_n]$ into three thirds and choose $[a_{n+1}, b_{n+1}]$ to be a third that does not contain x_{n+1}.

In this way we obtain a sequence of nested intervals

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq [a_3, b_3] \supseteq \cdots$$

with the property $x_n \notin [a_n, b_n]$, $n \in \mathbb{N}$.

By the Nested Interval Property (Theorem 1.12) we have $\bigcap_{n=1}^{\infty}[a_n, b_n] \neq \varnothing$. In other words, there exists a real number x with $x \in \bigcap_{n=1}^{\infty}[a_n, b_n]$. Since $x \in \mathbb{R}$, x is counted in our list, i.e. $x = x_m$ with some $m \in \mathbb{N}$. But then, by construction, $x = x_m \notin [a_m, b_m]$. It follows that $x \notin \bigcap_{n=1}^{\infty}[a_n, b_n]$ which is a contradiction. The contradiction shows that it is not possible to list the elements of $\mathbb{R}$ as a sequence, i.e., $\mathbb{R}$ is uncountable.

□

Since the union of two countable sets is countable, we arrive at the following conclusion.

Corollary 1.30 *The set of irrational numbers* $\mathbb{R} \setminus \mathbb{Q}$ *is uncountable.*

△

Chapter 2

Sequences and series

2.1 Limits of sequences

Definition 2.1 A *sequence* in a set A is a function $f : \mathbb{N} \to A$.

$\triangle$

It is common to use the notation $a_n = f(n)$, $n \in \mathbb{N}$. The elements a_n are called *terms* of the sequence. We will frequently write for a sequence $(a_n)_{n=1}^{\infty}$, $(a_n)_{n\in\mathbb{N}}$ or simply (a_n).

In this module we will be mainly concerned with real sequences $f : \mathbb{N} \to \mathbb{R}$. Examples of sequences are

$$\begin{aligned} (a_n)_{n=1}^{\infty} &= (1, -1, 1, -1, 1, -1, \ldots), \\ (b_n)_{n=1}^{\infty} &= (-1, 1, -1, 1, -1, 1, \ldots). \end{aligned}$$

Notice that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are two different sequences. For example, $a_1 = 1$ while $b_1 = -1$. Both sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ have the same set of values $\{-1, 1\}$.

In some cases, it is convenient to start a sequence with $n = 0$, or with any other $n = n_0 \in \mathbb{Z}$. As an example, consider the sequence

$$\left(\frac{1}{2^n}\right)_{n=0}^{\infty} = \left(1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \cdots\right).$$

It is possible to draw a graph of a function $f : \mathbb{N} \to \mathbb{R}$. Such a graph consists of isolated points. We will see examples below.

One of the most important concepts of analysis is that of convergence and of a limit. Intuitively, a sequence $(a_n)_{n=1}^{\infty}$ converges to a limit $a \in \mathbb{R}$ if its terms get closer and closer to a as n goes to infinity. A precise definition is given below; this is one of the most important definitions in this module — learn it.

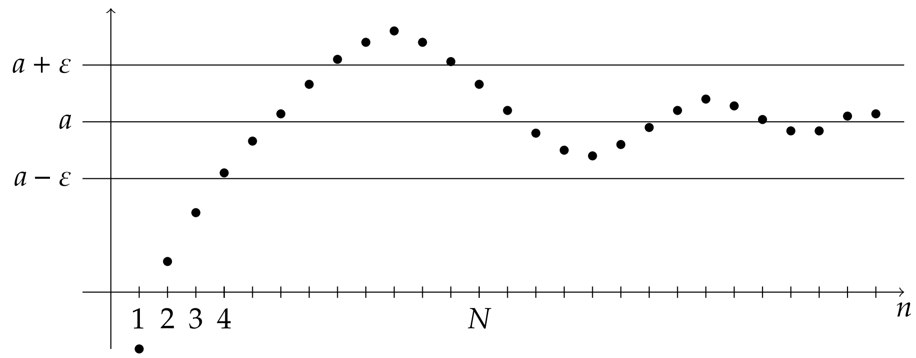

Figure 2.1: A graph of a converging sequence

Definition 2.2 We say that a sequence $(a_n)_{n=1}^{\infty}$ *converges* to a *limit* $a \in \mathbb{R}$ if

$$\forall\, \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall\, n \geq N : |a_n - a| < \varepsilon,$$

i.e. for any positive number ε there exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n \in \mathbb{N}$ with $n \geq N$. In this case we write $a = \lim_{n\to\infty} a_n$, or $a_n \to a$ as $n \to \infty$.

△

The definition means that for any $\varepsilon > 0$ all terms of the sequence but finitely many (namely, all terms a_n with $n \geq N$) lie between $a - \varepsilon$ and $a + \varepsilon$. We can visualize this behavior on the graph of a sequence — see Figure 2.1. For a smaller value of $\varepsilon > 0$, in general a larger value of $N \in \mathbb{N}$ is required. But no matter how small $\varepsilon > 0$ is, we can always find an $N \in \mathbb{N}$ with this property.

Example 2.3 Consider the sequence $\left(\frac{n+1}{n}\right)_{n=1}^{\infty}$.

The first few terms of the sequence are $\left(2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \cdots\right)$. We conjecture that the sequence converges to the limit 1. For the proof we consider

$$|a_n - 1| = \left|\frac{n+1}{n} - 1\right| = \left|1 + \frac{1}{n} - 1\right| = \frac{1}{n}.$$

Given $\varepsilon > 0$, by the Archimedean Property (Theorem 1.14) we can find a number $N \in \mathbb{N}$ such that $\frac{1}{\varepsilon} < N$. Then $\frac{1}{N} < \varepsilon$, and consequently for all $n \geq N$ we have

$$|a_n - 1| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

We conclude that $\lim_{n\to\infty} \frac{n+1}{n} = 1$.

Let us look at a couple of concrete values of ε. For example, for $\varepsilon = 0.1$ we can take $N = 11$, for $\varepsilon = 0.01$ we can take $N = 101$, for $\varepsilon = 0.005$ we can take $N = 201$, and so on.

△

Negating the statement in Definition 2.2, we can formulate a property that a sequence $(a_n)_{n=1}^{\infty}$ does not converge to $b \in \mathbb{R}$:

$$\exists \varepsilon > 0 \; \forall N \in \mathbb{N} \; \exists n \in \mathbb{N} : n \geq N \text{ and } |a_n - b| \geq \varepsilon.$$

As an example, let us show that the sequence $\left(\frac{n+1}{n}\right)_{n=1}^{\infty}$ does not converge to 0. We have

$$|a_n - 0| = \left|\frac{n+1}{n} - 0\right| = \frac{n+1}{n} > 1$$

for all $n \in \mathbb{N}$. We can choose $\varepsilon = 1$. Then for any $N \in \mathbb{N}$ we can find $n \geq N$ — for example, $n = N$ — with $|a_n - 0| > \varepsilon = 1$. Thus, $\left(\frac{n+1}{n}\right)_{n=1}^{\infty}$ does not converge to 0.

In fact, this follows already from $\lim_{n\to\infty} \frac{n+1}{n} = 1$, since a sequence cannot have two different limits. Namely, we have the following statement.

Proposition 2.4 *The limit of a sequence, if it exists, is unique.*

△

Proof. Suppose that the numbers $a, b \in \mathbb{R}$ are both limits of a sequence $(a_n)_{n=1}^{\infty}$. Given $\varepsilon > 0$, we can find $N_1, N_2 \in \mathbb{N}$ such that $|a_n - a| < \frac{\varepsilon}{2}$ for all $n \geq N_1$ and $|a_n - b| < \frac{\varepsilon}{2}$ for all $n \geq N_2$. Now take a number $n_0 \geq \max\{N_1, N_2\}$. By the triangle inequality we obtain

$$|a - b| \leq |a - a_{n_0}| + |a_{n_0} - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since this is true for any $\varepsilon > 0$, we must have $|a - b| = 0$, i.e. $a = b$.

□

Definition 2.5 We say that a sequence (a_n) is *divergent* if it does not converge to any $a \in \mathbb{R}$.

△

In other words, a sequence (a_n) is divergent if

$$\forall a \in \mathbb{R} \; \exists \varepsilon > 0 \; \forall N \in \mathbb{N} \; \exists n \in \mathbb{N} : n \geq N \text{ and } |a_n - a| \geq \varepsilon.$$

Example 2.6 Consider the sequence $\left(\frac{n^2}{n+1}\right)_{n=1}^{\infty}$.

We want to show that this sequence is divergent. Notice that for any $n \in \mathbb{N}$ we have $n \geq \frac{1}{2}(n+1)$. It follows that

$$\frac{n^2}{n+1} \geq \frac{n \cdot \frac{1}{2}(n+1)}{n+1} = \frac{1}{2}n.$$

Now take an arbitrary $a \in \mathbb{R}$ and $\varepsilon = 1$. By the Archimedean Property (Theorem 1.14) there exists $N \in \mathbb{N}$ with $N > 2(a+\varepsilon)$. Then for all $n \geq N$ we have

$$\frac{n^2}{n+1} \geq \frac{1}{2}n \geq \frac{1}{2}N > a + \varepsilon,$$

so that

$$\left|\frac{n^2}{n+1} - a\right| > \varepsilon \quad \text{for all } n \geq N.$$

It follows that $\lim_{n\to\infty} \frac{n^2}{n+1} \neq a$. Since this is true for any $a \in \mathbb{R}$, we conclude that the sequence $\left(\frac{n^2}{n+1}\right)_{n=1}^{\infty}$ diverges.

△

Example 2.7 Another example of a divergent sequence is

$$((-1)^n)_{n=0}^{\infty} = (1, -1, 1, -1, 1, -1, \ldots).$$

Take $\varepsilon = \frac{1}{2}$. It is quite easy to see that, for any $a \in \mathbb{R}$, at least one half of the terms of the sequence lies outside the interval $\left(a - \frac{1}{2}, a + \frac{1}{2}\right)$. (Make sure that you understand why this is the case.) Thus, a cannot be a limit of this sequence. Since this is true for any $a \in \mathbb{R}$, we see that the sequence diverges.

△

Definition 2.8 Let $(a_n)_{n=1}^{\infty}$ be a sequence in $\mathbb{R}$.

(1) We say that $(a_n)_{n=1}^{\infty}$ is *bounded above* if there exists a number $M \in \mathbb{R}$ such that $a_n \leq M$ for all $n \in \mathbb{N}$.

(2) We say that $(a_n)_{n=1}^{\infty}$ is *bounded below* if there exists a number $m \in \mathbb{R}$ such that $a_n \geq m$ for all $n \in \mathbb{N}$.

(3) We say that $(a_n)_{n=1}^{\infty}$ is *bounded* if it is bounded above and below, i.e. if there exist numbers $m, M \in \mathbb{R}$ such that $m \leq a_n \leq M$ for all $n \in \mathbb{N}$.

Notice that a sequence $(a_n)_{n=1}^{\infty}$ is bounded (bounded above, bounded below) if and only if its set of values $\{a_n : n \in \mathbb{N}\}$ is bounded (bounded above, bounded below).

$\triangle$

Proposition 2.9 *Any convergent sequence is bounded.*

$\triangle$

Proof. Consider a sequence $(a_n)_{n=1}^{\infty}$ with $\lim_{n\to\infty} a_n = a$. Take $\varepsilon = 1$. There exists $N \in \mathbb{N}$ such that $|a_n - a| < 1$ for all $n \geq N$. This is equivalent to saying that

$$a - 1 < a_n < a + 1 \quad \text{for all } n \geq N.$$

Now put $m = \min\{a_1, \dots, a_{N-1}, a-1\}$, $M = \max\{a_1, \dots, a_{N-1}, a+1\}$. By above we have

$$m \leq a_n \leq M \quad \text{for all } n \in \mathbb{N},$$

and this means that $(a_n)_{n=1}^{\infty}$ is bounded.

$\square$

Notice that the sequence $\left(\frac{n^2}{n+1}\right)_{n=1}^{\infty}$ in Example 2.6 is unbounded. Thus, it follows immediately from Proposition 2.9 that this sequence is divergent.

Remark The converse of Proposition 2.9 is not true: A bounded sequence need not converge. An example of a sequence that is bounded and divergent is $((-1)^n)_{n=0}^{\infty}$.

$\triangle$

2.2 Rules of working with limits

In the practice, only basic limits are calculated using the definition, and further limits are determined from the basic limits using rules for working with limits. We consider some of those rules in this section.

Theorem 2.10 (*The Squeeze Theorem*) *Let (a_n), (b_n) be two sequences with*

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L.$$

Suppose that (x_n) is a sequence such that

$$a_n \leq x_n \leq b_n \quad \textit{for all large values of } n.$$

Then the sequence (x_n) converges and $\lim_{n\to\infty} x_n = L$.

$\triangle$

Proof. Suppose that $a_n \le x_n \le b_n$ for all $n \ge N_0$, where $N_0 \in \mathbb{N}$. Given $\varepsilon > 0$, we can find $N_1, N_2 \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \ge N_1$ and $|b_n - L| < \varepsilon$ for all $n \ge N_2$.

Put $N = \max\{N_0, N_1, N_2\}$. For all $n \ge N$ we have

$$L - \varepsilon < a_n \le x_n \le b_n < L + \varepsilon,$$

which implies $|x_n - L| < \varepsilon$ for all $n \ge N$. This proves that $\lim_{n\to\infty} x_n = L$.

□

Example 2.11 Consider the sequence $\left(\frac{\sin n}{n}\right)_{n=1}^{\infty}$.

Since $-1 \le \sin n \le 1$, we have $-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$ for all $n \in \mathbb{N}$. We know that $\lim_{n\to\infty} \frac{1}{n} = \lim_{n\to\infty} \left(-\frac{1}{n}\right) = 0$. The Squeeze Theorem (Theorem 2.10) yields $\lim_{n\to\infty} \frac{\sin n}{n} = 0$.

△

Definition 2.12 A sequence (a_n) is called *eventually constant* if there are $a \in \mathbb{R}$ and $N \in \mathbb{N}$ such that $a_n = a$ for all $n \ge N$.

△

Proposition 2.13 *If a sequence is eventually constant, then it is convergent.*

△

Proof. Suppose that (a_n) is eventually constant with a and N as in Definition 2.12. Then for any $\varepsilon > 0$ we have

$$|a_n - a| = 0 < \varepsilon \quad \text{for all } n \ge N.$$

Thus, $\lim_{n\to\infty} a_n = a$.

□

Theorem 2.14 (*The Algebraic Limit Theorem*) *Suppose that* $\lim_{n\to\infty} a_n = a$ *and* $\lim_{n\to\infty} b_n = b$. *Then*

(1) $\lim_{n\to\infty}(ca_n) = ca$ *for all* $c \in \mathbb{R}$,

(2) $\lim_{n\to\infty}(a_n + b_n) = a + b$,

(3) $\lim_{n\to\infty}(a_n b_n) = ab$,

(4) $\lim_{n\to\infty} \frac{1}{b_n} = \frac{1}{b}$ *provided* $b \ne 0$,

(5) $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}$ *provided* $b \neq 0$.

△

Proof. (1) If $c = 0$, then $ca_n = 0$ for all $n \in \mathbb{N}$, and $\lim_{n\to\infty}(ca_n) = 0$ by Proposition 2.13.

Therefore assume that $c \neq 0$. For any $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - a| < \frac{\varepsilon}{|c|}$ for all $n \geq N$. Then we obtain for all such n

$$|ca_n - ca| = |c||a_n - a| < |c|\frac{\varepsilon}{|c|} = \varepsilon,$$

and the statement follows.

(2) Given $\varepsilon > 0$, there are $N_1, N_2 \in \mathbb{N}$ such that $|a_n - a| < \frac{\varepsilon}{2}$ for all $n \geq N_1$ and $|b_n - b| < \frac{\varepsilon}{2}$ for all $n \geq N_2$. Put $N = \max\{N_1, N_2\}$. For all $n \geq N$ we have by the triangle inequality

$$|(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

as desired.

(3) There is $N_1 \in \mathbb{N}$ such that $|b_n - b| < 1$ for all $n \geq N_1$. For such n we have

$$-(|b| + 1) \leq b - 1 < b_n < b + 1 \leq |b| + 1,$$

so that

$$|b_n| \leq |b| + 1, \quad n \geq N_1.$$

Given $\varepsilon > 0$, there are $N_2, N_3 \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2(|b| + 1)} \quad \text{for all } n \geq N_2,$$

$$|b_n - b| < \frac{\varepsilon}{2(|a| + 1)} \quad \text{for all } n \geq N_3.$$

Put $N = \max\{N_1, N_2, N_3\}$. For all $n \geq N$ we have

$$\begin{aligned} |a_nb_n - ab| &= |a_nb_n - ab_n + ab_n - ab| \leq |a_n - a||b_n| + |a||b_n - b| \\ &< \frac{\varepsilon}{2(|b| + 1)}(|b| + 1) + |a|\frac{\varepsilon}{2(|a| + 1)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

and this proves the statement.

(4) There is $N_1 \in \mathbb{N}$ such that $|b_n - b| < \frac{|b|}{2}$ for all $n \geq N_1$. For such n we have

$$|b| = |b - b_n + b_n| \leq |b - b_n| + |b_n| < \frac{|b|}{2} + |b_n|.$$

This implies $|b| - \frac{|b|}{2} < |b_n|$. Finally, we obtain

$$|b_n| > \frac{|b|}{2}, \quad n \geq N_1.$$

In particular, $b_n \neq 0$ for $n \geq N_1$. Moreover, we have

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \left|\frac{b - b_n}{bb_n}\right| < \frac{|b - b_n|}{|b| \cdot \frac{1}{2}|b|} = \frac{2|b - b_n|}{|b|^2}, \quad n \geq N_1.$$

Given $\varepsilon > 0$, there is $N_2 \in \mathbb{N}$ such that $|b_n - b| < \frac{|b|^2}{2}\varepsilon$ for all $n \geq N_2$. Put $N = \max\{N_1, N_2\}$. For all $n \geq N$ we have

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| < \frac{2|b - b_n|}{|b|^2} < \frac{2|b|^2\varepsilon}{2|b|^2} = \varepsilon,$$

which yields the statement.

(5) We write $\frac{a_n}{b_n} = a_n \frac{1}{b_n}$ and apply (4) and (3).

□

Remark The statements (1) and (2) of Theorem 2.14 imply the following: If (a_n), (b_n) are convergent sequences and $\alpha, \beta \in \mathbb{R}$, then the sequence $(\alpha a_n + \beta b_n)$ is convergent and

$$\lim_{n\to\infty}(\alpha a_n + \beta b_n) = \alpha \lim_{n\to\infty} a_n + \beta \lim_{n\to\infty} b_n.$$

Thus, the operation of taking the limits of convergent sequences is a linear operation.

△

In our next statement we show that the operation of taking the limits preserves the order on $\mathbb{R}$.

Theorem 2.15 (*The Order Limit Theorem*) *Suppose that $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$. If $a_n \leq b_n$ for all $n \in \mathbb{N}$, then $a \leq b$.*

△

Proof. Suppose, by contradiction, that $a > b$. Take $\varepsilon = \frac{a-b}{2} > 0$ and choose $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ and $|b - b_n| < \varepsilon$ for all $n \in \mathbb{N}$. But then

$$b_n < b + \varepsilon = b + \frac{a-b}{2} = \frac{a+b}{2} = a - \frac{a-b}{2} = a - \varepsilon < a_n, \quad n \geq N.$$

This contradicts the assumption $a_n \leq b_n$ for all $n \in \mathbb{N}$. Thus, we must have $a \leq b$.

□

Remark It is enough to assume in Theorem 2.15 that $a_n \leq b_n$ for all $n \geq N_0$, where $N_0 \in \mathbb{N}$.

△

Remark Notice that $a_n < b_n$ for all $n \in \mathbb{N}$ implies $a \leq b$, but does not imply $a < b$. As an example consider $(b_n)_{n=1}^\infty = \left(\frac{1}{n}\right)_{n=1}^\infty$ and the constant sequence $(a_n)_{n=1}^\infty = (0)_{n=1}^\infty$. We have $\frac{1}{n} > 0$ for all $n \in \mathbb{N}$, but $\lim_{n\to\infty} \frac{1}{n} = 0 = \lim_{n\to\infty} 0$.

△

2.3 The Monotone Convergence Theorem

We have seen at the end of Section 2.1 that a bounded sequence need not converge. However, if a bounded sequence is also monotone, it does converge. This statement is the content of this section.

Definition 2.16 We say that a sequence $(a_n)_{n=1}^\infty$ is

(1) *increasing*, if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$,

(2) *decreasing*, if $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$,

(3) *monotone*, if it is increasing or decreasing.

△

Notice that in some literature increasing sequences of Definition 2.16 (1) are called *non-decreasing*, and decreasing sequences of Definition 2.16 (2) are called *non-increasing*. If one wishes to emphasize that the inequalities $\leq, \geq$ are in fact strict, one uses the terminology of strict monotonicity: A sequence $(a_n)_{n=1}^\infty$ is called *strictly increasing* if $a_n < a_{n+1}$ for all $n \in \mathbb{N}$, and *strictly decreasing* if $a_n > a_{n+1}$ for all $n \in \mathbb{N}$.

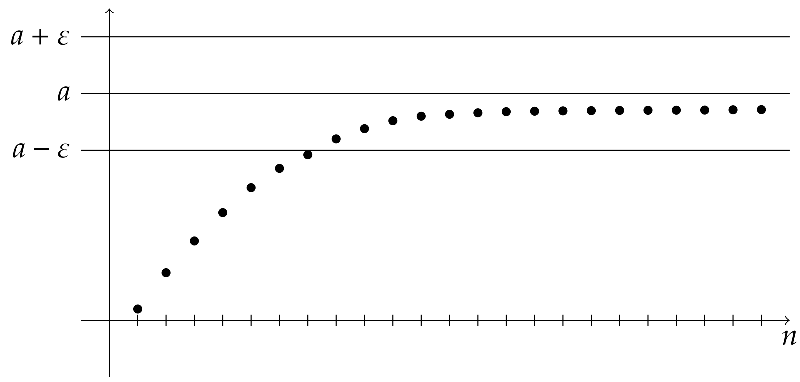

Figure 2.2: An increasing bounded sequence

Notice that a sequence $(a_n)_{n=1}^{\infty}$ is increasing if and only if $a_n \leq a_m$ for all $n, m \in \mathbb{N}$ with $n \leq m$. It is decreasing if and only if $a_n \geq a_m$ for all $n, m \in \mathbb{N}$ with $n \leq m$.

Assume that a sequence $(a_n)_{n=1}^{\infty}$ is bounded above. Then its set of values $\{a_n : n \in \mathbb{N}\}$ is bounded above. By The Completeness Axiom (Axiom 1.8), it has a supremum

$$\sup_{n\in\mathbb{N}} a_n = \sup\{a_n : n \in \mathbb{N}\} \in \mathbb{R}.$$

Similarly, if a sequence $(a_n)_{n=1}^{\infty}$ is bounded below, it has an infimum

$$\inf_{n\in\mathbb{N}} a_n = \inf\{a_n : n \in \mathbb{N}\} \in \mathbb{R}.$$

Theorem 2.17 (*The Monotone Convergence Theorem*) *If a sequence is increasing and bounded above, then it converges, and its limit is its supremum.*

△

For an illustration of this behavior, see Figure 2.2.

Proof of Theorem 2.17. Let $(a_n)_{n=1}^{\infty}$ be an increasing bounded above sequence, and put $s = \sup_{n\in\mathbb{N}} a_n$. Let $\varepsilon > 0$. By Proposition 1.11, we can find $N \in \mathbb{N}$ such that $a_N > s - \varepsilon$. Since the sequence $(a_n)_{n=1}^{\infty}$ is monotone and s is its supremum, we have

$$s - \varepsilon < a_N \leq a_n \leq s < s + \varepsilon \quad \text{for all } n \geq N,$$

which implies

$$|s - a_n| < \varepsilon \quad \text{for all } n \geq N.$$

It follows that $\lim_{n\to\infty} a_n = s$.

□

Corollary 2.18 *If a sequence is decreasing and bounded below, then it converges, and its limit is its infimum.*

△

Remark As usual in such statements regarding limits, it is sufficient to suppose that a sequence is *eventually monotone*. A sequence $(a_n)_{n=1}^{\infty}$ is called *eventually increasing* (*eventually decreasing*) if $a_n \le a_{n+1}$ ($a_n \ge a_{n+1}$) for all $n \ge N_0$ with some $N_0 \in \mathbb{N}$. A sequence that is eventually increasing and bounded above converges to its supremum. A sequence that is eventually decreasing and bounded below converges to its infimum.

△

Theorem 2.17 and Corollary 2.18 allow us to prove that a limit of a sequence exists without having to find its value. This can be very useful.

Example 2.19 Consider the sequence $(a_n)_{n=1}^{\infty}$ defined recursively by

$$a_0 = 0, \qquad a_n = \frac{1}{4}(a_{n-1}^2 + 1), \quad n \in \mathbb{N}.$$

We are going to show that this sequence is increasing and bounded above, and then Theorem 2.17 says that it is convergent.

Using induction, we show that $a_n < 1$ for all $n = 0, 1, 2, \ldots$. For the induction base, notice that $a_0 = 0 < 1$. Now suppose that $a_k < 1$ for some $k \in \mathbb{N} \cup \{0\}$. Then also $a_k^2 < 1$ and

$$a_{k+1} = \frac{1}{4}(a_k^2 + 1) < \frac{1}{4}(1 + 1) = \frac{1}{2} < 1.$$

By the Principle of Mathematical Induction we conclude that $a_n < 1$ for all $n \in \mathbb{N} \cup \{0\}$.

To show that $(a_n)_{n=1}^{\infty}$ is increasing, we will also use induction. For the induction base, notice that $a_0 = 0$, $a_1 = \frac{1}{4}$ and this $a_0 < a_1$. Now assume that $a_k < a_{k+1}$ for some $k \in \mathbb{N} \cup \{0\}$. Then also $a_k^2 < a_{k+1}^2$, and thus

$$a_{k+1} = \frac{1}{4}(a_k^2 + 1) < \frac{1}{4}(a_{k+1}^2 + 1) = a_{k+2}.$$

By the Principle of Mathematical Induction, $a_n < a_{n+1}$ for all $n \in \mathbb{N} \cup \{0\}$. Thus, the sequence $(a_n)_{n=1}^{\infty}$ is increasing (actually, we have shown that it is even strictly increasing).

By the Monotone Convergence Theorem (Theorem 2.17), the limit

$$\lim_{n\to\infty} a_n = a$$

exists. Notice that we have proven the existence of the limit without knowing its value.

In this particular example, we are even able to determine the value of the limit — but only as we know that it exists. Consider the equation

$$a_n = \frac{1}{4}(a_{n-1}^2 + 1), \quad n \in \mathbb{N}.$$

Applying the Algebraic Limit Theorem (Theorem 2.14), we can take limits at both sides to obtain

$$a = \frac{1}{4}(a^2 + 1).$$

Notice that this step is only possible since we know that the sequence $(a_n)_{n=1}^{\infty}$ converges! The above equation for a has two solutions $2+\sqrt{3}$ and $2-\sqrt{3}$. Of course only one of these numbers is the value of $\lim_{n\to\infty} a_n$ — the limit of a sequence, if exists, is unique (see Proposition 2.4). To decide which of the numbers $2+\sqrt{3}$, $2-\sqrt{3}$ is the value of the limit, we notice that $a_n < 1$ for all n, and thus by the Order Limit Theorem (Theorem 2.15) $a = \lim_{n\to\infty} a_n \leq 1$. However, $2+\sqrt{3} > 1$, and therefore cannot be the limit. We conclude that

$$\lim_{n\to\infty} a_n = 2 - \sqrt{3}.$$

$\triangle$

2.4 A first look at series

Definition 2.20 Let $(a_n)_{n=1}^{\infty}$ be a sequence. A formal expression on the form

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

is called an *(infinite) series*. The *partial sums* of the series $\sum_{n=1}^{\infty} a_n$ are the sums

$$s_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n, \quad n \in \mathbb{N}.$$

We say that the series $\sum_{n=1}^{\infty} a_n$ *converges* if the sequence of its partial sums $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$ converges. If $\sum_{n=1}^{\infty} a_n$ converges, we call the limit

$$s = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \sum_{k=1}^{n} a_k$$

the *sum* of the series and write $\sum_{n=1}^{\infty} a_n = s$.

If the sequence $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$ diverges, we say that the series $\sum_{n=1}^{\infty} a_n$ *diverges*.

$\triangle$

Note that writing the symbol $\sum_{n=1}^{\infty} a_n$ does not automatically mean that the series is supposed to be convergent. If the series $\sum_{n=1}^{\infty} a_n$ diverges, it can be still interesting to discuss its terms, its partial sums, and study the divergence of the sequence of partial sums.

Like with sequences, we can start the summation with any $n_0 \in \mathbb{Z}$ and consider series $\sum_{n=n_0}^{\infty} a_n$. Frequently we take $n_0 = 0$ like, for example, in the geometric series $\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$.

The next statement gives a simple but useful necessary condition for the convergence of a series.

Proposition 2.21 *If a series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.*

$\triangle$

Proof. Let $s_n = \sum_{k=1}^{n} a_k$ and $\sum_{k=1}^{\infty} a_k = s = \lim_{n\to\infty} s_n$. Notice that

$$a_n = \sum_{k=1}^{n} a_k - \sum_{k=1}^{n-1} a_k = s_n - s_{n-1}.$$

Taking the limits as $n \to \infty$, we obatain

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = s - s = 0.$$

$\square$

Remark The converse of Proposition 2.21 is not true. We will see in Example 2.29 an example of a divergent series with $\lim_{n\to\infty} a_n = 0$.

$\triangle$

Proposition 2.21 is frequently used in order to show that a series does not converge.

Example 2.22 The series $\sum_{n=0}^{\infty}(-1)^n$ diverges. Indeed, its terms $a_n = (-1)^n$ do not tend to zero as $n \to \infty$. The partial sums of this series are $s_0 = 1, s_1 = 0, s_2 = 1, s_3 = 0, \ldots$.

△

A simple consequence of the definition of convergence is the following useful statement about the tail of a convergent series.

Proposition 2.23 *If a series $\sum_{n=1}^{\infty} a_n$ converges, then for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that*

$$\left|\sum_{k=n+1}^{\infty} a_k\right| < \varepsilon \quad \textit{for all } n \geq N.$$

△

Proof. For a convergent series $\sum_{n=1}^{\infty} a_n$, put $s_n = \sum_{k=1}^{n} a_k$ and $s = \sum_{k=1}^{\infty} a_k$. Given $\varepsilon > 0$, by the definition of the limit $\lim_{n\to\infty} s_n = s$, there is $N \in \mathbb{N}$ such that

$$|s - s_n| < \varepsilon \quad \text{for any } n \geq N.$$

Noting that $s - s_n = \sum_{k=n+1}^{\infty} a_k$, we arrive at the desired statement.

□

Applying statements (1) and (2) of the Algebraic Limit Theorem (Theorem 2.14) to the sequences of the partial sums, we obtain the following results.

Proposition 2.24 (1) *If a series $\sum_{n=1}^{\infty} a_n$ is convergent and $c \in \mathbb{R}$, then the series $\sum_{n=1}^{\infty}(ca_n)$ is convergent, and*

$$\sum_{n=1}^{\infty}(ca_n) = c\sum_{n=1}^{\infty} a_n.$$

(2) *If series $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ are convergent, then the series $\sum_{n=1}^{\infty}(a_n + b_n)$ is convergent, and*

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

△

We conclude that if $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ are two convergent series and $\alpha, \beta \in \mathbb{R}$, then the series $\sum_{n=1}^{\infty}(\alpha a_n + \beta b_n)$ is convergent, and

$$\sum_{n=1}^{\infty}(\alpha a_n + \beta b_n) = \alpha\sum_{n=1}^{\infty} a_n + \beta\sum_{n=1}^{\infty} b_n.$$

Example 2.25 Consider the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

It holds $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. Thus, for the partial sums of this series we have

$$\begin{aligned}\sum_{n=1}^{m} \frac{1}{n(n+1)} &= \sum_{n=1}^{m} \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{m-1} - \frac{1}{m} + \frac{1}{m} - \frac{1}{m+1} \\ &= 1 - \frac{1}{m+1} \to 1, \quad m \to \infty.\end{aligned}$$

Thus, the series converges and

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

Series of this type are called *telescopic series*.

Notice that we could not write $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n} - \sum_{n=1}^{\infty} \frac{1}{n+1}$ because the series $\sum_{n=1}^{\infty} \frac{1}{n}$ and $\sum_{n=1}^{\infty} \frac{1}{n+1}$ are divergent (we will see this soon in Example 2.29).

△

Definition 2.26 A series of the form $\sum_{n=0}^{\infty} ar^n$, where $a, r \in \mathbb{R}$, is called a *geometric series*.

△

Examples of geometric series are $\sum_{n=0}^{\infty} (-1)^n = 1 - 1 + 1 - 1 + \cdots$, $\sum_{n=0}^{\infty} 2^n = 1 + 2 + 4 + 8 + \cdots$, $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2}\left(\frac{1}{2}\right)^n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$.

Theorem 2.27 (The Geometric Series) *Let $a \neq 0$. A geometric series $\sum_{n=0}^{\infty} ar^n$ converges if and only if $|r| < 1$. In this case*

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

△

Proof. If $a \neq 0$ and $|r| \geq 1$, then $|ar^n| \geq |a| \neq 0$, and thus the sequence $(ar^n)_{n=1}^{\infty}$ of the terms of the series does not converge to zero. By Proposition 2.21, the series $\sum_{n=0}^{\infty} ar^n$ diverges.

Now assume that $|r| < 1$ and put $s_n = \sum_{k=0}^{n} ar^k$, $n = 0, 1, 2, \ldots$. We can write

$$\begin{array}{ccccccccccccc} s_n & = & a & + & ar & + & ar^2 & + & \cdots & + & ar^n, & & \\ rs_n & = & & & ar & + & ar^2 & + & \cdots & + & ar^n & + & ar^{n+1}, \end{array}$$

so that

$$(1-r)s_n = a - ar^{n+1} = a(1-r^{n+1}),$$

and consequently

$$s_n = a\frac{1-r^{n+1}}{1-r}.$$

The property $|r| < 1$ implies that $\lim_{n\to\infty} r^{n+1} = 0$, and finally we obtain

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} a\frac{1-r^{n+1}}{1-r} = \frac{a}{1-r}.$$

This means that the series $\sum_{n=0}^{\infty} ar^n$ converges and its sum is $\frac{a}{1-r}$.

□

Example 2.28 We have

$$\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n = \sum_{n=0}^{\infty}\frac{1}{2}\left(\frac{1}{2}\right)^n = \frac{1}{2}\frac{1}{1-\frac{1}{2}} = \frac{1}{2}\cdot 2 = 1.$$

△

Another very important series will be considered in the next example.

Example 2.29 (*The Harmonic Series*) The series $\sum_{n=1}^{\infty}\frac{1}{n}$ is called the *harmonic series*.

The harmonic series $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges.

To prove this, we will show that the sequence of the partial sums $(s_n)_{n=1}^{\infty}$, $s_n = \sum_{k=1}^{n}\frac{1}{k}$, is unbounded. Then it follows by Proposition 2.9 that $(s_n)_{n=1}^{\infty}$ diverges.

We consider the partial sums with $n = 2^m$, $m \in \mathbb{N}$. We can estimate them as follows:

$$\begin{aligned}
s_{2^m} &= \sum_{k=1}^{2^m} \frac{1}{k} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots \\
&\qquad + \left(\frac{1}{2^{m-1}+1} + \frac{1}{2^{m-1}+2} + \cdots + \frac{1}{2^m}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^m} + \frac{1}{2^m} + \cdots + \frac{1}{2^m}\right) \\
&= 1 + \frac{1}{2} + 2 \cdot \frac{1}{2} + 4 \cdot \frac{1}{8} + \cdots + 2^{m-1} \cdot \frac{1}{2^m} \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2} = 1 + \frac{1}{2}m.
\end{aligned}$$

The sequence $\left(1 + \frac{1}{2}m\right)_{m=1}^{\infty}$ is unbounded. Thus, also $(s_{2^m})_{m=1}^{\infty}$ is unbounded, and consequently the sequence $(s_n)_{n=1}^{\infty}$ is unbounded.

We observe that the terms $\frac{1}{n}$ of the harmonic series decrease and tens to zero as $n \to \infty$. However, they decrease to zero not fast enough, and the partial sums of $\sum_{n=1}^{\infty} \frac{1}{n}$ eventually exceed any real number.

△

2.5 The comparison test for series

In this section we consider a very useful test for convergence and divergence of series with non-negative terms.

Theorem 2.30 (*The Comparison Test for Series*) *Let $(a_n)_{n=1}^{\infty}$, $(b_n)_{n=1}^{\infty}$ be two sequences satisfying $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}$.*

(1) *If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.*

(2) *If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.*

△

Proof. (1) Since $a_k \geq 0$, $b_k \geq 0$ for all $k \in \mathbb{N}$, the sequences $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$, $\left(\sum_{k=1}^{n} b_k\right)_{n=1}^{\infty}$ are increasing.

Suppose that the series $\sum_{n=1}^{\infty} b_n$ converges. This implies that the sequence $\left(\sum_{k=1}^{n} b_k\right)_{n=1}^{\infty}$ converges. By Proposition 2.9, the sequence $\left(\sum_{k=1}^{n} b_k\right)_{n=1}^{\infty}$ is bounded, i.e. there exists $M > 0$ such that $\sum_{k=1}^{n} b_k \leq M$ for all $n \in \mathbb{N}$. Now the assumption $a_k \leq b_k$ for all $k \in \mathbb{N}$ yields $\sum_{k=1}^{n} a_k \leq \sum_{k=1}^{n} b_k \leq M$ for all $n \in \mathbb{N}$.

We have shown that the increasing sequence $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$ is bounded above by M. By the Monotone Convergence Theorem (Theorem 2.17), the sequence $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$ converges, which means that the series $\sum_{n=1}^{\infty} a_n$ converges.

(2) To prove the second statement, assume that the series $\sum_{n=1}^{\infty} a_n$ diverges. If $\sum_{n=1}^{\infty} b_n$ would converge, then by (1) also $\sum_{n=1}^{\infty} a_n$ would converge. Thus, $\sum_{n=1}^{\infty} b_n$ must be divergent.

□

Remark (1) It is sufficient to assume in Theorem 2.30 that the inequalities $0 \leq a_n \leq b_n$ hold for all $n \geq N$ with some $N \in \mathbb{N}$.

(2) Under the assumptions of the theorem, it is possible that the series $\sum_{n=1}^{\infty} a_n$ converges and the series $\sum_{n=1}^{\infty} b_n$ diverges.

(3) The statements of the theorem only hold for series with non-negative terms.

△

Example 2.31 Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.
For all $n \in \mathbb{N}$ we have

$$0 \leq \frac{1}{n^2} = \frac{2}{n(n+n)} \leq \frac{2}{n(n+1)}.$$

From Example 2.25 we know that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges. By the first statement in Proposition 2.24, also the series $\sum_{n=1}^{\infty} \frac{2}{n(n+1)}$ converges. Finally, by the Comparison Test (Theorem 2.30), the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. Moreover,

$$0 \leq \sum_{n=1}^{\infty} \frac{1}{n^2} \leq 2 \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 2.$$

It can be shown that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

we will not go into details.

△

Example 2.32 Consider the series $\sum_{n=1}^{\infty} \left(\frac{\sin n}{n}\right)^2$.

The inequalities $0 \leq \sin^2 n \leq 1$ imply

$$0 \leq \left(\frac{\sin n}{n}\right)^2 \leq \frac{1}{n^2}, \quad n \in \mathbb{N}.$$

Since the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, it follows that also the series $\sum_{n=1}^{\infty} \left(\frac{\sin n}{n}\right)^2$ converges.

△

Example 2.33 Consider the series $\sum_{n=1}^{\infty} \frac{n^2+3}{n^3+2}$.

We have

$$\frac{n^2+3}{n^3+2} \geq \frac{n^2}{n^3+n^3} = \frac{1}{2n}, \quad n \geq 2.$$

Remember that the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (Example 2.29). By Proposition 2.24, also the series $\sum_{n=1}^{\infty} \frac{1}{2n}$ diverges. With the help of the Comparison Test, we conclude that the series $\sum_{n=1}^{\infty} \frac{n^2+3}{n^3+2}$ diverges.

△

Example 2.34 Consider the series $\sum_{n=0}^{\infty} \frac{1}{2+3^n}$.

For all $n = 0, 1, 2, \ldots$ we have $0 \leq \frac{1}{2+3^n} \leq \frac{1}{3^n} = \left(\frac{1}{3}\right)^n$. Since the geometric series $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$ converges, by the Comparison Test also the series $\sum_{n=0}^{\infty} \frac{1}{2+3^n}$ converges.

△

The Comparison Test in its essence allows us to figure out whether the terms of a series converge to zero fast enough (and then the series converges) or too slowly (and then the series diverges). The following result gives a convenient "scale" which we can compare a given series against.

Theorem 2.35 (*The p-series*) *The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ (called the p-series) converges if and only if $p > 1$.*

△

Proof. A. First suppose that $p \leq 1$. We have to show that the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges. This follows easily by comparison with the harmonic series: for all $n \in \mathbb{N}$ we have

$$0 < \frac{1}{n} \leq \frac{1}{n^p}.$$

Since the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, it follows that the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ with $p \leq 1$ diverges as well.

B. Now let us consider the case when $p > 1$. The sequence of the partial sums $\left(\sum_{k=1}^{n} \frac{1}{k^p}\right)_{n=1}^{\infty}$ is increasing. We are going to show that it is bounded above.

First we take $n = 2^m - 1$ with some $m \in \mathbb{N}$. We have

$$\begin{aligned}
s_{2^m-1} &= \sum_{k=1}^{2^m-1} \frac{1}{k^p} \\
&= 1 + \left(\frac{1}{2^p} + \frac{1}{3^p}\right) + \left(\frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p}\right) \\
&\qquad + \left(\frac{1}{8^p} + \cdots + \frac{1}{15^p}\right) + \cdots + \left(\frac{1}{(2^{m-1})^p} + \cdots + \frac{1}{(2^m-1)^p}\right) \\
&\leq 1 + \left(\frac{1}{2^p} + \frac{1}{2^p}\right) + \left(\frac{1}{4^p} + \frac{1}{4^p} + \frac{1}{4^p} + \frac{1}{4^p}\right) \\
&\qquad + \left(\frac{1}{8^p} + \cdots + \frac{1}{8^p}\right) + \cdots + \left(\frac{1}{(2^{m-1})^p} + \cdots + \frac{1}{(2^{m-1})^p}\right) \\
&= 1 + 2 \cdot \frac{1}{2^p} + 4 \cdot \frac{1}{4^p} + 8 \cdot \frac{1}{8^p} + \cdots + 2^{m-1} \cdot \frac{1}{(2^{m-1})^p} \\
&= 1 + \left(\frac{2}{2^p}\right)^1 + \left(\frac{2}{2^p}\right)^2 + \left(\frac{2}{2^p}\right)^3 + \cdots + \left(\frac{2}{2^p}\right)^{m-1} \\
&= \sum_{j=0}^{m-1} \left(\frac{2}{2^p}\right)^j.
\end{aligned}$$

Notice that $\frac{2}{2^p} < 1$, and therefore the geometric series $\sum_{j=0}^{\infty} \left(\frac{2}{2^p}\right)^j$ converges to a finite sum $M = \sum_{j=0}^{\infty} \left(\frac{2}{2^p}\right)^j \in \mathbb{R}$. We obtain

$$s_{2^m-1} = \sum_{k=1}^{2^m-1} \frac{1}{k^p} \leq \sum_{j=0}^{m-1} \left(\frac{2}{2^p}\right)^j \leq \sum_{j=0}^{\infty} \left(\frac{2}{2^p}\right)^j = M.$$

For an arbitrary $n \in \mathbb{N}$ we can find $m \in \mathbb{N}$ such that $n \leq 2^m - 1$. Consequently,

$$s_n \leq s_{2^m-1} \leq M \quad \text{for all } n \in \mathbb{N}.$$

This shows that the sequence $(s_n)_{n=1}^{\infty}$ is bounded above. By the Monotone Convergence Theorem (Theorem 2.17), it converges, i.e. the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent.

□

2.6 Subsequences and the Bolzano-Weierstrass Theorem

In this section we come back to studying sequences rather then series.

Definition 2.36 Let $(a_n)_{n=1}^{\infty}$ be a sequence, and let $n_1 < n_2 < n_3 < \cdots$ be a strictly increasing sequence of natural numbers. The sequence $(a_{n_k})_{k=1}^{\infty}$ is called a *subsequence* of the sequence $(a_n)_{n=1}^{\infty}$.

△

Thus, the terms of the subsequence are terms of the original sequence and they appear in the same order, but some of the terms may be left out. Any sequence is a subsequence of itself. Any sequence has infinitely many subsequences.

Example 2.37 Consider the sequence

$$\left((-1)^{n-1}\left(1 - \frac{1}{n}\right)\right)_{n=1}^{\infty} = \left(0, -\frac{1}{2}, \frac{2}{3}, -\frac{3}{4}, \frac{4}{5}, -\frac{5}{6}, \cdots\right).$$

As we mentioned above, this sequence has infinitely many subsequences. For example, the sequences

$$\left(0, \frac{2}{3}, \frac{4}{5}, \cdots\right) = \left(1 - \frac{1}{2k-1}\right)_{k=1}^{\infty}$$

and

$$\left(-\frac{1}{2}, -\frac{3}{4}, -\frac{5}{6}, \cdots\right) = \left(-\left(1 - \frac{1}{2k}\right)\right)_{k=1}^{\infty}$$

are two subsequences of it. Notice that the first of these subsequences converges to 1 while the second one converges to -1. The sequence $\left((-1)^{n-1}\left(1 - \frac{1}{n}\right)\right)_{n=1}^{\infty}$ itself diverges.

△

In this section we will discuss convergence of subsequences. Our first result is intuitively very clear.

Proposition 2.38 *Any subsequence of a converging sequence converges to the same limit.*

△

Proof. Let $(a_n)_{n=1}^{\infty}$ be a convergent sequence with $\lim_{n\to\infty} a_n = a$. By the definition of the limit, for any $\varepsilon > 0$ we can find $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N.$$

If $(n_k)_{k=1}^{\infty}$ is a strictly increasing sequence of natural numbers, then $n_k \geq k$ for all $k \in \mathbb{N}$. It follows that if $k \geq N$, then $n_k \geq k \geq N$ and thus

$$|a_{n_k} - a| < \varepsilon \quad \text{for all } k \geq N.$$

By definition, $\lim_{k\to\infty} a_{n_k} = a$.

□

The following corollaries are useful when we want to show that a given sequence does not converge.

Corollary 2.39 (1) *If a sequence has a divergent subsequence, then the sequence itself diverges.*

(2) *If a sequence has two subsequences that converge to different limits, than the sequence itself diverges.*

△

It is easy to see that if a sequence is bounded then all its subsequences are also bounded. We will now prove a surprising results that in this case there is a least one subsequence that is convergent.

Theorem 2.40 (*The Bolzano-Weierstrass Theorem*) *Every bounded sequence in* $\mathbb{R}$ *has a convergent subsequence.*

△

Proof. Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence. We can find a closed and bounded interval I_0 such that $a_n \in I_0$ for all $n \in \mathbb{N}$. Let the length of the interval I_0 be ℓ; we may assume that $\ell > 0$.

Using the midpoint of the interval I_0, we bisect I_0 into two closed and bounded intervals of length $\frac{1}{2}\ell$. At least one of these intervals contains infinitely many terms of $(a_n)_{n=1}^{\infty}$. We choose this interval to be I_1.

We continue in the same manner. In the k-th step we obtain a closed and bounded interval I_k of length $\left(\frac{1}{2}\right)^k \ell$ that contains infinitely many terms of $(a_n)_{n=1}^{\infty}$. In the $(k+1)$-st step we use the midpoint of I_k to bisect it into two closed and bounded intervals of length $\left(\frac{1}{2}\right)^{k+1} \ell$. At least one of these intervals contains infinitely many terms of $(a_n)_{n=1}^{\infty}$, and we take this interval to be I_{k+1}.

In this way we construct a sequence of closed bounded intervals $(I_k)_{k=1}^{\infty}$ such that

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots,$$

the interval I_k has length $\left(\frac{1}{2}\right)^k \ell$, and each interval I_k contains infinitely many terms of $(a_n)_{n=1}^{\infty}$.

By the Nested Interval Property (Theorem 1.12), $\bigcap_{k=1}^{\infty} I_k \neq \emptyset$. Take $a \in \bigcap_{k=1}^{\infty} I_k$.[1]

We will now construct a subsequence $(a_{n_k})_{k=1}^{\infty}$ that converges to a. We know that I_1 contains infinitely many terms of $(a_n)_{n=1}^{\infty}$. We choose $n_1 \in \mathbb{N}$ such that $a_{n_1} \in I_1$. Furthermore, I_2 contains infinitely many terms of $(a_n)_{n=1}^{\infty}$. Thus we can choose $n_2 > n_1$ such that $a_{n_2} \in I_2$. We continue in the same manner. In the k-th step we choose $n_k > n_{k-1}$ such that $a_{n_k} \in I_k$. In this way we construct a subsequence $(a_{n_k})_{k=1}^{\infty}$ with the property that $a_{n_k} \in I_k$, $k \in \mathbb{N}$.

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $\left(\frac{1}{2}\right)^N \ell < \varepsilon$. Since $a \in I_k$ and $a_{n_k} \in I_k$, we have

$$|a_{n_k} - a| \leq \left(\frac{1}{2}\right)^k \ell \leq \left(\frac{1}{2}\right)^N \ell < \varepsilon \quad \text{for all } k \geq N.$$

This proves that $\lim_{k\to\infty} a_{n_k} = a$.

□

We finish this section by a useful property whose proof is left as an exercise.

Proposition 2.41 *Let $(a_n)_{n=1}^{\infty}$ be a sequence. If $\lim_{k\to\infty} a_{2k} = \lim_{k\to\infty} a_{2k-1} = L$, then $\lim_{n\to\infty} a_n = L$.*

△

[1]We will not use this in the proof, but in fact $\bigcap_{k=1}^{\infty} I_k = \{a\}$. Indeed, suppose that $b \in \bigcap_{k=1}^{\infty} I_k$. Then $a, b \in I_k$ for any $k \in \mathbb{N}$, and since the length of I_k is $\left(\frac{1}{2}\right)^k \ell$, we have that $|a - b| \leq \left(\frac{1}{2}\right)^k \ell$ for all $k \in \mathbb{N}$. Since $\left(\frac{1}{2}\right)^k \ell \to 0$ as $k \to \infty$, it follows that $|a - b| = 0$, i.e. $a = b$.

Proof. Exercise. □

2.7 Cauchy sequences

Definition 2.42 We say that a sequence $(a_n)_{n=1}^{\infty}$ is a *Cauchy sequence* if

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n, m \geq N : |a_n - a_m| < \varepsilon,$$

i.e. for any positive number ε there exists $N \in \mathbb{N}$ such that $|a_n - a_m| < \varepsilon$ for all $n, m \in \mathbb{N}$ with $n, m \geq N$. △

The definition of a Cauchy sequence must remind you of the definition of a convergent sequence (Definition 2.2). Indeed, we will see in this section that convergent sequences and Cauchy sequences are closely related.

Proposition 2.43 *Every Cauchy sequence is bounded.* △

Proof. Let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence. Using the definition and taking $\varepsilon = 1$, we see that there exists $N \in \mathbb{N}$ such that $|a_n - a_k| < 1$ for all $n, k \geq N$. In particular,

$$|a_n - a_N| < 1 \quad \text{for all } n \geq N.$$

Put $m = \min\{a_1, \dots, a_{N-1}, a_N - 1\}$, $M = \max\{a_1, \dots, a_{N-1}, a_N + 1\}$. By above we have

$$m \leq a_n \leq M \quad \text{for all } n \in \mathbb{N}.$$

□

Proposition 2.43 and its proof should remind you of the statement and the proof of Proposition 2.9. Modifying the proof of Theorem 2.14 in a similar way, one can show that scalar multiples, sums, products and (under sensible assumptions) quotients of Cauchy sequences are Cauchy sequences. Try to establish these properties as an exercise!

Let us now investigate a relation between the properties of a sequence to be Cauchy and to converge.

Lemma 2.44 *Every convergent sequence is a Cauchy sequence.* △

Proof. Let $(a_n)_{n=1}^{\infty}$ be a convergent sequence with $\lim_{n\to\infty} a_n = a$.

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N.$$

But then for all $n, m \geq N$ we have

$$|a_n - a_m| = |(a_n - a) - (a_m - a)| \leq |a_n - a| + |a_m - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which means that $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence.

□

In the theorem below we will show that the converse statement is true in $\mathbb{R}$, i.e. every Cauchy sequence in $\mathbb{R}$ converges in $\mathbb{R}$. Thus, a sequence is convergent in $\mathbb{R}$ if and only if it is a Cauchy sequence. However, it should be noticed that this property does not hold in other spaces. While Lemma 2.44 is always true (for example, also for sequences in $\mathbb{Q}$), the fact that every Cauchy sequence has a limit is a property of $\mathbb{R}$. This statement does not hold, for example, in $\mathbb{Q}$: One can construct a Cauchy sequence in $\mathbb{Q}$ that does not converge in $\mathbb{Q}$ — take, say, a sequence of rational numbers that converge to $\sqrt{2}$. To some extent, the statement that every Cauchy sequence converges in $\mathbb{R}$ is equivalent to the Completeness Axiom.[2]

Theorem 2.45 (*The Cauchy Criterion*) *In $\mathbb{R}$, a sequence converges if and only if it is a Cauchy sequence.*

△

Proof. We have proven in Lemma 2.44 that every convergent sequence is a Cauchy sequence. Thus, we need to prove the converse statement.

Let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence. By Proposition 2.43, it is bounded. Therefore, by the Bolzano-Weierstrass Theorem (Theorem 2.40), $(a_n)_{n=1}^{\infty}$ has a convergent subsequence $(a_{n_k})_{k=1}^{\infty}$. Put $a = \lim_{k\to\infty} a_{n_k}$.

We will show that $a = \lim_{n\to\infty} a_n$. Take an arbitrary $\varepsilon > 0$. There is $K \in \mathbb{N}$ such that

$$|a_{n_k} - a| < \frac{\varepsilon}{2} \quad \text{for all } k \geq K.$$

Since $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence, there is $N \in \mathbb{N}$ such that

$$|a_n - a_m| < \frac{\varepsilon}{2} \quad \text{for all } n, m \geq N.$$

[2] You can find a discussion on the exact relationship between these properties in Section 2.6 of Abbott's textbook "Understanding Analysis".

Now choose $k_0 \in \mathbb{N}$ such that $k_0 \geq K$ and $n_{k_0} \geq N$. Then for all $m \geq N$ we have

$$|a_m - a| \leq |a_m - a_{n_{k_0}}| + |a_{n_{k_0}} - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

and this proves that $\lim_{n\to\infty} a_n = a$.

□

Theorem 2.45 has an implication for series.

Corollary 2.46 (*The Cauchy Criterion for Series*) *In $\mathbb{R}$, a series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that*

$$|a_{n+1} + a_{n+2} + \cdots + a_m| < \varepsilon \quad \textit{for any } m > n \geq N.$$

△

Proof. By above, the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the sequence of the partial sums $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$ is a Cauchy sequence. If $m > n$, then

$$\sum_{k=1}^{m} a_k - \sum_{k=1}^{n} a_k = a_{n+1} + a_{n+2} + \cdots + a_m.$$

Thus, $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$ is a Cauchy sequence if and only if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\left|\sum_{k=1}^{m} a_k - \sum_{k=1}^{n} a_k\right| = |a_{n+1} + a_{n+2} + \cdots + a_m| < \varepsilon \quad \text{for any } m > n \geq N.$$

□

2.8 Absolute and conditional convergence of series

We have seen that the Comparison Test (Theorem 2.30) only can be applied to series with non-negative terms. In the general situation, sometimes it may help to consider the series $\sum_{n=1}^{\infty} |a_n|$ along with $\sum_{n=1}^{\infty} a_n$.

Proposition 2.47 *If the series $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ converges as well.*

△

Proof. We are going to use the Cauchy Criterion (Corollary 2.46). If the series $\sum_{n=1}^{\infty} |a_n|$ converges, then for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_{n+1}| + |a_{n+2}| + \cdots + |a_m| < \varepsilon \quad \text{for all } m > n \geq N.$$

But the triangle inequality gives

$$|a_{n+1} + a_{n+2} + \cdots + a_m| \leq |a_{n+1}| + |a_{n+2}| + \cdots + |a_m| < \varepsilon \quad \text{for all } m > n \geq N,$$

and the Cauchy Criterion applied to series $\sum_{n=1}^{\infty} a_n$ yields that $\sum_{n=1}^{\infty} a_n$ converges. □

The converse of Proposition 2.47 does not hold, we will see examples below. This motivates the following definition.

Definition 2.48 (1) We say that a series $\sum_{n=1}^{\infty} a_n$ *converges absolutely* if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

(2) If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ diverges, we say that the series $\sum_{n=1}^{\infty} a_n$ *converges conditionally*.

△

An example of an absolutely convergent series is $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$. Indeed, we have seen in Example 2.31 that $\sum_{n=1}^{\infty} \left|\frac{(-1)^{n-1}}{n^2}\right| = \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

An example of a conditionally convergent series is $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$. We already know that the harmonic series $\sum_{n=1}^{\infty} \left|\frac{(-1)^{n-1}}{n}\right| = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges, so that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ cannot be absolutely convergent. Its convergence follows from Theorem 2.49 below. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is called the *alternating harmonic series*.

We consider, more generally, series of the form

$$\sum_{n=0}^{\infty} (-1)^n a_n, \quad \text{where } a_n \geq 0,\ n \in \mathbb{N}.$$

Such series are called *alternating series*. They have the form

$$\sum_{n=0}^{\infty} (-1)^n a_n = a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + \cdots.$$

The name comes from the fact that the signs of the terms alternate.

Theorem 2.49 *(The Alternating Series Test, or the Leibniz Test)* *Let $(a_n)_{n=0}^{\infty}$ be a sequence such that $a_0 \geq a_1 \geq a_2 \geq \cdots \geq a_n \geq a_{n+1} \geq \cdots$ and $\lim_{n\to\infty} a_n = 0$. Then*

(1) *the alternating series $\sum_{n=0}^{\infty}(-1)^n a_n$ converges,*

(2) *the partial sums $s_m = \sum_{n=0}^{m}(-1)^n a_n$ approximate the sum $s = \sum_{n=0}^{\infty}(-1)^n a_n$ within the error*

$$|s - s_m| \leq a_{m+1}.$$

△

Notice that the assumptions $a_0 \geq a_1 \geq a_2 \geq \cdots \geq a_n \geq a_{n+1} \geq \cdots$ and $\lim_{n\to\infty} a_n = 0$ imply that $a_n \geq 0$, $n \in \mathbb{N}$.

Proof. We will consider the subsequences $(s_{2k})_{k=0}^{\infty}$ and $(s_{2k+1})_{k=0}^{\infty}$ of the sequence of partial sums $(s_m)_{m=0}^{\infty}$. We have

$$\begin{aligned} s_{2k+2} - s_{2k} &= \sum_{n=0}^{2k+2}(-1)^n a_n - \sum_{n=0}^{2k}(-1)^n a_n \\ &= (-1)^{2k+2}a_{2k+2} + (-1)^{2k+1}a_{2k+1} = a_{2k+2} - a_{2k+1} \leq 0. \end{aligned}$$

Thus, the sequence $(s_{2k})_{k=0}^{\infty}$ is decreasing. Similarly,

$$\begin{aligned} s_{2k+1} - s_{2k-1} &= \sum_{n=0}^{2k+1}(-1)^n a_n - \sum_{n=0}^{2k-1}(-1)^n a_n \\ &= (-1)^{2k+1}a_{2k+1} + (-1)^{2k}a_{2k} = -a_{2k+1} + a_{2k} \geq 0, \end{aligned}$$

and the sequence $(s_{2k+1})_{k=0}^{\infty}$ is increasing. Furthermore,

$$s_{2k+1} - s_{2k} = \sum_{n=0}^{2k+1}(-1)^n a_n - \sum_{n=0}^{2k}(-1)^n a_n = (-1)^{2k+1}a_{2k+1} \leq 0,$$

so that $s_{2k+1} \leq s_{2k}$ for all k. Taking into consideration that $(s_{2k})_{k=0}^{\infty}$ is decreasing and $(s_{2k+1})_{k=0}^{\infty}$ is increasing, we obtain that

$$s_1 \leq s_{2k+1} \leq s_{2k} \leq s_0 \quad \text{for all } k = 0, 1, 2, \dots .$$

We have shown that the sequence $(s_{2k})_{k=0}^{\infty}$ is decreasing and bounded below by s_1. At the same time, the sequence $(s_{2k+1})_{k=0}^{\infty}$ is increasing and bounded above by s_0. By the Monotone Convergence Theorem (Theorem 2.17), both sequences are convergent.

Put $s = \lim_{k\to\infty} s_{2k}$. As $s_{2k+1} = s_{2k} + (-1)^{2k+1}a_{2k+1} = s_{2k} - a_{2k+1}$, we have

$$\lim_{k\to\infty} s_{2k+1} = \lim_{k\to\infty} s_{2k} - \lim_{k\to\infty} a_{2k+1} = s - 0 = s.$$

Thus, both sequences $(s_{2k})_{k=0}^{\infty}$ and $(s_{2k+1})_{k=0}^{\infty}$ converge to the same limit

$$s = \sup_{k=0,1,\dots} s_{2k+1} = \inf_{k=0,1,\dots} s_{2k}.$$

It follows by Proposition 2.41 that $\lim_{m\to\infty} s_m = s$. In other words, the series $\sum_{n=0}^{\infty}(-1)^n a_n$ converges.

The above consideration implies that

$$s_1 \le s_3 \le \cdots \le s_{2k+1} \le \cdots \le s \le \cdots \le s_{2k} \le \cdots \le s_2 \le s_0.$$

From here we see that

$$0 \le s - s_{2k+1} \le s_{2k+2} - s_{2k+1} = a_{2k+2}$$

and

$$0 \le s_{2k} - s \le s_{2k} - s_{2k+1} = a_{2k+1}.$$

Together this gives

$$|s - s_m| \le a_{m+1}.$$

□

Example 2.50 It follows by the Alternating Convergence Test (Theorem 2.49) that the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges. We have mentioned above that it converges conditionally.

With the help of a Taylor series, one can show that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$, we do not go into details.

Using the second statement in Theorem 2.49, one can estimate how many terms of the alternating series one should take in order to calculate its sum within a given accuracy. Imagine, for example, that we want to calculate the value of $\ln 2$ using the alternating harmonic series within the accuracy of 0.001. We should choose m such that[3]

$$\left|\ln 2 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{(-1)^{m-1}}{m}\right)\right| \le \frac{1}{m+1} < 0.001.$$

[3]The rule that is easy to remember is that for estimating the difference between the sum of a series and its partial sum, we take the first term of the series that was not used in the partial sum.

We should take $m = 1000$.

In practice, other series are used to calculate $\ln 2$, because — as we just have seen — the alternating harmonic series converges very slowly.

△

Remark In the Alternating Series Test (Theorem 2.49) we assume that the sequence (a_n) is decreasing and that $\lim_{n\to\infty} a_n = 0$. Both assumptions are necessary for the theorem to hold. We already know (Proposition 2.21) that the assumption $\lim_{n\to\infty} a_n = 0$ is necessary for any series to converge. The example below shows that the assumption of the monotonicity of the sequence (a_n) cannot be removed.

△

Example 2.51 Consider the alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}a_n$, where

$$a_n = \begin{cases} \frac{4}{n^2}, & \text{if } n \text{ is even,} \\ \frac{2}{n+1}, & \text{if } n \text{ is odd.} \end{cases}$$

Notice that $\lim_{n\to\infty} a_n = 0$.

We consider the partial sums s_{2m}, $m \in \mathbb{N}$. For an odd n we write $n = 2k - 1$ and for an even n we write $n = 2k$ with $k \in \mathbb{N}$. We have

$$s_{2m} = \sum_{n=1}^{2m}(-1)^{n-1}a_n = \sum_{k=1}^{m}\frac{2}{(2k-1)+1} - \sum_{k=1}^{m}\frac{4}{(2k)^2} = \sum_{k=1}^{m}\frac{1}{k} - \sum_{k=1}^{m}\frac{1}{k^2}.$$

We know that the sequence $\left(\sum_{k=1}^{m}\frac{1}{k^2}\right)_{m=1}^{\infty}$ converges and the sequence $\left(\sum_{k=1}^{m}\frac{1}{k}\right)_{m=1}^{\infty}$ diverges. If follows that the sequence $(s_{2m})_{m=1}^{\infty}$ diverges. This implies that the series $\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ diverges.

△

There are several further convergence tests for conditionally convergent series such as Abel's Test and Dirichtlet's Test. These tests are generalizations of the Alternating Series Test, but they can be applied also to series that are not alternating. We will not discuss these tests in our course, you can find them in Abbott's textbook "Understanding Analysis".

2.9 The ratio and the root tests

In this section we discuss two tests for absolute convergence that are frequently used in the practice. Both these tests are based on the comparison with a geometric series.

Theorem 2.52 (*The Ratio Test*) *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers such that the limit*

$$L = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$$

exists. Then the following holds.

(1) *If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.*

(2) *If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.*

(3) *If $L = 1$, then the test is inconclusive.*

△

Proof. (1) Suppose that $L < 1$. Take $r \in \mathbb{R}$ such that $L < r < 1$. By the definition of the limit with $\varepsilon = r - L > 0$, there exists $N \in \mathbb{N}$ such that

$$\frac{|a_{n+1}|}{|a_n|} < r \quad \text{for all } n \geq N.$$

This implies $|a_{N+1}| < |a_N|r$, $|a_{N+2}| < |a_{N+1}|r < |a_N|r^2$, and so on. In general,

$$|a_{N+j}| < |a_N|r^j, \quad j \in \mathbb{N};$$

this can be shown by induction. Since $\sum_{j=0}^{\infty} |a_N|r^j$ is a converging geometric series, it follows by the Comparison Test (Theorem 2.30) that the series $\sum_{n=N}^{\infty} |a_n|$ converges. Hence, the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(2) Now assume that $L > 1$. Using again the definition of the limit with $\varepsilon = L - 1 > 0$, we conclude that there exists $N \in \mathbb{N}$ such that

$$\frac{|a_{n+1}|}{|a_n|} > 1 \quad \text{for all } n \geq N.$$

But then $|a_{n+1}| > |a_n| > 0$ for all $n \geq N$, so that (a_n) does not converge to 0. Thus, the series $\sum_{n=1}^{\infty} a_n$ diverges.

(3) If $L = 1$, then the test is inconclusive. For the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ we have

$$L = \lim_{n\to\infty} \frac{1/(n+1)}{1/n} = \lim_{n\to\infty} \frac{n}{n+1} = 1$$

and the series is divergent. For the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ we also have $L = 1$ and the series is conditionally convergent. Finally, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ we have

$$L = \lim_{n\to\infty} \frac{1/(n+1)^2}{1/n^2} = \lim_{n\to\infty} \frac{n^2}{(n+1)^2} = 1$$

and the series is absolutely convergent.

□

Example 2.53 Consider the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$.

Clearly, all terms of the series are positive, so we do not need to care for the absolute values. We have

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1} n!}{(n+1)! 2^n} = \frac{2}{n+1}$$

and

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{2}{n+1} = 0 < 1.$$

It follows that the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges.

△

Example 2.54 Consider the series $\sum_{n=1}^{\infty} \frac{n(n+1)}{1+2^n}$.

Again, all terms are positive. We have

$$\begin{aligned} L &= \lim_{n\to\infty} \frac{(n+1)(n+2)}{1+2^{n+1}} \cdot \frac{1+2^n}{n(n+1)} = \lim_{n\to\infty} \frac{n+2}{n} \cdot \lim_{n\to\infty} \frac{1+2^n}{1+2^{n+1}} \\ &= 1 \cdot \lim_{n\to\infty} \frac{\frac{1}{2^n}+1}{\frac{1}{2^n}+2} = \frac{1}{2} < 1, \end{aligned}$$

and thus the series $\sum_{n=1}^{\infty} \frac{n(n+1)}{1+2^n}$ converges.

△

Theorem 2.55 (*The Root Test*) *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers such that the limit*

$$L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}}$$

exists. Then the following holds.

(1) *If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.*

(2) *If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.*

(3) *If $L = 1$, then the test is inconclusive.*

△

Proof. (1) Suppose that $L < 1$. Take $r \in \mathbb{R}$ such that $L < r < 1$. By the definition of the limit with $\varepsilon = r - L > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n|^{\frac{1}{n}} < r \quad \text{for all } n \geq N.$$

This implies that

$$|a_n| < r^n \quad \text{for all } n \geq N.$$

Since the geometric series $\sum_{n=1}^{\infty} r^n$ converges, by the Comparison Test (Theorem 2.30), the series $\sum_{n=1}^{\infty} |a_n|$ converges, i.e. the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(2) Now assume that $L > 1$. Using again the definition of the limit with $\varepsilon = L - 1 > 0$, we conclude that there exists $N \in \mathbb{N}$ such that

$$|a_n|^{\frac{1}{n}} > 1 \quad \text{for all } n \geq N.$$

But then $|a_n| > 1$ for all $n \geq N$. In particular, (a_n) does not converge to 0. Thus, the series $\sum_{n=1}^{\infty} a_n$ diverges.

(3) The same three examples as in the proof of Theorem 2.52 show that the test is inconclusive when $L = 1$.

□

Example 2.56 Consider the series $\sum_{n=1}^{\infty} \frac{2^{3n+4}}{n^n}$.

We have

$$a_n = \frac{2^{3n+4}}{n^n} = \frac{2^4 \cdot (2^3)^n}{n^n} = 16\left(\frac{8}{n}\right)^n.$$

By Proposition 2.24, it suffices to show that the series $\sum_{n=1}^{\infty} \left(\frac{8}{n}\right)^n$ converges. We apply the Root Test and see that

$$L = \lim_{n\to\infty} \left(\left(\frac{8}{n}\right)^n\right)^{\frac{1}{n}} = \lim_{n\to\infty} \frac{8}{n} = 0 < 1.$$

Thus, the series $\sum_{n=1}^{\infty} \frac{2^{3n+4}}{n^n}$ converges.

△

2.10 Rearrangements

Due to the commutative law of addition $a + b = b + a$, $a, b \in \mathbb{R}$, we know that the sum of finitely many terms does not depend on the order of summation. In this section we investigate the question whether this is also true for infinite series.

Given a series $\sum_{n=1}^{\infty} a_n$, we consider a series whose terms are the same, but their order is different. The formal definition is the following.

Definition 2.57 A series $\sum_{n=1}^{\infty} b_n$ is called a *rearrangement* of the series $\sum_{n=1}^{\infty} a_n$ if there exists a bijection $f : \mathbb{N} \to \mathbb{N}$ such that $b_{f(n)} = a_n$, $n \in \mathbb{N}$.

△

Clearly, any series is a rearrangement of itself. If $\sum_{n=1}^{\infty} b_n$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$, then $\sum_{n=1}^{\infty} a_n$ is a rearrangement of $\sum_{n=1}^{\infty} b_n$. If $\sum_{n=1}^{\infty} b_n$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} c_n$ is a rearrangement of $\sum_{n=1}^{\infty} b_n$, then $\sum_{n=1}^{\infty} c_n$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$. These properties follow from the properties of bijections.

The main result of this section says that the sum of a series does not depend on the order of summation if the series is absolutely convergent.

Theorem 2.58 *If a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then any rearrangement of $\sum_{n=1}^{\infty} a_n$ converges to the same sum.*

△

Proof. Let $\sum_{n=1}^{\infty} b_n$ be a rearrangement of $\sum_{n=1}^{\infty} a_n$. Denote by $s = \sum_{n=1}^{\infty} a_n$ the sum of the series $\sum_{n=1}^{\infty} a_n$. Put

$$s_m = \sum_{n=1}^{m} a_n, \quad t_m = \sum_{n=1}^{m} b_n, \quad m \in \mathbb{N}.$$

We know that $\lim_{m\to\infty} s_m = s$. Our aim is to show that $\lim_{m\to\infty} t_m = s$.

Let $\varepsilon > 0$. By the definition of convergence, there exists $N_1 \in \mathbb{N}$ such that

$$|s_m - s| < \frac{\varepsilon}{2} \quad \text{for all } m \geq N_1.$$

Since also the series $\sum_{n=1}^{\infty} |a_n|$ converges, by Proposition 2.23 there is $N_2 \in \mathbb{N}$ such that

$$\sum_{n=m+1}^{\infty} |a_n| < \frac{\varepsilon}{2} \quad \text{for all } m \geq N_2.$$

Put $N = \max\{N_1, N_2\}$. Since $\sum_{n=1}^{\infty} b_n$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$, the terms $a_1, \dots, a_N$ will all appear in $\sum_{n=1}^{\infty} b_n$. We want to take m large enough so that

$a_1, \dots, a_N$ all appear in t_m. Let $f : \mathbb{N} \to \mathbb{N}$ be a bijection such that $a_n = b_{f(n)}$, $n \in \mathbb{N}$. Put

$$M = \max\{f(1), f(2), \dots, f(N)\}.$$

Then for any $m \geq M$ the partial sum t_m includes all terms $a_1, \dots, a_N$. The difference $t_m - s_N$ consists of some terms a_j whose indices satisfy $j > N$. Thus,

$$|t_m - s_N| \leq \sum_{j=N+1}^{\infty} |a_j| < \frac{\varepsilon}{2}.$$

Therefore for all $m \geq M$ we have

$$|t_m - s| \leq |t_m - s_N| + |s_N - s| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This implies that $\lim_{m\to\infty} t_m = s$, i.e. $\sum_{n=1}^{\infty} b_n = s$.

□

The content of the following remark is not included in the obligatory part of the course, but it gives you an important information about the behavior of rearrangements.

Remark We can now ask ourselves whether the property described in Theorem 2.58 is true for any series. It turns out that the statement does not hold when the series is conditionally convergent. Moreover, there is the following perhaps surprising result.

Theorem 2.59 (*The Riemann Rearrangement Theorem*) *If a series $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, then for any $x \in \mathbb{R}$ there is a rearrangement of $\sum_{n=1}^{\infty} a_n$ whose sum is x.*

△

We will not give a formal proof of this result. However, we will consider an example that demonstrates the method.

We know that the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

is conditionally convergent, and its sum is $\ln 2 = 0.693147\dots$. The series built from the positive terms

$$\sum_{k=1}^{\infty} \frac{1}{2k-1} = 1 + \frac{1}{3} + \frac{1}{5} + \cdots$$

as well as the series built from the absolute values of the negative terms

$$\sum_{k=1}^{\infty} \frac{1}{2k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \cdots$$

both diverge, their partial sums increase to ∞.

Now let us, for example, take $x = 1.1$. We will construct a rearrangement of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ whose sum is 1.1. In the first step we add positive terms until the sum exceeds $x = 1.1$:

$$1 + \frac{1}{3} = 1.333333\ldots > 1.1.$$

In the second step we add negative terms until the resulting sum is smaller than $x = 1.1$:

$$1 + \frac{1}{3} - \frac{1}{2} = 0.833333\ldots < 1.1.$$

In the next step we again add positive terms until the sum is larger that $x = 1.1$:

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} = 1.176190\ldots > 1.1.$$

Then we add negative terms again until the sum falls below $x = 1.1$:

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} = 0.926190\ldots < 1.1.$$

The next steps are

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} = 1.128210\ldots > 1.1,$$

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} = 0.961544\ldots < 1.1,$$

and so on. Since the terms of the series go to 0, it is intuitively clear that the partial sums of the rearrangement built this way will converge to $x = 1.1$.

$\triangle$

Chapter 3

The topology of $\mathbb{R}$

3.1 Open sets

Definition 3.1 Given $x \in \mathbb{R}$ and $\varepsilon > 0$, the *ε-neighborhood* of x is the set

$$V_\varepsilon(x) = \{y \in \mathbb{R} : |x - y| < \varepsilon\} = (x - \varepsilon, x + \varepsilon).$$

△

Definition 3.2 A set $G \subseteq \mathbb{R}$ is called *open* if for each $x \in G$ there is $\varepsilon > 0$ such that $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq G$.

△

Notice that we regard the empty set as being open.

Example 3.3 An interval (a, b) with $a, b \in \mathbb{R}$, $a < b$ is open. Indeed, consider an arbitrary $x \in (a, b)$. Take $\varepsilon = \min\{x - a, b - x\} > 0$. Then $(x - \varepsilon, x + \varepsilon) \subseteq (a, b)$. Indeed, if $y \in (x - \varepsilon, x + \varepsilon)$, then

$$y > x - \varepsilon \geq x - (x - a) = a \quad \text{and} \quad y < x + \varepsilon \leq x + (b - x) = b,$$

so that $y \in (a, b)$.

△

Similarly, the sets $(-\infty, a)$, (a, ∞) with $a \in \mathbb{R}$ are open. Also the set $\mathbb{R}$ is open; in this case any ε-neighborhood of any point $x \in \mathbb{R}$ is in $\mathbb{R}$.

Example 3.4 An interval $[c, d]$ with $c, d \in \mathbb{R}$, $c \leq d$ is not open. Indeed, the point $c \in [c, d]$ does not possess an ε-neighborhood $V_\varepsilon(c)$ that lies in $[c, d]$. Namely, for any $\varepsilon > 0$ the point $c - \frac{1}{2}\varepsilon$ lies in $V_\varepsilon(c) = (c - \varepsilon, c + \varepsilon)$, but $c - \frac{1}{2}\varepsilon \notin [c, d]$. Thus, $V_\varepsilon(c) \nsubseteq [c, d]$.

△

Theorem 3.5 (1) *The union of any collection of open sets is open.*

(2) *The intersection of a finite collection of open sets is open.*

△

Proof. (1) Let $\{G_i : i \in I\}$ be a collection of open sets, with I being the index set of arbitrary cardinality. If $\bigcup_{i\in I} G_i = \emptyset$, then we are done, so assume that $\bigcup_{i\in I} G_i \neq \emptyset$. Take $x \in \bigcup_{i\in I} G_i$. Then $x \in G_j$ for some $j \in I$. Since G_j is open, there is an ε-neighborhood $V_\varepsilon(x) \subseteq G_j$. But this implies that also $V_\varepsilon(x) \subseteq \bigcup_{i\in I} G_i$. Thus, $\bigcup_{i\in I} G_i$ is open.

(2) Let $\{G_1, G_2, \dots, G_n\}$ be a finite collection of open sets. If $\bigcap_{i=1}^{n} G_i = \emptyset$, then we are done, so assume that $\bigcap_{i=1}^{n} G_i \neq \emptyset$. Take $x \in \bigcap_{i=1}^{n} G_i$. Then $x \in G_j$ for all $j = 1, 2, \dots, n$. Since each G_j is open, there exists $\varepsilon_j > 0$ such that $V_{\varepsilon_j}(x) \subseteq G_j$. Put $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n\} > 0$. We claim that $V_\varepsilon(x) \subseteq \bigcap_{i=1}^{n} G_i$. Indeed, since $\varepsilon \leq \varepsilon_j$, $V_\varepsilon(x) \subseteq V_{\varepsilon_j}(x) \subseteq G_j$ for each $j = 1, 2, \dots, n$. Thus, $\bigcap_{i=1}^{n} G_i$ is open.

□

Remark Statement (2) is in general not true for an intersection of infinitely many open sets. As an example, consider $G_k = \left(-\frac{1}{k}, 1 + \frac{1}{k}\right)$, $k \in \mathbb{N}$. Each G_k is open, but $\bigcap_{k\in\mathbb{N}} G_k = [0, 1]$ is not open.

△

The next notion and the following statement will prove to be useful later.

Definition 3.6 Let $A \subseteq \mathbb{R}$. We say that a sequence $(x_n)_{n=1}^{\infty}$ is *eventually in* A, if there is $N \in \mathbb{N}$ such that $x_n \in A$ for all $n \geq N$.

△

Proposition 3.7 *Let G be an open set. If a sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in G$, then $(x_n)_{n=1}^{\infty}$ is eventually in G.*

△

Proof. Since G is open, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq G$. In other words, for $y \in \mathbb{R}$ we have

$$|y - x| < \varepsilon \implies y \in G.$$

Since $\lim_{n\to\infty} x_n = x$, there is $N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$. By above, $x_n \in G$ for all $n \geq N$. This means that $(x_n)_{n=1}^{\infty}$ is eventually in G.

□

3.2 Closed sets

Definition 3.8 Let $A \subseteq \mathbb{R}$.

(1) A point $x \in \mathbb{R}$ is called a *limit point* of A, if any ε-neighborhood $V_\varepsilon(x)$ of the point x contains a point of A different from x.

(2) The set A' of all limit points of A is called the *derived set* of A.

△

Other terms for limit points are *cluster points* or *accumulation points*. The name "limit points" is explained by the following characterization of limit points.

Proposition 3.9 *A point x is a limit point of a set A if and only if there is a sequence $(x_n)_{n=1}^\infty$ in $A \setminus \{x\}$ with $\lim_{n\to\infty} x_n = x$.*

△

Proof. A. Suppose that x is a limit point of the set A. For each $n \in \mathbb{N}$, the neighborhood $V_{1/n}(x) = \left(x - \frac{1}{n}, x + \frac{1}{n}\right)$ contains at least one point of A that is different from x. Let x_n be such a point. Note that $|x_n - x| < \frac{1}{n}$. Then the sequence $(x_n)_{n=1}^\infty$ lies in $A \setminus \{x\}$ and $\lim_{n\to\infty} x_n = x$.

B. Now assume that a sequence $(x_n)_{n=1}^\infty$ lies in $A \setminus \{x\}$ and $\lim_{n\to\infty} x_n = x$. We want to show that x is a limit point of A. Let $\varepsilon > 0$. By the definition of the limit, we can find $N \in \mathbb{N}$ such that $|x - x_n| < \varepsilon$ for all $n \geq N$. But then $x_N \in (x - \varepsilon, x + \varepsilon) = V_\varepsilon(x)$ and $x_N \neq x$. Since this is true for any $\varepsilon > 0$, x is a limit point of A.

□

Example 3.10 Let $A = (0,1] \cup \{2\}$. We claim that $A' = [0,1]$.

If $x \in (-\infty, 0)$, then x cannot be a limit point of A, because $(-\infty, 0)$ is an open set and by Proposition 3.7 any sequence converging to x must be eventually in $(-\infty, 0)$. But $(-\infty, 0)$ does not contain any points from $A \setminus \{x\}$. By exactly the same argument, $x \in (1, \infty)$ cannot be a limit point of A. Notice that also $x = 2$ is not a limit point of A since $(1, \infty)$ does not contain any points from $A \setminus \{2\}$.

Every point $x \in (0,1]$ is a limit point of A. For example, the sequence $\left(x\frac{n}{n+1}\right)_{n=1}^\infty$ lies in $A \setminus \{x\}$ and converges to x. Finally, $x = 0$ is a limit point of A. In this case we can take the sequence $\left(\frac{1}{n}\right)_{n=1}^\infty$, it lies in $A \setminus \{0\}$ and converges to 0.

△

We have seen in the previous example that limit points of A may belong or not belong to A. On the other hand, not all points of A are limit points of A.

Definition 3.11 If $x \in A$ and x is not a limit point of A, then x is called an *isolated point* of A.

△

Thus, each point of A is a limit point of A or an isolated point of A. In the above example of $A = (0,1] \cup \{2\}$ the point $x = 2$ is an isolated point of A and all points $x \in (0,1]$ are limit points of A.

Example 3.12 Consider the set $A = \mathbb{N}$. Any convergent sequence in $\mathbb{N}$ is eventually constant. This implies that $\mathbb{N}$ has no limit points. Each point in $\mathbb{N}$ is an isolated point.

△

Example 3.13 Consider the set $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. The only limit point of A is $0 \notin A$. Each point $\frac{1}{n}$, $n \in \mathbb{N}$ is an isolated point of $\mathbb{N}$.

△

Definition 3.14 A set $F \subseteq \mathbb{R}$ is called *closed* if it contains all its limit points, i.e. if $F' \subseteq F$.

△

Example 3.15 The set $[a,b]$ with $a,b \in \mathbb{R}$, $a \leq b$ is closed. Indeed, for any convergent sequence $(x_n)_{n=1}^{\infty}$ in $[a,b]$ we have $a \leq x_n \leq b$, $n \in \mathbb{N}$, and thus by Theorem 2.15 $a \leq \lim_{n\to\infty} x_n \leq b$. Thus, all limits of all converging sequences in $[a,b]$ and, in particular, all limit points are in $[a,b]$.

△

The sets $\varnothing$, $\mathbb{R}$, $(-\infty,a]$ and $[a,\infty)$ with $a \in \mathbb{R}$ are closed. The set (a,b) is not closed as it does not contain its limit points a and b.

Notice that "closed" does not mean "not open" and vice versa. Thus, the sets $\varnothing$ and $\mathbb{R}$ are simultaneously open and closed. On the other hand, the sets $(0,1]$ and $[0,1)$ are neither open nor closed.

Theorem 3.16 *A set $A \subseteq \mathbb{R}$ is open if and only if its complement $\mathbb{R} \setminus A$ is closed.*

△

Proof. Denote $B = \mathbb{R} \setminus A$.

A. Assume that A is open. We want to show that B is closed, i.e. all its limit points are in B. If $x \in A$, then x cannot be a limit point of B, because by Proposition 3.7 any sequence with limit x must be eventually in A, and this is impossible for a sequence in $B \setminus \{x\}$. Thus, if x is a limit point of B, then $x \notin A$, i.e. $x \in B$.

B. Now assume that B is closed. We want to show that A is open. Take $x \in A$. Since B is closed, it contains all its limit points. Therefore x cannot be a limit point of B. By the definition of a limit point there must exist a ε-neighborhood $V_\varepsilon(x)$ that does not contain points from $B \setminus \{x\}$. Since also $x \notin B$, we conclude that $V_\varepsilon(x)$ does not contain any points of B. Hence, $V_\varepsilon(x) \subseteq A$. This means that A is open.

□

Theorem 3.17 (1) *The intersection of any collection of closed sets is closed.*

(2) *The union of a finite collection of closed sets is closed.*

△

Proof. Both statements follow from Theorem 3.5 and De Morgan's Laws

$$\mathbb{R} \setminus \left(\bigcup_{i \in I} A_i \right) = \bigcap_{i \in I} (\mathbb{R} \setminus A_i), \quad \mathbb{R} \setminus \left(\bigcap_{i \in I} A_i \right) = \bigcup_{i \in I} (\mathbb{R} \setminus A_i).$$

□

Definition 3.18 The set $\overline{A} = A \cup A'$ is called a *closure* of the set A.

△

Theorem 3.19 *Let $A \subseteq \mathbb{R}$. The closure $\overline{A}$ is closed. Moreover, $\overline{A}$ is the smallest closed set containing A.*

△

Proof. A. We show that $\overline{A} = A \cup A'$ is closed. Let x be a limit point of $\overline{A}$. We have to show that $x \in \overline{A}$.

If $x \in A$, then we are done since $A \subseteq \overline{A}$. Therefore suppose that $x \notin A$.

Let $(x_n)_{n=1}^{\infty}$ be a sequence in $\overline{A} \setminus \{x\}$ that converges to x. For each $j \in \mathbb{N}$, there is $N_j \in \mathbb{N}$ such that

$$|x_n - x| < \frac{1}{2j} \quad \text{for all } n \geq N_j.$$

It follows that we can choose $n_1 \in \mathbb{N}$ such that $|x_{n_1} - x| < \frac{1}{2}$, then choose $n_2 > n_1$ such that $|x_{n_2} - x| < \frac{1}{2\cdot 2}$, then choose $n_3 > n_2$ such that $|x_{n_3} - x| < \frac{1}{2\cdot 3}$, and so on. In this way we obtain a subsequence $(x_{n_j})_{j=1}^{\infty}$, lying in $\overline{A} \setminus \{x\}$, with

$$|x_{n_j} - x| < \frac{1}{2j}, \quad j \in \mathbb{N}.$$

Now we will construct a sequence $(a_j)_{j\in\mathbb{N}}$ in A that converges to x. If $x_{n_j} \in A$, we put $a_j = x_{n_j}$. If $x_{n_j} \notin A$, then $x_{n_j} \in A'$, i.e., x_{n_j} is a limit point of A. There is a sequence in A that converges to x_{n_j}. We choose a_j to be a term of this sequence with

$$|x_{n_j} - a_j| < \frac{1}{2j}.$$

Notice that the above estimate holds also in the case when $a_j = x_{n_j}$.

The constructed sequence $(a_j)_{j=1}^{\infty}$ lies in A and converges to x. Indeed,

$$|a_j - x| \le |a_j - x_{n_j}| + |x_{n_j} - x| < \frac{1}{2j} + \frac{1}{2j} = \frac{1}{j}, \quad j \in \mathbb{N}.$$

Thus, $x \in A'$.

The above shows that any limit point of $\overline{A}$ lies in $A \cup A' = \overline{A}$. Thus, $\overline{A}$ is closed.

B. Let us now prove that $\overline{A}$ is the smallest closed set containing A. Clearly, $A \subseteq \overline{A}$ and $\overline{A}$ is closed.

Let F be a closed set with $A \subseteq F$. We want to show that $\overline{A} = A \cup A' \subseteq F$.

The inclusion $A \subseteq F$ is given, so that we have to show that $A' \subseteq F$. Let $x \in A'$. Then x is a limit point of A, so that there is a sequence $(x_n)_{n=1}^{\infty}$ in A with $x = \lim_{n\to\infty} x_n$. But $A \subseteq F$ which means that $(x_n)_{n=1}^{\infty}$ is a sequence in F. Thus, x is a limit point of F. Since F is closed, it contains all its limit points. Hence, $x \in F$, and we are done.

□

Proposition 3.20 *Let A be a nonempty set in $\mathbb{R}$.*

(1) *If A is bounded above, then* $\sup A \in \overline{A}$.

(2) *If A is bounded below, then* $\inf A \in \overline{A}$.

△

Proof. We prove (1); a proof of (2) is similar.

Let A be a nonempty set in $\mathbb{R}$ that is bounded above. By the Completeness Axiom (Axiom 1.8), A has a least upper bound $\sup A = a$. If $a \in A$, then $a \in \overline{A}$ by the inclusion $A \subseteq \overline{A}$. If $a \notin A$, then for each $n \in \mathbb{N}$ by Proposition 1.11 we can find $a_n \in A$ with $a - \frac{1}{n} < a_n < a$. We obtain a sequence $(a_n)_{n=1}^{\infty}$ in $A \setminus \{a\}$ that converges to a. Thus, $a = \sup A$ is a limit point of A, and therefore $a = \sup A \in \overline{A}$.

□

3.3 Compact sets

Definition 3.21 A set $K \subseteq \mathbb{R}$ is called *compact* if every sequence in K has a subsequence that converges to a point in K.

△

Theorem 3.22 *A set $K \subseteq \mathbb{R}$ is compact if and only if K is closed and bounded.*

△

Proof. A. Suppose that K is compact.

We first show that K is closed. Let x be a limit point of K. Then x is a limit of a sequence $(x_n)_{n=1}^{\infty}$ in K. Since K is compact, $(x_n)_{n=1}^{\infty}$ contains a subsequence that converges to an element of K. But all subsequences of a convergent sequence converge to the same limit (see Proposition 2.38). Thus, $x \in K$. This shows that all limit points of K belong to K, i.e., K is closed.

Now let us prove that K is bounded. Assume, by contradiction, that K is unbounded. Then for each $n \in \mathbb{N}$ we can find a point in K, say x_n, with $|x_n| > n$. Consider the sequence $(x_n)_{n=1}^{\infty}$ in K. Clearly, any subsequence of $(x_n)_{n=1}^{\infty}$ is unbounded and therefore divergent. Thus, $(x_n)_{n=1}^{\infty}$ does not contain a convergent subsequence which contradicts the definition of a compact set. We conclude that K must be bounded.

B. Now assume that K is closed and bounded. We want to show that K is compact.

Let $(x_n)_{n=1}^{\infty}$ be an arbitrary sequence in K. Since K is bounded, $(x_n)_{n=1}^{\infty}$ is bounded. By the Bolzano-Weierstrass Theorem (Theorem 2.40), $(x_n)_{n=1}^{\infty}$ has a convergent subsequence. Since K is closed, its limit belongs to K. Thus, K is compact.

□

Examples of compact sets are intervals $[a, b]$ with $a, b \in \mathbb{R}$, $a \leq b$. Also notice that any finite set in $\mathbb{R}$ is compact.

Chapter 4

Limits of functions and continuity

4.1 Limits of functions

Definition 4.1 Let $A \subseteq \mathbb{R}$ be a nonempty set and $f : A \to \mathbb{R}$. Let a be a limit point of A. We say that f converges to a *limit* $L \in \mathbb{R}$ as $x \to a$ if

$$\forall\, \varepsilon > 0\, \exists \delta > 0\, \forall\, x \in A : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon,$$

i.e. for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all $x \in A$ with $0 < |x - a| < \delta$. We write $\lim_{x \to a} f(x) = L$.

△

For an illustration for this definition, please see Figure 4.1.

It is not difficult to prove that if a function f has a limit at a point a, then the limit is unique (compare with Proposition 2.4).

Notice that we do not require that $a \in A$. In particular, $f(a)$ need not be defined. However, f must be defined at points in A that are arbitrary close to a. This is ensured by assuming that a is a limit point of A.

The condition $0 < |x - a|$ in the definition of the limit means that we do not consider the value of the function f at the point a, even if it exists. In particular, it can happen that $f(a) \neq L$.

Typically, the set A in the definition is the domain of the function f. However, in some situations it can be useful to restrict A. For example, we can ask what happens if we approach the point a from one side, say, from the left. In this case we take A to be the set of points x from the domain of f with $x < a$. If we approach a from the right, we take A to be the set of points x from the domain of f with $x > a$. In this case we talk about *one-sided limits*.

For example, consider the function $f(x) = \frac{x}{|x|}$. This function is not defined at

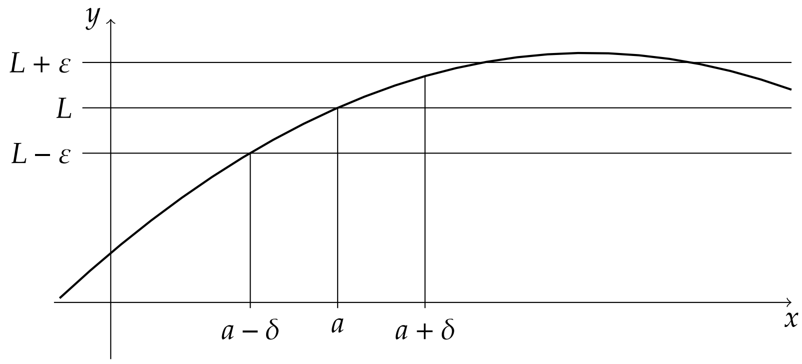

Figure 4.1: A limit of a function

$x = 0$. The right-sided limit of f at $x = 0$ is

$$\lim_{\substack{x\to 0\\ x>0}} f(x) = \lim_{\substack{x\to 0\\ x>0}} \frac{x}{x} = 1.$$

We also write $\lim_{x\to 0+} f(x) = 1$. The left-sided limit at $x = 0$ is

$$\lim_{\substack{x\to 0\\ x<0}} f(x) = \lim_{\substack{x\to 0\\ x<0}} \frac{x}{-x} = -1.$$

We also write $\lim_{x\to 0-} f(x) = -1$.

Now let us consider some examples how to use Definition 4.1.

Example 4.2 We have $\lim_{x\to 2}(x^2+3) = 7$.

Here we consider the function $f(x) = x^2 + 3$, its domain $A = \mathbb{R}$ and $a = 2$ which is a limit point of A. We have

$$|f(x) - 7| = |x^2 + 3 - 7| = |x^2 - 4| = |x-2| \cdot |x+2|.$$

Since we are interested in the behavior of f for x close to 2, let us assume that $|x-2| < 1$, i.e. $1 < x < 3$. For such x we have $3 < x+2 < 5$, so that $3 < |x+2| < 5$ and

$$|f(x) - 7| < 5|x-2|.$$

Given $\varepsilon > 0$, chose $\delta = \min\left\{1, \frac{\varepsilon}{5}\right\}$. Then $0 < |x-2| < \delta$ implies that

$$|f(x) - 7| < 5|x-2| < 5 \cdot \frac{\varepsilon}{5} = \varepsilon.$$

Thus, by definition, $\lim_{x\to 2}(x^2+3) = 7$.

△

Example 4.3 Let $g : \mathbb{R} \to \mathbb{R}$, and suppose that $\lim_{x\to 6} g(x) = \sqrt{2}$. We will prove that

$$\lim_{x\to 6} \frac{1}{1+g(x)^2} = \frac{1}{3}.$$

By the definition of the limit, we have to show the following:

$$\forall\, \varepsilon > 0\ \exists \delta > 0\ \forall\, x \in \mathbb{R} : 0 < |x-6| < \delta \implies \left|\frac{1}{1+g(x)^2} - \frac{1}{3}\right| < \varepsilon.$$

We have

$$\left|\frac{1}{1+g(x)^2} - \frac{1}{3}\right| = \left|\frac{3-1-g(x)^2}{3(1+g(x)^2)}\right| = \frac{|2-g(x)^2|}{3(1+g(x)^2)} = \frac{\left|\sqrt{2}-g(x)\right| \cdot \left|\sqrt{2}+g(x)\right|}{3(1+g(x)^2)}.$$

Since $g(x)^2 \geq 0$, we have $1+g(x)^2 \geq 1$, and $\frac{1}{1+g(x)^2} \leq 1$. It follows that

$$\left|\frac{1}{1+g(x)^2} - \frac{1}{3}\right| \leq \frac{1}{3} \cdot \left|\sqrt{2}-g(x)\right| \cdot \left|\sqrt{2}+g(x)\right|.$$

We know that $\lim_{x\to 6} g(x) = \sqrt{2}$. Let us estimate $\left|\sqrt{2}+g(x)\right|$. We take in the definition of the limit, say, $\varepsilon = 1$. There exists $\delta_1 > 0$ such that

$$0 < |x-6| < \delta_1 \implies \left|g(x) - \sqrt{2}\right| < 1.$$

For such x we have

$$\sqrt{2} - 1 < g(x) < \sqrt{2} + 1,$$

and therefore

$$0 < 2\sqrt{2} - 1 < g(x) + \sqrt{2} < 2\sqrt{2} + 1 < 5.$$

We conclude that

$$0 < |x-6| < \delta_1 \implies \left|g(x) + \sqrt{2}\right| < 5.$$

Thus, for $x \in \mathbb{R}$ with $0 < |x-6| < \delta_1$ we have

$$\left|\frac{1}{1+g(x)^2} - \frac{1}{3}\right| < \frac{5}{3} \cdot \left|\sqrt{2}-g(x)\right|.$$

Given $\varepsilon > 0$, there exists $\delta_2 > 0$ such that

$$0 < |x-6| < \delta_2 \implies \left|g(x) - \sqrt{2}\right| < \frac{3}{5}\varepsilon.$$

Now put $\delta = \min\{\delta_1, \delta_2\} > 0$. We have

$$0 < |x-6| < \delta \implies \left|\frac{1}{1+g(x)^2} - \frac{1}{3}\right| < \frac{5}{3} \cdot \left|\sqrt{2} - g(x)\right| < \frac{5}{3} \cdot \frac{3}{5}\varepsilon = \varepsilon$$

as required. Thus, by definition $\lim_{x\to 6} \frac{1}{1+g(x)^2} = \frac{1}{3}$.

△

It is easy to prove following statements immediately from the definition.

Proposition 4.4 *If $f(x) = c$ with some $c \in \mathbb{R}$ for all $x \in A$, and if a is a limit point of A, then $\lim_{x\to a} f(x) = c$.*

△

Proposition 4.5 *If $f(x) = x$ for all $x \in A$, and if a is a limit point of A, then $\lim_{x\to a} f(x) = a$.*

△

The following result says that if a function f has a positive (negative) limit at a point a, then there is an open neighborhood around a such that $f(x)$ is positive (negative) for all x in this neighborhood, $x \neq a$.

Lemma 4.6 *Suppose that $\lim_{x\to a} f(x) = L$.*

(1) *If $L > 0$, then $\exists \delta > 0\ \forall x \in A : 0 < |x-a| < \delta \implies f(x) > 0$.*

(2) *If $L < 0$, then $\exists \delta > 0\ \forall x \in A : 0 < |x-a| < \delta \implies f(x) < 0$.*

△

Proof. We will prove (1); the proof of (2) is similar.

Let $L > 0$. In the definition of the limit, take $\varepsilon = \frac{L}{2} > 0$. There exists $\delta > 0$ such that $0 < |x-a| < \delta \implies |f(x) - L| < \frac{L}{2}$. Then, in particular, for such x

$$f(x) > L - \frac{L}{2} = \frac{L}{2} > 0.$$

□

We have seen that limits of sequences preserve the arithmetic operations (Theorem 2.14). The same is true for the functional limits. We omit the proof as it is similar to the proof of Theorem 2.14. Please prove the statements as an exercise.

Theorem 4.7 (*The Algebraic Limit Theorem for Functional Limits*)
If $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$, *then*

(1) $\lim_{x\to a} cf(x) = cL$ *for any* $c \in \mathbb{R}$,

(2) $\lim_{x\to a}(f(x) + g(x)) = L + M$,

(3) $\lim_{x\to a}(f(x)g(x)) = LM$,

(4) $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$, *provided* $M \neq 0$.

△

Notice that in (4) the assumption $M \neq 0$ guarantees by Lemma 4.6 that $g(x) \neq 0$ in a certain neighborhood of a except, perhaps, at $x = a$. Therefore, the quotient $\frac{f}{g}$ is defined in this neighborhood for $x \neq a$.

Also the Order Limit Theorem for sequences (Theorem 2.15) has a analogue for the functional limits, with a similar proof.

Theorem 4.8 (*The Order Limit Theorem for Functional Limits*) *Let* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$. *If* $f(x) \leq g(x)$ *for all* $x \in A$, $x \neq a$, *then* $L \leq M$.

△

In the next theorem we will give an equivalent definition of a functional limit in terms of sequences. It can be used to carry over properties of limits of sequences to limits of functions, such as Theorem 4.7 and Theorem 4.8, thus given a second proof of these statements.

Theorem 4.9 *Let* $f : A \to \mathbb{R}$, *and let* a *be a limit point of* A. *The following statements are equivalent:*

(1) $\lim_{x\to a} f(x) = L$,

(2) *for any sequence* $(x_n)_{n=1}^{\infty}$ *in* $A \setminus \{a\}$ *with the property* $\lim_{n\to\infty} x_n = a$ *we have* $\lim_{n\to\infty} f(x_n) = L$.

△

Proof. A. Suppose that $\lim_{x\to a} f(x) = L$. Let $\varepsilon > 0$ be given. By the definition of the limit, there exists $\delta > 0$ such that $0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$.

Now let $(x_n)_{n=1}^{\infty}$ be a sequence in $A \setminus \{a\}$ such that $\lim_{n\to\infty} x_n = a$. By the definition of the limit of a sequence, there exists $N \in \mathbb{N}$ such that $|x_n - a| < \delta$ for all $n \geq N$. But then by above $|f(x_n) - L| < \varepsilon$. This implies that $\lim_{n\to\infty} f(x_n) = L$.

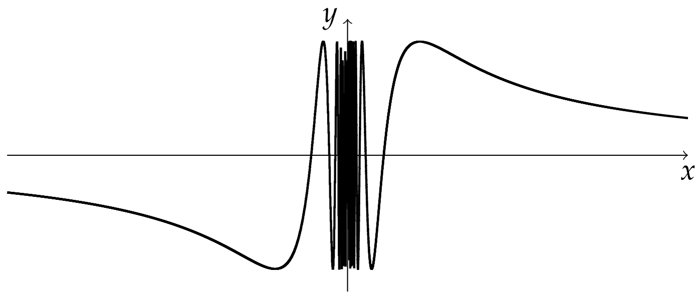

Figure 4.2: A graph of the function $f(x) = \sin\left(\frac{1}{x}\right)$

B. Now assume that $\lim_{n\to\infty} f(x_n) = L$ for any sequence $(x_n)_{n=1}^{\infty}$ in $A \setminus \{a\}$ with $\lim_{n\to\infty} x_n = a$.

Suppose, by contradiction, that $\lim_{x\to a} f(x) \neq L$. Then there is a number $\varepsilon > 0$ such that for each $n \in \mathbb{N}$ there is a point $x_n \in A \setminus \{a\}$ with

$$0 < |x_n - a| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - L| \geq \varepsilon.$$

Consider the sequence $(x_n)_{n=1}^{\infty}$. We have that $\lim_{n\to\infty} x_n = a$ but $\lim_{n\to\infty} f(x_n) \neq L$, a contradiction. The contradiction shows that $\lim_{x\to a} f(x) = L$.

□

Theorem 4.9 gives us useful criteria for divergence.

Corollary 4.10 *(The Sequential Test for Divergence)*

(1) *If there exists a sequence $(x_n)_{n=1}^{\infty}$ in $A \setminus \{a\}$ such that $(x_n)_{n=1}^{\infty}$ converges to a, but $(f(x_n))_{n=1}^{\infty}$ diverges, then $\lim_{x\to a} f(x)$ does not exist.*

(2) *If there exist two sequences $(x_n)_{n=1}^{\infty}$ and $(y_n)_{n=1}^{\infty}$ in $A \setminus \{a\}$ that both converge to a, such that $\lim_{n\to\infty} f(x_n) \neq \lim_{n\to\infty} f(y_n)$, then $\lim_{x\to a} f(x)$ does not exist.*

△

Example 4.11 Consider the function $f(x) = \sin\left(\frac{1}{x}\right)$, $x \neq 0$. Its graph is shown in Figure 4.2.

The limit $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist. Indeed, take $x_n = \frac{1}{2\pi n}$ and $y_n = \frac{1}{2\pi n + \frac{\pi}{2}}$, $n \in \mathbb{N}$. Clearly, $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n = 0$. We have $f(x_n) = \sin(2\pi n) = 0$, $n \in \mathbb{N}$, so that $\lim_{n\to\infty} f(x_n) = 0$. On the other hand, $f(y_n) = \sin\left(2\pi n + \frac{\pi}{2}\right) = 1$ for all $n \in \mathbb{N}$, and $\lim_{n\to\infty} f(y_n) = 1$. The statement follows by Corollary 4.10.

△

Theorem 4.12 (*The Squeeze Theorem for Functional Limits*) *Let $f, g, h : A \to \mathbb{R}$, and let a be a limit point of A. Suppose that $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$ and*

$$g(x) \le f(x) \le h(x) \quad \text{for all } x \in A.$$

Then $\lim_{x\to a} f(x) = L$.

△

This statement can be proven either directly, or using Theorem 4.9 and the Squeeze Theorem for sequences (Theorem 2.10).

We finish this section by a brief discussion of infinite limits. First we define infinite limits at a finite point.

Definition 4.13 Let $f : A \to \mathbb{R}$, and let a be a limit point of A.

(1) We say that $\lim_{x\to a} f(x) = \infty$, if

$$\forall M > 0\ \exists \delta > 0\ \forall x \in A : 0 < |x - a| < \delta \implies f(x) > M.$$

(2) Similarly, we say that $\lim_{x\to a} f(x) = -\infty$, if

$$\forall M > 0\ \exists \delta > 0\ \forall x \in A : 0 < |x - a| < \delta \implies f(x) < -M.$$

△

Notice that in both cases the limit $\lim_{x\to a} f(x)$ does not exist; the symbols ∞ and $-\infty$ are not real numbers.

Finally, we define limits of a function at ∞ and $-\infty$.

Definition 4.14 Let $f : A \to \mathbb{R}$ and $L \in \mathbb{R}$.

(1) Assume that A is not bounded above. We say that $\lim_{x\to\infty} f(x) = L$, if

$$\forall \varepsilon > 0\ \exists M > 0\ \forall x \in A : x > M \implies |f(x) - L| < \varepsilon.$$

(2) Assume that A is not bounded below. We say that $\lim_{x\to-\infty} f(x) = L$, if

$$\forall \varepsilon > 0\ \exists M > 0\ \forall x \in A : x < -M \implies |f(x) - L| < \varepsilon.$$

△

Combining ideas from Definition 4.13 and Definition 4.14, one can give definitions for infinite limits at ∞ and $-\infty$, for example, for $\lim_{x\to\infty} f(x) = \infty$. We let the details to the reader.

4.2 Continuity

Definition 4.15 Let $f : A \to \mathbb{R}$, and let $a \in A$. We say that f is *continuous at a*, if

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A : |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

If f is continuous at each point $a \in A$, we say that f is *continuous on A*.

△

The definition above looks very similar to the definition of a limit. One difference is that now we consider $a \in A$. If $a \in A$ is a limit point of A, the definition above means exactly that

$$\lim_{x \to a} f(x) = f(a).$$

However, if a is an isolated point of A, then $\lim_{x \to a} f(x)$ is not defined. However, it follows from the definition that if a is an isolated point of A, then f is always continuous at a. This is because we can take $\delta > 0$ so small that the only point $x \in A$ with $|x - a| < \delta$ is the point $x = a$.

The Algebraic Limit Theorem (Theorem 4.7) yields that sums, products and quotients (under sensible assumptions) of continuous functions are continuous.

It follows that any polynomial function is continuous on $\mathbb{R}$, and any rational function is continuous on its domain.

Theorem 4.16 (*Composition of Continuous Functions*) *Let $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$. Assume that $f(A) = \{f(x) : x \in A\} \subseteq B$. If f is continuous at $a \in A$ and g is continuous at $f(a) \in B$, then the composition $g \circ f : A \to \mathbb{R}$ is continuous at a.*

△

Notice that the assumption $f(A) = \{f(x) : x \in A\} \subseteq B$ guarantees that the composition $g \circ f(x) = g(f(x))$ is defined on A.

Proof. Let $\varepsilon > 0$. By the continuity of g at the point $b = f(a)$, there exists $\eta > 0$ such that

$$\forall y \in B : |y - f(a)| < \eta \implies |g(y) - g(f(a))| < \varepsilon.$$

On the other hand, f is continuous at a, and therefore there exists $\delta > 0$ such that

$$\forall x \in A : |x - a| < \delta \implies |f(x) - f(a)| < \eta.$$

Combining the two formulas with $y = f(x)$, we obtain

$$\forall x \in A : |x - a| < \delta \implies |g(f(x)) - g(f(a))| < \varepsilon$$

as desired.

□

Theorem 4.9 gives the following criterion for continuity.

Theorem 4.17 *(The Sequential Criterion for Continuity)* *Let $f : A \to \mathbb{R}$ and $a \in A$. The following statements are equivalent:*

(1) *f is continuous at a,*

(2) *for any sequence $(x_n)_{n=1}^{\infty}$ in A with $\lim_{n\to\infty} x_n = a$ we have $\lim_{n\to\infty} f(x_n) = f(a)$.*

△

Corollary 4.18 *If there is a sequence $(x_n)_{n=1}^{\infty}$ in A such that $\lim_{n\to\infty} x_n = a$, but the sequence $(f(x_n))_{n=1}^{\infty}$ does not converge to $f(a)$, then f is not continuous at a.*

△

In the remaining part of this section we will study images and preimages of sets under continuous functions. First we recall the definitions.

Definition 4.19 Let $f : X \to Y$.

(1) The *image* of a set $A \subseteq X$ is the set

$$f(A) = \{f(x) : x \in A\} \subseteq Y.$$

(2) The *preimage* of a set $B \subseteq Y$ is the set

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \subseteq X.$$

△

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function on $\mathbb{R}$. It is possible to show that the preimage of any open set is open, and the preimage of any closed set is closed. We will not go into the details. Notice, however, that images are not well-behaved.

Example 4.20 Consider the function $f(x) = \frac{1}{1+x^2}$ that is continuous on $\mathbb{R}$. Observe that $f(\mathbb{R}) = (0,1]$, so that $\mathbb{R}$ is closed but its image $f(\mathbb{R}) = (0,1]$ is not closed. We also see that $f((-1,1)) = \left(\frac{1}{2},1\right]$, where $(-1,1)$ is open, but its image $\left(\frac{1}{2},1\right]$ is not open.

△

Example 4.21 Consider the function $f(x) = \frac{x}{1-x^2}$ that is continuous on $(-1,1)$. Observe that $f((-1,1)) = \mathbb{R}$, so that the set $(-1,1)$ is bounded but its image $f((-1,1)) = \mathbb{R}$ is unbounded.

△

In light of these examples it is perhaps surprising that images of compact, i.e. closed and bounded, sets under continuous functions are compact.

Theorem 4.22 *If f is continuous on a compact set K, then $f(K)$ is compact.*

△

Proof. Let $(y_n)_{n=1}^{\infty}$ be an arbitrary sequence in $f(K)$. To prove that $f(K)$ is compact, we have to show that $(y_n)_{n=1}^{\infty}$ has a subsequence $(y_{n_k})_{k=1}^{\infty}$ that converges to some $y \in f(K)$.

For each $n \in \mathbb{N}$, let $x_n \in K$ be such that $y_n = f(x_n)$. Consider the sequence $(x_n)_{n=1}^{\infty}$ in K. Since K is compact, it has a subsequence $(x_{n_k})_{k=1}^{\infty}$ that converges to some $x \in K$. Since f is continuous on K and, in particular, at x, we can apply Theorem 4.17. We conclude that $(f(x_{n_k}))_{k=1}^{\infty}$ converges to $y = f(x) \in f(K)$.

□

One consequence of this fact is the following well-known statement.

Theorem 4.23 (*The Extreme Value Theorem*) *If f is continuous on a compact set K, then f attains its maximum value and its minimum value on K. In other words, there exist $u, v \in K$ such that $f(u) = \min_{x \in K} f(x)$ and $f(v) = \max_{x \in K} f(x)$.*

△

Proof. By Theorem 4.22 $f(K)$ is compact, i.e. closed and bounded. Let $m = \inf f(K)$ and $M = \sup f(K)$. By Proposition 3.20 we have that $m, M \in \overline{f(K)} = f(K)$. Thus, there are $u, v \in K$ such that $m = f(u)$, $M = f(v)$.

□

Another well-known result is the following statement.

Theorem 4.24 (*The Intermediate Value Theorem*) *Let f be continuous on a closed bounded interval $[a,b]$. Let $y \in \mathbb{R}$ be a value between $f(a)$ and $f(b)$. Then there exists $c \in [a,b]$ such that $y = f(c)$.*

△

Proof. If $f(a) = f(b)$ then there is nothing to prove. So suppose that $f(a) \neq f(b)$. Without loss if generality, we may assume that $f(a) < f(b)$ (if this is not the case, consider $-f$ instead of f).

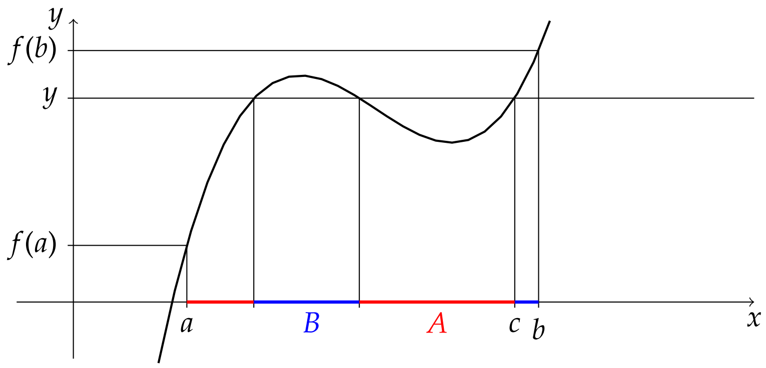

Figure 4.3: The proof of the Intermediate Value Theorem

Take $y \in \mathbb{R}$ with $f(a) < y < f(b)$. Put

$$A = \{x \in [a, b] : f(x) \le y\},$$
$$B = \{x \in [a, b] : f(x) > y\}.$$

We have $a \in A$, $b \in B$. See Figure 4.3 for an illustration.

By the Completeness Axiom (Axiom 1.8), the set A has a least upper bound c. Since $[a, b]$ is closed, $c \in [a, b]$. We will show that $f(c) = y$. Take a sequence $(x_n)_{n=1}^{\infty}$ in A with $\lim_{n\to\infty} x_n = c$. Notice that $f(x_n) \le y$ for all $n \in \mathbb{N}$. Since f is continuous, $f(c) = \lim_{n\to\infty} f(x_n) \le y$.

On the other hand, $c < b$, because $y < f(b)$. Therefore, for n large enough we have $c + \frac{1}{n} \in [a, b]$. Since c is the least upper bound of A, we obtain that $c + \frac{1}{n} \in B$, so that $f\left(c + \frac{1}{n}\right) > y$. But $\lim_{n\to\infty} \left(c + \frac{1}{n}\right) = c$, and the continuity of f yields $f(c) = \lim_{n\to\infty} f\left(c + \frac{1}{n}\right) \ge y$. The two inequalities $f(c) \le y$ and $f(c) \ge y$ together give $f(c) = y$ as desired.

$\square$

Essentially, the Intermediate Value Theorem tell us that the image of an interval under a continuous function is an interval.

4.3 Uniform Continuity

Recall that a function f is continuous on a set $A \subseteq \mathbb{R}$ if

$$\forall\, \varepsilon > 0\ \forall\, a \in A\ \exists \delta > 0\ \forall\, x \in A : |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

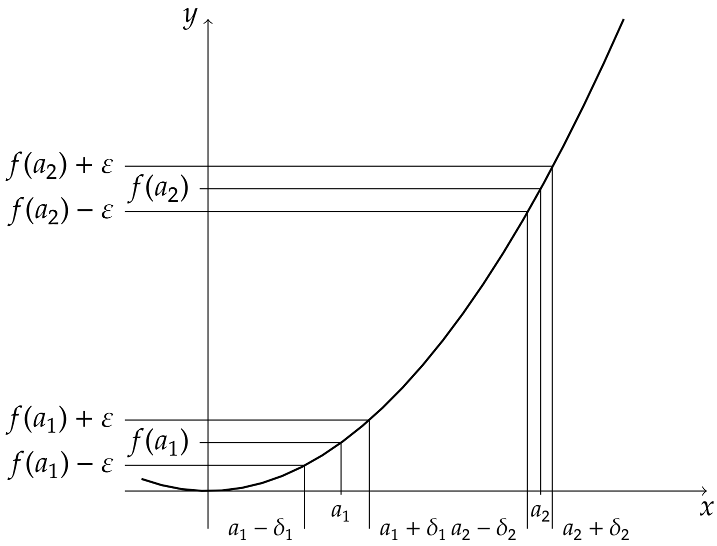

Figure 4.4: The function $f(x) = x^2$

In this definition, the number $\delta > 0$ depends on both $\varepsilon > 0$ and the point $a \in A$. In general, a choice of $\delta > 0$ for one point $a \in A$ might not work for other points. If it happens, however, that, given $\varepsilon > 0$, we can take $\delta > 0$ that works for all $a \in A$, we say that f is uniformly continuous on A. We will give a formal definition after considering examples.

Example 4.25 Consider the function $f(x) = 2x + 3$ on $\mathbb{R}$. Let $a \in \mathbb{R}$. We have

$$|f(x) - f(a)| = |(2x + 3) - (2a + 3)| = 2 \cdot |x - a|.$$

Given $\varepsilon > 0$, take $\delta = \frac{\varepsilon}{2}$. Then for any $a \in \mathbb{R}$ we have $|x-a| < \delta \implies |f(x)-f(a)| < \varepsilon$. The function $f(x) = 2x + 3$ is absolutely continuous on $\mathbb{R}$.

△

Example 4.26 Consider the function $f(x) = x^2$ on $\mathbb{R}$. The graph of this function is shown in Figure 4.4. We see that the graph of this function gets steeper for larger x, and the value of $\delta > 0$ depends on a. The function $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$. We will give a formal proof of this fact below.

△

Definition 4.27 Let $f : A \to \mathbb{R}$. We say that f is *uniformly continuous* on A, if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A : |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

△

Of course, if f is uniformly continuous on A, then it is also continuous on A. The additional property given by the uniform continuity that now $\delta > 0$ in the definition of continuity does not depend on a point in A.

Let us build a negation of the formula in the definition of uniform continuity. A function $f : A \to \mathbb{R}$ is *not* uniformly continuous on A, if

$$\exists \varepsilon > 0 \, \forall \delta > 0 \, \exists x, y \in A : |x - y| < \delta \text{ and } |f(x) - f(y)| \geq \varepsilon.$$

Taking $\delta_n = \frac{1}{n}$, $n \in \mathbb{N}$ in the above formula, we obtain two sequences of points $(x_n)_{n=1}^{\infty}$ and $(y_n)_{n=1}^{\infty}$ in A such that $|x_n - y_n| < \frac{1}{n}$ and $|f(x_n) - f(y_n)| \geq \varepsilon$. We arrive at the following statement.

Proposition 4.28 *A function $f : A \to \mathbb{R}$ is not uniformly continuous on A if and only if there exists a number $\varepsilon > 0$ and a pair of sequences $(x_n)_{n=1}^{\infty}$, $(y_n)_{n=1}^{\infty}$ in A such that $\lim_{n\to\infty}(x_n - y_n) = 0$ but $|f(x_n) - f(y_n)| > \varepsilon$ for all $n \in \mathbb{N}$.*

△

Example 4.29 This is a continuation of Example 4.26. We will now use Proposition 4.28 to show that the function $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Consider the sequences $x_n = \sqrt{n+1}$, $y_n = \sqrt{n}$, $n \in \mathbb{N}$. We have

$$\lim_{n\to\infty}(x_n - y_n) = \lim_{n\to\infty}\left(\sqrt{n+1} - \sqrt{n}\right) = \lim_{n\to\infty}\frac{1}{\sqrt{n+1} + \sqrt{n}} = 0,$$

but

$$f(x_n) - f(y_n) = \left(\sqrt{n+1}\right)^2 - \left(\sqrt{n}\right)^2 = (n+1) - n = 1.$$

This shows that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

△

The sum of two uniformly continuous functions is uniformly continuous (Exercise). Interestingly, the product of two uniformly continuous functions need not be uniformly continuous. For example, it is easy to see that the function $g(x) = x$ is uniformly continuous on $\mathbb{R}$. However, as we have seen above, the function $f(x) = x^2 = g(x) \cdot g(x)$ is not uniformly continuous on $\mathbb{R}$.

Again, compact sets play a special role.

Theorem 4.30 *If f is continuous on a compact set K, then f is uniformly continuous on K.*

△

Proof. We will prove this statement by contradiction. Suppose that f is continuous, but not uniformly continuous on K. By Proposition 4.28 there exists $\varepsilon > 0$ and a pair of sequences $(x_n)_{n=1}^{\infty}$, $(y_n)_{n=1}^{\infty}$ in K such that $\lim_{n\to\infty}(x_n - y_n) = 0$ but $|f(x_n) - f(y_n)| > \varepsilon$ for all $n \in \mathbb{N}$.

Since K is compact, the sequence $(x_n)_{n=1}^{\infty}$ has a subsequence $(x_{n_k})_{k=1}^{\infty}$ that converges to some $x \in K$. We have

$$\lim_{k\to\infty} y_{n_k} = \lim_{k\to\infty}(y_{n_k} - x_{n_k} + x_{n_k}) = \lim_{k\to\infty}(y_{n_k} - x_{n_k}) + \lim_{k\to\infty} x_{n_k} = 0 + x = x.$$

Thus, the sequence $(y_{n_k})_{k=1}^{\infty}$ also converges to x.

Since f is continuous at x, we have by Theorem 4.17

$$\lim_{k\to\infty}(f(x_{n_k}) - f(y_{n_k})) = \lim_{k\to\infty} f(x_{n_k}) - \lim_{k\to\infty} f(y_{n_k}) = f(x) - f(x) = 0,$$

but this is impossible since by assumption $|f(x_{n_k}) - f(y_{n_k})| > \varepsilon$ for all $k \in \mathbb{N}$. This contradiction shows that f is uniformly continuous on K.

□

Chapter 5

Derivatives

You are familiar with derivatives from your Calculus course. In this short chapter we will give a definition of the derivative and present several important theorems that involve derivatives.

In this chapter we will mostly consider functions on an open interval (a, b), where $a, b \in \mathbb{R}$ and $a < b$.

Definition 5.1 Let $f : (a, b) \to \mathbb{R}$. We will say that f is *differentiable at $x_0 \in (a, b)$* if the limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. If this is the case, we call the value

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

the *derivative* of the function f at the point x_0. If f is differentiable at every point $x_0 \in (a, b)$, we say that f is *differentiable on (a, b)*.

△

Sums, products, quotients (under sensible conditions) and compositions of differentiable functions are differentiable. For the derivatives we have rules such as the sum rule, the product rule, the quotient rule, the chain rule, etc. We will not repeat them here.

Also the following results must be known to you.

Theorem 5.2 *Let $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$. If f is differentiable at x_0, then f is continuous at x_0.*

△

Proof. Exercise.

□

The converse of this statement is not true. For example, the function $f(x) = |x|$ is continuous but not differentiable at $x = 0$.

Definition 5.3 Let $f : (a, b) \to \mathbb{R}$.

(1) We say that f has a *local maximum* at a point $c \in (a, b)$ if

$$\exists \delta > 0\ \forall x \in (a, b) : |x - c| < \delta \implies f(x) \leq f(c).$$

(2) Similarly, we say that f has a *local minimum* at a point $c \in (a, b)$ if

$$\exists \delta > 0\ \forall x \in (a, b) : |x - c| < \delta \implies f(x) \geq f(c).$$

(3) We say that f has a *local extremum* at a point $c \in (a, b)$ if f has a local maximum or a local minimum at c.

△

Theorem 5.4 (*Fermat's Theorem*) *Let $f : (a, b) \to \mathbb{R}$ be differentiable on a open interval (a, b). If f has a local extremum at a point $c \in (a, b)$, then $f'(c) = 0$.*

△

Proof. Suppose that $f'(c) < 0$. We will show that f cannot have a local extremum at c.

By Lemma 4.6, $f'(c) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c} < 0$ implies that there exists $\delta > 0$ such that

$$\frac{f(x) - f(c)}{x - c} < 0$$

for all $x \in (a, b)$ with $0 < |x - c| < \delta$. If $c < x < c + \delta$, then $x - c > 0$, and thus $f(x) - f(c) < 0$, i.e. $f(x) < f(c)$. This means that c cannot be a local minimum. On the other hand, if $c - \delta < x < c$, then $x - c < 0$, and thus $f(x) - f(c) > 0$, i.e. $f(x) > f(c)$. Thus, c cannot be a local maximum either.

The case when $f'(c) > 0$ can be considered similarly.

□

Notice that the converse of Fermat's Theorem is not true. For example, for the function $f(x) = x^3$ we have $f'(0) = 0$, but $x = 0$ is neither a local maximum nor a local minimum of f.

Theorem 5.5 (*Rolle's Theorem*) *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.*

△

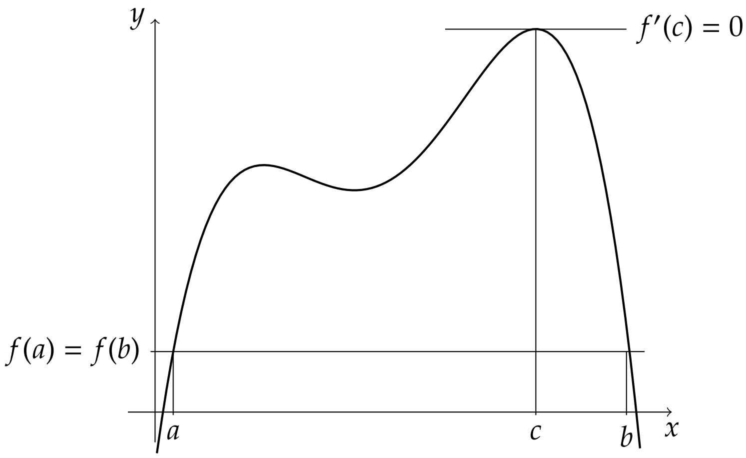

Figure 5.1: Rolle's Theorem

For an illustration, see Figure 5.1.

Proof. If f is constant on $[a, b]$, that $f'(x) = 0$ for all $x \in (a, b)$, and we are done.

Therefore assume that f is not constant. By the Extreme Value Theorem (Theorem 4.23), f attains its minimum and its maximum on $[a, b]$. Notice that $\max_{x\in[a,b]} f(x) > \min_{x\in[a,b]} f(x)$ as f is not constant. This means that $f(a) = f(b)$ is different from at least one of the values $\max_{x\in[a,b]} f(x)$, $\min_{x\in[a,b]} f(x)$. We conclude that f has at least one local extremum $c \in (a, b)$. By Fermat's Theorem (Theorem 5.4), $f'(c) = 0$ as desired.

□

Theorem 5.6 *(The Mean Value Theorem)* *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). There exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

△

Geometrically this theorem says that there is a point $c \in (a, b)$ such that the tangent line to the graph of f at the point $(c, f(c))$ is parallel to the secant line through the points $(a, f(a))$ and $(b, f(b))$. For an illustration, see Figure 5.2.

Proof. The equation of the secant line through the points $(a, f(a))$ and $(b, f(b))$ is

$$y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

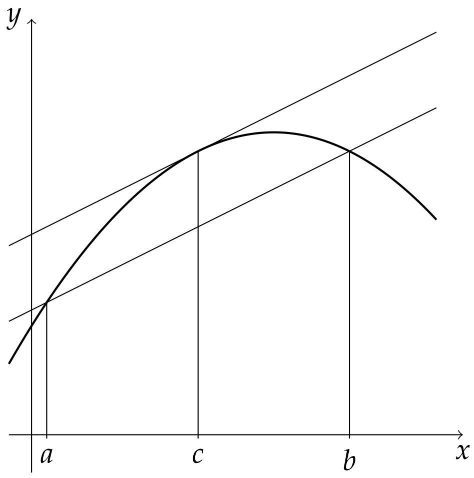

Figure 5.2: Mean Value Theorem

Consider the function

$$F(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(x - a)\right).$$

The function F is continuous on $[a, b]$ and differentiable on (a, b) with

$$F'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Moreover, $F(a) = F(b) = 0$. By Rolle's Theorem (Theorem 5.5) there exists a point $c \in (a, b)$ with

$$F'(c) = f'(c) - \frac{f(b) - f(a)}{b - a} = 0.$$

We obtain that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

as desired.

□

An important application of the Mean Value Theorem is the statement that if $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$. This result is frequently used when integrating (remember the integration constant) or when solving differential equations. We will not go into detail here.

Instead, we establish a more general version of the Mean Value Theorem. Assume that we have two functions f and g that are continuous on $[a,b]$ and differentiable on (a,b). Applying the Mean Value Theorem to each function separately, we see that there exist two points $c,d \in (a,b)$ with

$$f'(c) = \frac{f(b)-f(a)}{b-a}, \quad g'(d) = \frac{g(b)-g(a)}{b-a}.$$

If we further assume that $f(b) \neq f(a)$, we obtain that

$$\frac{g'(d)}{f'(c)} = \frac{g(b)-g(a)}{f(b)-f(a)}.$$

The statement we are going to prove says that we can find a single point $c \in (a,b)$ such that

$$\frac{g'(c)}{f'(c)} = \frac{g(b)-g(a)}{f(b)-f(a)}.$$

This statement has also a geometric interpretation. Consider in the plane the parametric curve with the equation $x = f(t)$, $y = g(t)$, $t \in [a,b]$. The slope of the secant line joining the points $A(f(a),g(a))$ and $B(f(b),g(b))$ is $\frac{g(b)-g(a)}{f(b)-f(a)}$. The theorem says that there is a point $C(f(c),g(c))$, $c \in (a,b)$, such that the tangent line to the curve at the point C is parallel to the secant line through A and B. For an illustration, see Figure 5.3.

Theorem 5.7 (*The Generalized Mean Value Theorem*) *Let $f,g : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). There exists $c \in (a,b)$ such that*

$$(f(b)-f(a))g'(c) = f'(c)(g(b)-g(a)).$$

△

Proof. Consider the function

$$G(x) = (f(b)-f(a))g(x) - (g(b)-g(a))f(x).$$

The function G is continuous on $[a,b]$ and differentiable on (a,b) with

$$G'(x) = (f(b)-f(a))g'(x) - (g(b)-g(a))f'(x).$$

Moreover, $G(a) = G(b) = f(b)g(a) - f(a)g(b)$. By Rolle's Theorem (Theorem 5.5) there exists a point $c \in (a,b)$ with

$$G'(c) = (f(b)-f(a))g'(c) - (g(b)-g(a))f'(c) = 0,$$

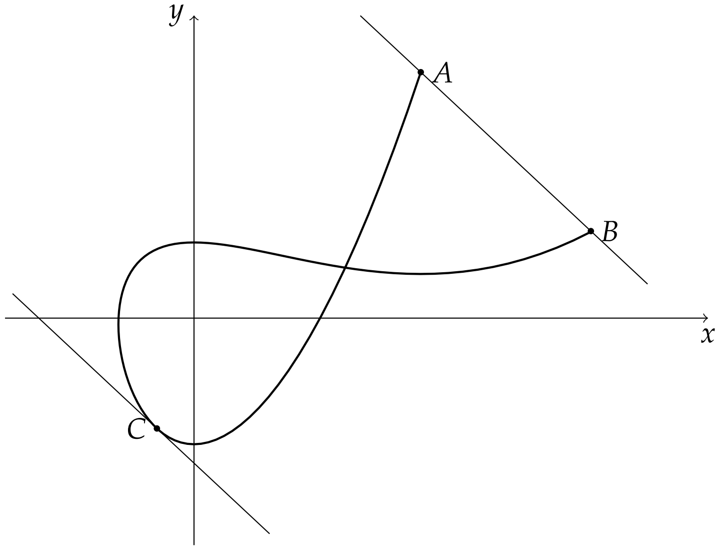

Figure 5.3: Generalized Mean Value Theorem

i.e.,

$$(f(b) - f(a))g'(c) = f'(c)(g(b) - g(a)).$$

□

As an application of Theorem 5.7 we will prove the well-known l'Hospital's Rule. We will consider the $\frac{0}{0}$ case; other variants such as the $\frac{\infty}{\infty}$ case and the limits at ∞ and $-\infty$ can be deduced from it.

Theorem 5.8 *(l'Hospital's Rule, $\frac{0}{0}$ case)* *Let $f, g : (a,b) \to \mathbb{R}$ be differentiable on (a,b). Assume that for $c \in (a,b)$ we have $f(c) = g(c) = 0$, and that $f'(x) \neq 0$ for all $x \in (a,b) \setminus \{c\}$. If $\lim_{x\to c} \frac{g'(x)}{f'(x)}$ exists, then*

$$\lim_{x\to c} \frac{g(x)}{f(x)} = \lim_{x\to c} \frac{g'(x)}{f'(x)}.$$

△

Proof. Let $x \in (a,b) \setminus \{c\}$. By applying the Generalized Mean Value Theorem (Theorem 5.7) to the functions f and g at the points c and x, we conclude that there is a point c_x that lies strictly between x and c such that

$$\frac{g'(c_x)}{f'(c_x)} = \frac{g(x) - g(c)}{f(x) - f(c)} = \frac{g(x)}{f(x)}.$$

The point c_x depends on x. Since it lies between c and x, we have $\lim_{x\to c} c_x = c$. It follows that

$$\lim_{x\to c} \frac{g'(c_x)}{f'(c_x)} = \lim_{x\to c} \frac{g'(x)}{f'(x)}.$$

Thus, the limit $\lim_{x\to c} \frac{g(x)}{f(x)}$ exists and

$$\lim_{x\to c} \frac{g(x)}{f(x)} = \lim_{x\to c} \frac{g'(x)}{f'(x)}.$$

□

Chapter 6

Sequences of functions

6.1 Pointwise and uniform convergence

In this chapter we will consider sequences and series of functions. This section is devoted to two types of convergence of sequences of functions.

Definition 6.1 Let $A \subseteq \mathbb{R}$ be a nonempty set. For each $n \in \mathbb{N}$, let $f_n : A \to \mathbb{R}$. We say that the sequence $(f_n)_{n=1}^{\infty}$ *converges pointwise* on A to a function $f : A \to \mathbb{R}$ if

$$\lim_{n\to\infty} f_n(x) = f(x) \quad \text{for all } x \in A.$$

In this case we say that f is the *pointwise limit* of the sequence $(f_n)_{n=1}^{\infty}$.

△

Example 6.2 Consider the sequence of functions $f_n(x) = x^n$, $n \in \mathbb{N}$, defined on $\mathbb{R}$.

For a fixed $x \in \mathbb{R}$ with $|x| > 1$, the sequence $(x^n)_{n=1}^{\infty}$ is unbounded, and thus $\lim_{n\to\infty} x^n$ does not exist. For $x = -1$, this is the sequence $(-1, 1, -1, 1, \dots)$, and again $\lim_{n\to\infty} x^n$ does not exist. If $-1 < x < 1$, then $\lim_{n\to\infty} x^n = 0$. Finally, if $x = 1$, then the sequence is the constant sequence $(1, 1, 1, 1, \dots)$, and $\lim_{n\to\infty} x^n = 1$.

Thus, the sequence $(f_n)_{n=1}^{\infty}$ converge pointwise on $(-1, 1]$, and

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0, & -1 < x < 1, \\ 1, & x = 1. \end{cases}$$

Notice that the limit function is discontinuous on $(-1, 1]$.

△

Example 6.3 For $n \in \mathbb{N}$ and $x \in \mathbb{R}$, let

$$f_n(x) = \sum_{k=0}^{n-1} x^k = \begin{cases} \frac{1-x^n}{1-x}, & x \neq 1, \\ n, & x = 1. \end{cases}$$

The sequence $(f_n)_{n=1}^{\infty}$ converge pointwise to $f(x) = \frac{1}{1-x}$ if $x \in (-1,1)$ and diverges for $x \leq -1$ or $x \geq 1$.

△

Example 6.4 For $n \in \mathbb{N}$ and $x \geq 0$, let $f_n(x) = x^{\frac{1}{n}}$. If $x > 0$, then $\lim_{n\to\infty} f_n(x) = 1$, and $\lim_{n\to\infty} f_n(0) = 0$. Thus, the pointwise limit of $(f_n)_{n=1}^{\infty}$ on $[0,\infty)$ is

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0, & x = 0, \\ 1, & x > 0. \end{cases}$$

△

These examples show, in particular, that the pointwise limit of a sequence of continuous functions need not be continuous.

Notice that a sequence $(f_n)_{n=1}^{\infty}$ converges pointwise on A to a function f, if

$$\forall\, \varepsilon > 0\ \forall\, x \in A\ \exists N \in \mathbb{N}\ \forall\, n \geq N : |f_n(x) - f(x)| < \varepsilon.$$

The number N here depends on $\varepsilon > 0$ and on $x \in A$.

A stronger condition is obtained if we require that N depends only on $\varepsilon > 0$ and does not depend on $x \in A$. Recall that we have used a similar idea to define uniform continuity in Section 4.3.

Definition 6.5 Let $A \subseteq \mathbb{R}$ be a nonempty set. For each $n \in \mathbb{N}$, let $f_n : A \to \mathbb{R}$. We say that the sequence $(f_n)_{n=1}^{\infty}$ *converges uniformly* on A to a function $f : A \to \mathbb{R}$ if

$$\forall\, \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall\, x \in A\ \forall\, n \geq N : |f_n(x) - f(x)| < \varepsilon.$$

In this case we say that f is the *uniform limit* of the sequence $(f_n)_{n=1}^{\infty}$.

△

Uniform convergence implies pointwise convergence. If the uniform limit exists, then it is also the pointwise limit. The converse is not true, in general.

Example 6.6 This is a continuation of Example 6.3. Consider $f_n(x) = \frac{1-x^n}{1-x}$, $x \in (-1,1)$, $n \in \mathbb{N}$. For any $r \in (0,1)$, the sequence $(f_n)_{n=1}^{\infty}$ converges uniformly on $[-r,r]$ to $f(x) = \frac{1}{1-x}$. Indeed, for $x \in [-r,r]$ we have

$$|f_n(x) - f(x)| = \left| \frac{1-x^n}{1-x} - \frac{1}{1-x} \right| = \frac{|x|^n}{1-x} \leq \frac{r^n}{1-r}.$$

We know that $\frac{r^n}{1-r} \to 0$ as $n \to \infty$. Thus, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\frac{r^n}{1-r} < \varepsilon$ for all $n \geq N$. For such n and for all $x \in [-r, r]$ we obtain

$$|f_n(x) - f(x)| < \varepsilon.$$

It follows that the convergence is uniform on $[-r, r]$.

Notice that f is uniformly continuous on $[-r, r]$.

$\triangle$

Theorem 6.7 *Let $(f_n)_{n=1}^{\infty}$ be a sequence of continuous functions on $A \subseteq \mathbb{R}$ that converges uniformly on A to a function f. Then f is continuous.*

$\triangle$

Proof. Let $a \in A$. We will show that f is continuous at a.

Given $\varepsilon > 0$, by the uniform convergence there exists $N \in \mathbb{N}$ such that

$$|f_N(t) - f(t)| < \frac{\varepsilon}{3} \quad \text{for all } t \in A.$$

Since f_N is continuous at a, there is $\delta > 0$ such that for $x \in A$

$$|x - a| < \delta \implies |f_N(x) - f_N(a)| < \frac{\varepsilon}{3}.$$

Now take $x \in A$ with $|x - a| < \delta$. For such x we have

$$|f(x) - f(a)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus, f is continuous at a.

$\square$

Example 6.8 This is a continuation of Example 6.4. We have seen that the sequence $\left(x^{\frac{1}{n}}\right)_{n=1}^{\infty}$ converges pointwise on $[0, \infty)$ to the function

$$f(x) = \begin{cases} 0, & x = 0, \\ 1, & x > 0. \end{cases}$$

Each of the functions $f_n(x) = x^{\frac{1}{n}}$ is continuous on $[0, \infty)$, but the function $f(x) = \lim_{n\to\infty} f_n(x)$ is not. It follows that the convergence is not uniform.

$\triangle$

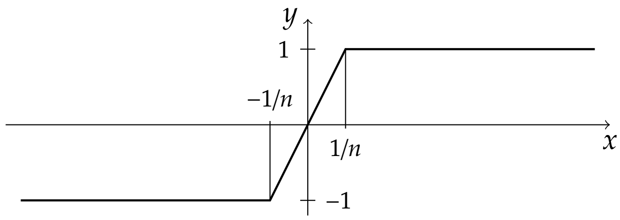

Figure 6.1: A graph of the function f_n

Example 6.9 For $n \in \mathbb{N}$ and $x \in \mathbb{R}$ we define

$$f_n(x) = \begin{cases} -1, & x \le -\frac{1}{n}, \\ nx, & -\frac{1}{n} < x < \frac{1}{n}, \\ 1, & x \ge \frac{1}{n}. \end{cases}$$

A graph of this function is shown in Figure 6.1.

The pointwise limit of the sequence $(f_n)_{n=1}^{\infty}$ is

$$f(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0. \end{cases}$$

As in the previous example, each function f_n is continuous on $\mathbb{R}$, but the limit function f is not. We conclude that the convergence is not uniform.

△

Recall the Cauchy Criterion (Theorem 2.45) for the convergence of sequences in $\mathbb{R}$. There is a corresponding variant for the uniform convergence of sequences of functions. Prove it as an exercise!

Theorem 6.10 (*The Cauchy Criterion for Uniform Convergence*) *A sequence of functions $(f_n)_{n=1}^{\infty}$ converges uniformly on a set $A \subseteq \mathbb{R}$ if and only if*

$$\forall\, \varepsilon > 0\; \exists N \in \mathbb{N}\; \forall\, n, m \ge N\; \forall\, x \in A : |f_n(x) - f_m(x)| < \varepsilon.$$

△

Now suppose that the sequence $(f_n)_{n=1}^{\infty}$ converges pointwise to a function f on a certain set A, and that each function f_n is differentiable. In general it is not true that the limit function f is differentiable, or that the derivative of the limit function f is equal to the limit of f_n'. This is not surprising: recall that even the property of continuity is not preserved under pointwise limits. However, the desired properties are valid in some situations. The theorem below states the conditions that guarantee that one can interchange the differentiation and the limit.

Theorem 6.11 (*The Differential Limit Theorem*) *Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions that are defined and differentiable on $[a,b]$. Suppose that*

(1) *$(f_n)_{n=1}^{\infty}$ converges pointwise on $[a,b]$ to a function $f : [a,b] \to \mathbb{R}$,*

(2) *$(f_n')_{n=1}^{\infty}$ converges uniformly on $[a,b]$ to a function $g : [a,b] \to \mathbb{R}$.*

Then f is differentiable on $[a,b]$ and $f' = g$. In particular,

$$f'(x) = \lim_{n\to\infty} f_n'(x), \quad x \in [a,b].$$

△

Proof. Let $c \in [a,b]$ be an arbitrary point. We want to show that $f'(c)$ exists and that $f'(c) = g(c)$. In other words, we want to show that

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = g(c).$$

Let $\varepsilon > 0$. We have to find $\delta > 0$ such that for any $x \in [a,b]$

$$0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - g(c) \right| < \varepsilon.$$

Since $\lim_{n\to\infty} f_n'(c) = g(c)$, there is $N_1 \in \mathbb{N}$ such that

$$|f_n'(c) - g(c)| < \frac{\varepsilon}{3} \quad \text{for all } n \geq N_1.$$

Since (f_n') converges uniformly on $[a,b]$, by Theorem 6.10 there is $N_2 \in \mathbb{N}$ such that

$$|f_n'(x) - f_m'(x)| < \frac{\varepsilon}{3} \quad \text{for all } n, m \geq N_2 \text{ and } x \in [a,b].$$

Put $N = \max\{N_1, N_2\}$. We have

$$|f_N'(c) - g(c)| < \frac{\varepsilon}{3},$$

$$|f_N'(x) - f_m'(x)| < \frac{\varepsilon}{3} \quad \text{for all } m \geq N \text{ and } x \in [a,b].$$

The function f_N is differentiable at c, and thus there is $\delta > 0$ such that for $x \in [a,b]$
$$0 < |x-c| < \delta \implies \left| \frac{f_N(x) - f_N(c)}{x-c} - f_N'(c) \right| < \frac{\varepsilon}{3}.$$
We will now show that this $\delta > 0$ satisfies our requirements.

Let $x \in [a,b]$ with $0 < |x-c| < \delta$, and let $m \geq N$. By the Mean Value Theorem (Theorem 5.6), there is a point z lying between x and c such that
$$\frac{(f_m(x) - f_N(x)) - (f_m(c) - f_N(c))}{x-c} = f_m'(z) - f_N'(z).$$
But by the above $|f_m'(z) - f_N'(z)| < \frac{\varepsilon}{3}$, so that
$$\left| \frac{(f_m(x) - f_N(x)) - (f_m(c) - f_N(c))}{x-c} \right| = \left| \frac{f_m(x) - f_m(c)}{x-c} - \frac{f_N(x) - f_N(c)}{x-c} \right| < \frac{\varepsilon}{3}.$$
Taking the limit as $m \to \infty$, we obtain
$$\left| \frac{f(x) - f(c)}{x-c} - \frac{f_N(x) - f_N(c)}{x-c} \right| \leq \frac{\varepsilon}{3}.$$
Finally, the inequalities above and the triangle inequality give
$$\begin{aligned} \left| \frac{f(x) - f(c)}{x-c} - g(c) \right| &\leq \left| \frac{f(x) - f(c)}{x-c} - \frac{f_N(x) - f_N(c)}{x-c} \right| \\ &\quad + \left| \frac{f_N(x) - f_N(c)}{x-c} - f_N'(c) \right| + |f_N'(c) - g(c)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$
for all $x \in [a,b]$ with $0 < |x-c| < \delta$ as desired.

□

6.2 Series of functions

Definition 6.12 Let $(f_k)_{k=1}^{\infty}$ be a sequence of functions $f_k : A \to \mathbb{R}$, and let $f : A \to \mathbb{R}$. We say that the series $\sum_{k=1}^{\infty} f_k$

(1) *converges pointwise* on A to f if the sequence of partial sums $\left(\sum_{k=1}^{n} f_k(x)\right)_{n=1}^{\infty}$ converges to $f(x)$ for each $x \in A$,

(2) *converges uniformly* on A to f if the sequence of partial sums $\left(\sum_{k=1}^{n} f_k\right)_{n=1}^{\infty}$ converges to f uniformly on A.

△

As in the case of series of real numbers, we say that a series $\sum_{k=1}^{\infty} f_k$ *converges absolutely* if the series $\sum_{k=1}^{\infty} |f_k|$ converges.

Example 6.13 Let $f_k(x) = x^k$, $k \in \mathbb{N} \cup \{0\}$, $x \in \mathbb{R}$. The series $\sum_{k=0}^{\infty} f_k(x) = \sum_{k=0}^{\infty} x^k$ does not converge at the whole of $\mathbb{R}$ because there exist values of $x \in \mathbb{R}$, for which $\sum_{k=0}^{\infty} x^k$ diverges.

The series $\sum_{k=0}^{\infty} x^k$ converges pointwise on the open interval $(-1, 1)$ to $f(x) = \frac{1}{1-x}$. Moreover, $\sum_{k=0}^{\infty} x^k$ converges absolutely and uniformly on any interval $[-r, r]$, where $r \in (0, 1)$. We will show this in a moment. Notice that $\sum_{k=0}^{\infty} x^k$ does not converge on $[-1, 1]$ as it does not converge for $x = -1$ and $x = 1$.

△

Note that a series series $\sum_{k=1}^{\infty} f_k$ converges uniformly on a set A to a function f if and only if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall x \in A \; \forall n \geq N : \left| f(x) - \sum_{k=1}^{n} f_k(x) \right| < \varepsilon.$$

The next statement contains a very useful test for the absolute and uniform convergence of a series.

Theorem 6.14 (*The Weierstrass M-Test*) *Suppose that for each $k \in \mathbb{N}$ there is a number $M_k \geq 0$ such that*

$$|f_k(x)| \leq M_k \quad \textit{for all } x \in A.$$

If $\sum_{k=1}^{\infty} M_k$ converges, then $\sum_{k=1}^{\infty} f_k$ converges absolutely and uniformly on A.

△

Proof. By the Comparison Test (Theorem 2.30), the series $\sum_{k=1}^{\infty} |f_k(x)|$ converges for each $x \in A$. Thus, $\sum_{k=1}^{\infty} f_k(x)$ converges absolutely for every $x \in A$. It remains to prove that the convergence is uniform.

Put $f(x) = \sum_{k=1}^{\infty} f_k(x)$, $x \in A$; this function is well-defined on A. Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $\sum_{k=n+1}^{\infty} M_k < \varepsilon$ for all $n \geq N$. But then for any $n \geq N$ and any $x \in A$ we have

$$\left| f(x) - \sum_{k=1}^{n} f_k(x) \right| = \left| \sum_{k=n+1}^{\infty} f_k(x) \right| \leq \sum_{k=n+1}^{\infty} |f_k(x)| \leq \sum_{k=n+1}^{\infty} M_k < \varepsilon.$$

This proves the uniform convergence.

□

Example 6.15 This is a continuation of Example 6.13. The absolute and uniform convergence of the series $\sum_{k=0}^{\infty} x^k$ on $[-r, r]$ with $r \in (0, 1)$ follows from the Weierstrass M-Test with $M_k = r^k$, $k \in \mathbb{N} \cup \{0\}$.

△

Example 6.16 Consider the series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$, $x \in \mathbb{R}$. You may recognize the Maclaurin series of the function e^x.

Take $r > 0$ and consider the interval $[-r, r]$. We have

$$\left|\frac{x^k}{k!}\right| \leq \frac{r^k}{k!} = M_k, \quad x \in [-r, r],\ k \in \mathbb{N} \cup \{0\}.$$

The series $\sum_{k=0}^{\infty} \frac{r^k}{k!}$ converges by the Ratio Test (Theorem 2.52), indeed,

$$L = \lim_{k\to\infty} \frac{r^{k+1}}{(k+1)!} \frac{k!}{r^k} = \lim_{k\to\infty} \frac{r}{k+1} = 0 < 1.$$

Thus, by the Weierstrass M-Test, the series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges absolutely and uniformly on $[-r, r]$.

Notice that the partial sums $\sum_{k=0}^{n} \frac{x^k}{k!}$ are polynomials. In particular, they are continuous. It follows from Theorem 6.7 that e^x is continuous on $[-r, r]$ with any $r > 0$. Consequently, e^x is continuous on $\mathbb{R}$. Furthermore, each partial sum is differentiable with

$$\left(\sum_{k=0}^{n} \frac{x^k}{k!}\right)' = \sum_{k=1}^{n} \frac{x^{k-1}}{(k-1)!} = \sum_{k=0}^{n-1} \frac{x^k}{k!}.$$

By above, the sequence of the derivatives $\left(\sum_{k=0}^{n-1} \frac{x^k}{k!}\right)_{n=1}^{\infty}$ converges uniformly on $[-r, r]$ to e^x. By Theorem 6.11, e^x is differentiable and $(e^x)' = e^x$.

△

Example 6.17 Consider the series $\sum_{k=0}^{\infty} \frac{\sin(kx)}{2^k}$. This is an example of a Fourier series. We have

$$\left|\frac{\sin(kx)}{2^k}\right| \leq \frac{1}{2^k} = M_k \quad \text{for all } x \in \mathbb{R} \text{ and } k \in \mathbb{N} \cup \{0\}.$$

The series $\sum_{k=0}^{\infty} \frac{1}{2^k}$ is a convergent geometric series. Hence, by the Weierstrass M-Test the series $\sum_{k=0}^{\infty} \frac{\sin(kx)}{2^k}$ converges absolutely and uniformly on $\mathbb{R}$. Moreover, its sum is continuous on $\mathbb{R}$.

△

6.3 Power series

Definition 6.18 Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers and $x_0 \in \mathbb{R}$. The series

$$\sum_{n=0}^{\infty} a_n(x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \cdots$$

is called a *power series*.

△

In case when $x_0 = 0$, a power series has the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots .$$

For each $n \in \mathbb{N}$, the partial sum

$$p_n(x) = \sum_{k=0}^{n} a_k(x-x_0)^k = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots + a_n(x-x_0)^n$$

is a polynomial of degree at most n.

For a fixed $x \in \mathbb{R}$, the sequence of the partial sums $(p_n(x))_{n=1}^{\infty}$ might or might not converge. It will always converge at $x = x_0$, because $p_n(x_0) = a_0$ for all $n \in \mathbb{N}$. On the other hand, if $a_n = 0$ for all $n \geq N$ with some $N \in \mathbb{N}$, then the series converges for all $x \in \mathbb{R}$. Another example of a power series that converges for all $x \in \mathbb{R}$ is $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

We are interested in investigating the *domain of convergence* of the power series $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ which is the set

$$\left\{ x \in \mathbb{R} : \sum_{n=0}^{\infty} a_n(x-x_0)^n \text{ converges} \right\}.$$

We will see that that the domain of convergence of a power series is always an interval centered at x_0. It can be the whole of $\mathbb{R}$, the single point $\{x_0\}$, or an interval of the form (x_0-R, x_0+R), $[x_0-R, x_0+R]$, $[x_0-R, x_0+R)$, or $(x_0-R, x_0+R]$ with some $R > 0$.

Example 6.19 Consider the series $\sum_{n=0}^{\infty} n!x^n$. It converges at $x = 0$. However, if $x \neq 0$, then it diverges by the Ratio Test (Theorem 2.52):

$$L = \lim_{n\to\infty} \frac{(n+1)!x^{n+1}}{n!x^n} = \lim_{n\to\infty} (n+1)x = \infty.$$

The domain of convergence is $\{0\}$.

△

Example 6.20 The power series $\sum_{n=0}^{\infty} x^n$ converges when $|x| < 1$ and diverges when $|x| \geq 1$. Its domain of convergence is $(-1, 1)$.

△

Example 6.21 Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{n}$. We know that it diverges if $x = 1$ (the harmonic series) and converges if $x = -1$ (the alternating harmonic series). To study its convergence for $x \in \mathbb{R}$ with $|x| \neq 1$, we apply the Ratio Test (Theorem 2.52):

$$L = \lim_{n\to\infty} \left|\frac{x^{n+1}}{n+1}\right| \left|\frac{n}{x^n}\right| = |x| \lim_{n\to\infty} \frac{n}{n+1} = |x| \begin{cases} < 1, & \text{if } |x| < 1, \\ > 1, & \text{if } |x| > 1. \end{cases}$$

Thus, the series converges if $|x| < 1$ and diverges if $|x| > 1$. The domain of convergence is $[-1, 1)$.

△

Theorem 6.22 (1) *Suppose that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges for a certain $x_1 \in \mathbb{R}$ with $x_1 \neq x_0$. Let $0 < r < |x_1 - x_0|$. Then $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely and uniformly on $[x_0 - r, x_0 + r]$.*

(2) *If $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ diverges for some $x_2 \in \mathbb{R}$, then it diverges for all $x \in \mathbb{R}$ with $|x - x_0| > |x_2 - x_0|$.*

△

This theorem implies that the domain of convergence is centered at x_0.

Proof. We first prove (1). Since the series $\sum_{n=0}^{\infty} a_n(x_1 - x_0)^n$ converges, we have $\lim_{n\to\infty} a_n(x_1 - x_0)^n = 0$. By the definition of the limit, there exists $N \in \mathbb{N}$ such that $|a_n(x_1 - x_0)^n| < 1$ for all $n \geq N$. It follows that

$$|a_n| < \frac{1}{|x_1 - x_0|^n} \quad \text{for all } n \geq N.$$

Now let $0 < r < |x_1 - x_0|$ and $x \in [x_0 - r, x_0 + r]$. Using the inequality $|x - x_0| \leq r$, we obtain

$$|a_n(x - x_0)^n| = |a_n| \cdot |x - x_0|^n \leq \frac{r^n}{|x_1 - x_0|^n} \quad \text{for all } n \geq N.$$

Notice that $\left|\frac{r}{x_1 - x_0}\right| < 1$, and therefore the geometric series $\sum_{n=0}^{\infty} \frac{r^n}{|x_1 - x_0|^n}$ converges. By the Weierstrass M-Test (Theorem 6.14) we conclude that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely and uniformly on $[x_0 - r, x_0 + r]$.

Statement (2) follows from (1). Indeed, suppose, by contradiction, that the series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges for some $x \in \mathbb{R}$ with $|x - x_0| > |x_2 - x_0|$. Then by (1) it would converge also at x_2, which contradicts the assumption.

□

Definition 6.23 The *radius of convergence* of a power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ is the number

$$R = \sup\left\{|x - x_0| : \sum_{n=0}^{\infty} a_n(x - x_0)^n \text{ converges}\right\}.$$

If $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges for all $x \in \mathbb{R}$, we put $R = \infty$.

△

Notice that

$$R = \inf\left\{|x - x_0| : \sum_{n=0}^{\infty} a_n(x - x_0)^n \text{ diverges}\right\}.$$

If $R = 0$, the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges only at $x = x_0$. If $R > 0$, then the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely and uniformly on any interval $[x_0 - r, x_0 + r]$ with $0 < r < R$. It converges absolutely on $(x_0 - R, x_0 + R)$, but might not converge uniformly on this interval. It might or might not converge at the points $x_0 - R$ and $x_0 + R$.

The partial sums $\sum_{k=0}^{n} a_k(x - x_0)^k$ are polynomials and therefore differentiable. It can be shown (we will not do this in our course) that $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ and the differentiated series $\sum_{n=1}^{\infty} na_n(x - x_0)^{n-1}$ have the same radius of convergence. The Differential Limit Theorem (Theorem 6.11) says that $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ is differentiable on $(x_0 - R, x_0 + R)$, and can be differentiated term-by-term.

We finish this section by presenting two useful formulas for the radius of convergence.

Theorem 6.24 *Let $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ be a power series with the radius of convergence R.*

(1) *If* $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$ *exists, then*

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|.$$

(2) *If* $\lim_{n\to\infty}|a_n|^{\frac{1}{n}}$ *exists, then*

$$\frac{1}{R} = \lim_{n\to\infty}|a_n|^{\frac{1}{n}}.$$

△

Proof. To prove (1), we will use the Ratio Test (Theorem 2.52). Put

$$L = \lim_{n\to\infty}\frac{|a_{n+1}(x-x_0)^{n+1}|}{|a_n(x-x_0)^n|} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right||x-x_0|.$$

We have

$$\begin{aligned} L<1 &\iff |x-x_0| < \frac{1}{\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|}, \\ L>1 &\iff |x-x_0| > \frac{1}{\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|}. \end{aligned}$$

This gives

$$R = \frac{1}{\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|}.$$

The proof of (2) is similar. It uses the Root Test (Theorem 2.55). We have

$$L = \lim_{n\to\infty}|a_n(x-x_0)^n|^{\frac{1}{n}} = \lim_{n\to\infty}|a_n|^{\frac{1}{n}}|x-x_0|.$$

By a similar argument we conclude that

$$R = \frac{1}{\lim_{n\to\infty}|a_n|^{\frac{1}{n}}}.$$

□

6.4 Taylor series

Let $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ be a power series with radius of convergence $R > 0$. For $0 < r < R$, the series converges absolutely and uniformly on $[x_0 - r, x_0 + r]$ to a function f. This defines f on the interval $(x_0 - R, x_0 + R)$. Let us establish a connection between the function f and the coefficients $(a_n)_{n=0}^{\infty}$.

We have

$$f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \cdots.$$

Substituting $x = x_0$ in this formula gives $f(x_0) = a_0$.

As discussed in the previous section, the function f is differentiable on $(x_0 - R, x_0 + R)$, and the differentiated series $\sum_{n=1}^{\infty} na_n(x-x_0)^{n-1}$ converges absolutely and uniformly on $[x_0 - r, x_0 + r]$ to $f'(x)$. Thus, for $x \in (x_0 - R, x_0 + R)$ we have

$$f'(x) = a_1 + a_2(x-x_0) + a_3(x-x_0)^2 + a_4(x-x_0)^3 + \cdots.$$

Substituting $x = x_0$ in this formula gives $f'(x_0) = a_1$.

Using induction, one can prove that f can be differentiated as many times as we please, and for $m \in \mathbb{N}$ the differentiated series

$$\sum_{n=m}^{\infty} n(n-1)\cdots(n-m+1)a_n(x-x_0)^{n-m}$$

converges absolutely and uniformly on $[x_0 - r, x_0 + r]$ to $f^{(m)}(x)$. Thus, for $x \in (x_0 - R, x_0 + R)$ we have

$$f^{(m)}(x) = m(m-1)\cdots 1a_m + (m+1)m\cdots 2a_{m+1}(x-x_0) + \cdots.$$

Substituting $x = x_0$ in this formula, we see that

$$a_m = \frac{f^{(m)}(x_0)}{m!}, \quad m \in \mathbb{N} \cup \{0\}.$$

Thus,

$$f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n, \quad x \in (x_0 - R, x_0 + R).$$

We recognize that the latter series is the *Taylor series* of f at the point x_0. In the special case when $x_0 = 0$ it is also called the *Maclaurin series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

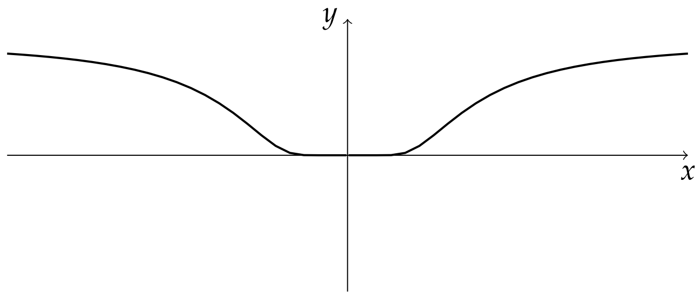

Figure 6.2: A graph of the function g_0

Let us summarize what we just did. We started with a power series that converges on the interval $(x_0 - R, x_0 + R)$ and defined the function f to be its sum. Then we saw that the power series must be the Taylor series of f.

Now let us instead suppose that a function $f : (x_0 - R, x_0 + R) \to \mathbb{R}$ is given and that f has derivatives of all orders on $(x_0 - R, x_0 + R)$. We can then formally write down the Taylor series of f

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

Unfortunately, it may happen that this series does not converge to f on $(x_0 - R, x_0 + R)$. For a particular $x \in (x_0 - R, x_0 + R)$ it can happen that the series diverges, or that it converges to a value different from $f(x)$.

Example 6.25 Consider the function

$$g_0(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

A graph of this function is shown in Figure 6.2.

It is possible to prove that $g_0^{(n)}(0) = 0$ for all $n \in \mathbb{N} \cup \{0\}$. It follows that its Taylor series at $x_0 = 0$ is

$$\sum_{n=0}^{\infty} \frac{g_0^{(n)}(0)}{n!} x^n = 0 \quad \text{for all } x \in \mathbb{R}.$$

It does not converge to $g_0(x)$ if $x \neq 0$.

This example can be generalized. Assume that a function f has a power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ that converges to f on $(x_0 - R, x_0 + R)$. Take $\alpha \in \mathbb{R}$ and consider the function

$$g(x) = \begin{cases} f(x) + \alpha e^{-\frac{1}{(x-x_0)^2}}, & x \neq x_0, \\ f(x_0), & x = x_0. \end{cases}$$

By above, $f^{(n)}(x_0) = g^{(n)}(x_0)$ for all $n \in \mathbb{N} \cup \{0\}$. This means that f and g have the same Taylor series at x_0. Since $\alpha \in \mathbb{R}$ can be chosen arbitrarily, we see that there are infinitely many functions having the same Taylor series at x_0.

△

Let us now address the following task. Let $f : (x_0 - R, x_0 + R) \to \mathbb{R}$ be a given function that has derivatives of all orders on $(x_0 - R, x_0 + R)$. How can we check whether its Taylor series converges to f? And it if converges, how fast?

For $n \in \mathbb{N}$, the partial sum of the Taylor series

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

is called the *Taylor polynomial* of order n. We study the *reminder*

$$R_n(x) = f(x) - T_n(x).$$

For a given x, the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$ converges to $f(x)$ if and only if

$$\lim_{n\to\infty} R_n(x) = 0.$$

There are several different formulas for the reminder $R_n(x)$. We are going to prove one of them.

Theorem 6.26 *(Taylor's Theorem with Lagrange's Form of the Reminder)* *Let $f : (x_0 - R, x_0 + R) \to \mathbb{R}$ be a function that is $n+1$ times differentiable on $(x_0 - R, x_0 + R)$. For any $x \in (x_0 - R, x_0 + R)$, there exists a point c_x that lies between x and x_0 such that*

$$R_n(x) = \frac{f^{(n+1)}(c_x)}{(n+1)!}(x - x_0)^{n+1}.$$

△

According to Taylor's Theorem,

$$\begin{aligned} f(x) =& f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f^{(3)}(x_0)}{3!}(x - x_0)^3 + \cdots \\ &+ \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \frac{f^{(n+1)}(c_x)}{(n+1)!}(x - x_0)^{n+1}. \end{aligned}$$

Proof. Let $x \in (x_0 - R, x_0 + R)$ be a fixed number. For $t \in (x_0 - R, x_0 + R)$, consider the functions

$$F(t) = (t - x)^{n+1}, \quad G(t) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k.$$

We have

$$F(x_0) = (x_0 - x)^{n+1}, \quad F(x) = 0, \quad F'(t) = (n+1)(t - x)^n.$$

Moreover,

$$G(x_0) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k, \quad G(x) = f(x)$$

and

$$\begin{aligned} G'(t) &= \sum_{k=0}^{n} \left(\frac{f^{(k+1)}(t)}{k!}(x - t)^k - \frac{f^{(k)}(t)}{k!} k (x - t)^{k-1} \right) \\ &= \sum_{k=0}^{n} \frac{f^{(k+1)}(t)}{k!}(x - t)^k - \sum_{k=0}^{n-1} \frac{f^{(k+1)}(t)}{k!}(x - t)^k = \frac{f^{(n+1)}(t)}{n!}(x - t)^n. \end{aligned}$$

By the Generalized Mean Value Theorem (Theorem 5.7), there is a point c_x between x and x_0 such that

$$(F(x) - F(x_0))G'(c_x) = F'(c_x)(G(x) - G(x_0)).$$

Substituting the above formulas, we obtain

$$\begin{aligned} &\left(0 - (x_0 - x)^{n+1}\right) \frac{f^{(n+1)}(c_x)}{n!}(x - c_x)^n \\ &\qquad = (n+1)(c_x - x)^n \underbrace{\left(f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right)}_{=R_n(x)}, \end{aligned}$$

so that

$$R_n(x) = \frac{-(-1)^{n+1}(x - x_0)^{n+1} f^{(n+1)}(c_x)(x - c_x)^n}{n!(n+1)(-1)^n (x - c_x)^n} = \frac{f^{(n+1)}(c_x)}{(n+1)!}(x - x_0)^{n+1}$$

as desired.

□

Corollary 6.27 (*Taylor's Inequality*) *If $|f^{(n+1)}(x)| \le M$ for all $x \in [x_0 - r, x_0 + r]$ with some $M > 0$, then*

$$|R_n(x)| \le \frac{Mr^{n+1}}{(n+1)!}, \quad x \in [x_0 - r, x_0 + r].$$

△

Example 6.28 Consider the function $f(x) = \sin x$, $x \in \mathbb{R}$.

We know that its Taylor series at the point 0 is

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}.$$

We are going to discuss the convergence of this series.

We can use the Ratio Test (Theorem 2.52) to show that this series converges absolutely for all $x \in \mathbb{R}$. Indeed,

$$L = \lim_{k\to\infty} \frac{|x|^{2k+3}}{(2k+3)!} \frac{(2k+1)!}{|x|^{2k+1}} = |x|^2 \lim_{k\to\infty} \frac{1}{(2k+3)(2k+2)} = 0 < 1$$

for all $x \in \mathbb{R}$.

Our next aim is to show that the sum of the series is indeed $\sin x$. We are going to use Corollary 6.27. All derivatives of the function $f(x) = \sin x$ have the form $\pm \sin x$ or $\pm \cos x$. Thus, we definitely have

$$|f^{(n)}(x)| \le 1 \quad \text{for all } x \in \mathbb{R} \text{ and } n \in \mathbb{N}.$$

This gives us

$$|R_n(x)| \le \frac{|x|^{n+1}}{(n+1)!} \quad \text{for all } x \in \mathbb{R} \text{ and } n \in \mathbb{N}.$$

We conclude that

$$\lim_{n\to\infty} R_n(x) = 0 \quad \text{for any } x \in \mathbb{R}.$$

This implies that

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}, \quad x \in \mathbb{R}.$$

The above estimate for $|R_n(x)|$ can be also used to determine the number of summands needed to calculate $\sin x$ for a given x with a given precision. As an example, let us calculate $\sin 1$ within 4 digits. For, we need to find n such that

$$|R_n(1)| < \frac{1}{2} \cdot 10^{-4}.$$

Using the estimate $|R_n(1)| \leq \frac{1}{(n+1)!}$, we see that it suffices to find n such that

$$\frac{1}{(n+1)!} < \frac{1}{2} \cdot 10^{-4},$$

or, equivalently,

$$(n+1)! > 20\,000.$$

Calculating 1!, 2! and so on, we find that $8! = 40\,320$, so we take $n = 7$. The desired approximation is

$$\sin 1 \approx 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} = 0.8415.$$

$\triangle$

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